The Macroeconomic Impact of Sustainable Investing

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Abstract

We investigate the impact of sustainable investing—the investing practice that accounts for environmental/societal/governance factors—in a macroeconomic model with dynamic general equilibrium forces. Preference-driven sustainable investing impacts firms' cost of capital and investors' saving incentives, resulting in a dynamic feedback loop between productive capital reallocation and investor wealth ownership. We investigate the magnitude and timing of the impact in a calibrated quantitative model where sustainable preferences are disciplined by estimated portfolio return differentials. We finally present a policy implementation exercise that highlights the potential and the limits of sustainable investing as a market-driven mechanism to deliver policy-relevant macroeconomic impact.

Keywords: sustainable investing, ESG, portfolio choice, impact investing, carbon emissions, climate finance

JEL codes: E10, G11, O40, Q50

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1 Introduction

Sustainable investing refers to a range of strategies in which financial investors include environmental, social responsibility, and corporate governance considerations in their investment decisions. An important rationale for such practices has been attributed to the nonpecuniary returns investors may perceive by aligning their portfolio allocations with their values, as emphasized by Starks (2023). The appeal of the so-called preference-driven sustainable investing rests on a simple but far-reaching premise: financial markets can translate investors' preferences into long-lasting economic outcomes. In this paper we put that premise to the test and ask: how can preference-driven sustainable investing ultimately impact the allocation of productive capital in the macroeconomy?

Recent empirical studies document that investors value characteristics of their holdings beyond their financial payoffs, and are thus willing to sacrifice expected returns.[1] Similarly, a growing literature has shown that investors' willingness to trade pecuniary returns for nonpecuniary benefits can be capitalized into asset prices, generate cross-sectional differences in required returns, and affect firms' cost of capital.[2] Yet the broader promise of sustainable investing is ultimately a macroeconomic one: for preference-induced changes in portfolio allocations and asset prices to generate a lasting impact, they must interact with firms' investment decisions, households' saving behavior, and the evolution of wealth across investors. Much less is known about these forces once they are jointly determined in general equilibrium over time. This paper aims at filling this gap.

We study these forces in a dynamic macroeconomic model with heterogeneous firms and investors. Our model features *Advocate* investors, who derive nonpecuniary returns from holding firms with characteristics they value, and *Benchmark* investors, who care only about pecuniary returns. In equilibrium, Advocates accept lower pecuniary returns on firms whose characteristics they value more, which lowers those firms' cost of capital and increases their investment. This is the *composition* effect that typically emerges in models of sustainable investing impact. Because Advocates derive utility from the characteristics of the wealth they hold, however, wealth carries a nonpecuniary return in addition to its future consumption value. This generates a *nonpecuniary saving motive*: other things equal, Advocates have a stronger incentive to accumulate financial wealth. Their relative wealth therefore evolves according to the balance between stronger saving

[1] See Riedl and Smeets (2017); Bauer et al. (2021); Barber et al. (2021). We discuss this literature below.

[2] A non-exhaustive list of recent contributions includes Berk and van Binsbergen (2025); Gillan et al. (2021); Humphrey et al. (2021); Idzorek et al. (2021); Kedia et al. (2021); Atz et al. (2023); Berg et al. (2022); Caramichael and Rapp (2024). See Pástor et al. (2025) for a survey of the sustainable-investing literature.

motives and lower financial returns. As the wealth share of Advocates changes, so does the set of firms for which they are the marginal investors, shifting the ownership threshold and therefore the strength and reach of the composition effect over time. We refer to this mechanism as the *wealth–ownership feedback*.

Our model thus turns sustainable investing into an endogenous dynamic process: preferences affect asset prices and capital allocation today, while the resulting saving and return differentials determine the financial capacity of Advocates to affect capital allocation tomorrow. This wealth–ownership feedback determines how persistent and how far-reaching the composition effect becomes, and it provides the link from individual sustainable-investing preferences to the evolving allocation of productive capital in the macroeconomy.

We present a calibrated version of the model to assess the potential macroeconomic impact of sustainable investing. We discipline the strength of investors' nonpecuniary preferences using empirical estimates of the return investors are willing to forgo to hold assets with valued sustainability characteristics, and parameterize the cross-sectional distribution of firm characteristics using observed firm-level data. The calibrated model allows us to quantify not only the eventual reallocation of productive capital generated by investor preferences, but also the speed at which that reallocation takes place as Advocate investors accumulate wealth and expand the set of firms whose cost of capital they influence. The quantitative exercises show that sustainable investing can generate a substantial expansion of productive capacity, concentrated at firms with highly valued characteristics. Asset prices respond immediately, while the wealth–ownership feedback gradually broadens the real effects as Advocate investors accumulate wealth. We compare two applications in which investors value firms' ESG characteristics and lower emissions intensity, respectively. The emissions application shows that a substantial expansion of productive capital need not translate into a reduction in total emissions.

We finally present a policy implementation exercise where we ask whether the portfolio and saving choices of Advocate investors could implement the capital allocation of a carbon-emission target reduction policy, both in the steady state and along the dynamic transition path. Our results show that the wealth-ownership feedback combined with the presence of Benchmark investors prevents sustainable investing from implementing the policy allocation, underscoring the limits of the promise carried by such strategies to deliver a market-driven solution to complex societal problems.

Connection to the Literature The literature on sustainable investing is extensively surveyed in Starks (2023) and Pástor et al. (2025). These reviews distinguish financial motives for incorporating sustainability into investment decisions from nonpecuniary preferences for sustainable holdings, and discuss their implications for portfolio choice and asset prices. We contribute to the literature by focusing on how nonpecuniary preferences translate into real economic outcomes when saving, investor wealth, and productive capital allocation evolve jointly.

Our analysis is motivated by the empirical evidence that nonpecuniary motives influence investment decisions and can lead investors to accept lower financial returns in exchange for holding assets consistent with their values, see Riedl and Smeets (2017), Bauer et al. (2021), and Heeb et al. (2023). Estimates of the associated financial tradeoffs vary substantially across assets, institutional settings, investor populations, and time periods, as documented by Barber et al. (2021) and Baker et al. (2024).[3] Motivated by this evidence, our quantitative analysis takes an initial annual pecuniary return differential of 20 basis points for Advocate investors as an illustrative benchmark, with sensitivity analysis at 10 and 40 basis points.

Our paper is connected to Pástor et al. (2021), with whom our model shares the mechanism through which heterogeneous sustainability preferences affect required returns and redirect productive capital toward firms valued by sustainable investors. They study this mechanism through portfolio choice under risk, whereas our setting is deterministic; in both cases, nonpecuniary tastes generate a willingness to accept lower financial returns for holding valued assets.[4] Our contribution is to endogenize the saving decisions and wealth distribution that determine the strength of this composition effect over time. Sustainable investors earn lower pecuniary returns but also face a nonpecuniary incentive to accumulate wealth. The interaction of these forces changes their rela-

[3] Riedl and Smeets (2017) link socially responsible fund holdings to social preferences and social signaling, while Bauer et al. (2021) find support for sustainable pension policies even among participants who expect lower financial returns. Heeb et al. (2023) document willingness to pay for sustainable investments but limited sensitivity to the absolute magnitude of their environmental impact. Quantitatively, Barber et al. (2021) estimate that investors accept expected internal rates of return 2.5–3.7 percentage points lower for impact-oriented venture capital funds. The earlier version of Baker, Egan, and Sarkar, *How Do Investors Value ESG?* (Baker et al., 2022), reports average willingness to pay of 20 basis points annually for an ESG mandate. Their revised paper, *Demand for ESG* (Baker et al., 2024), estimates a U.S. premium rising from 3 basis points in 2019 to 17 in 2020 before falling to −33 in 2023, while European willingness to pay remains positive. Giglio et al. (2025) further show that ethical motives influence ESG portfolio choices, but participation and allocations remain strongly associated with investors' return expectations, including among those reporting ethical motives.

[4] In Pástor et al. (2021), risk disciplines portfolio demand by keeping portfolio positions well defined, while sustainability tastes generate differences in expected returns beyond those explained by market risk, also known as "alphas." In our deterministic setting, short-sale and borrowing constraints keep portfolio choices well defined, and differences in financial returns reflect nonpecuniary valuations, akin the alphas in Pástor et al. (2021). The resulting portfolio return differential between Advocate and Benchmark investors is the relevant counterpart for our analysis of relative wealth accumulation. We do consider a version of our model with risk as part of robustness extensions and we show that our main endogenous mechanism operates under risk as well.

tive financial capacity, which feeds back into ownership, required returns, and firms' investment. This wealth–ownership feedback makes the magnitude and persistence of capital reallocation an equilibrium outcome.

Our paper relates to dynamic macro-finance models of the green transition. Hambel et al. (2024) study how asset diversification interacts with climate policy and physical climate risks, showing that diversification can initially support, but eventually impede, further decarbonization. Hambel and van der Ploeg (2025) show how uncertainty about future climate policy generates carbon premiums, affects capital allocation, and strengthens precautionary saving. These papers analyze the green transition through the financial consequences of climate risks and policy in economies with a representative household. Our focus is on taste-driven sustainable investing: investors derive nonpecuniary benefits from their holdings, and their consumption, saving, and portfolio choices jointly determine how these preferences translate into productive capital allocation over time.

Our paper also relates to the literature on sustainability preferences, climate risk, and asset prices such as Pedersen et al. (2021); Karydas and Xepapadeas (2022); Bolton and Kacperczyk (2021, 2023); Hsu et al. (2023); Zhang (2025). A common theme in these contributions is how environmental characteristics enter investors' financial decisions and are reflected in asset prices and returns, through exposure to climate and regulatory risks, information about firm fundamentals, or sustainability preferences. We complement this literature by isolating how sustainability preferences affect saving incentives even in the absence of uncertainty, generating an endogenous feedback between investor wealth, firm ownership, and productive capital.

Finally, Chittaro et al. (2025) provide a closely related analysis of taste-driven sustainable investing and its connection to carbon taxation. They characterize the asset-return schedule that replicates a carbon tax and show how investors without environmental preferences limit the return penalties that markets can impose on polluting firms. Their framework allows aggregate saving to respond to nonpecuniary returns, but pools resources across investor types and assigns green investors a fixed share of investable wealth. Our framework makes this share endogenous, allowing differences in saving and portfolio returns to change the financial capacity of both sustainable and conventional investors. Building on their connection between asset returns and carbon taxation, we develop a policy implementation exercise which examines how this evolving wealth distribution shapes the ability of sustainable investing to achieve an emissions target, both in the long run and during the transition.

The rest of the paper is organized as follows. Section 2 introduces the key features of our model and isolates the composition and nonpecuniary saving motive in the context of a two-period economy. Section 3 studies the wealth-ownership feedback and its dynamic impact on capital composition in a fully dynamic economy. Section 4 presents our main quantitative analysis and results. Section 5 studies the policy implementation exercise. The Appendix contains the derivations of the main results of the paper, the model extensions, and the computational details.

2 Sustainability Preferences, Capital Composition, and Saving

We begin with a two-period economy to illustrate how sustainable-investing preferences affect both the allocation of productive capital and investors' saving decisions. Advocate investors are willing to accept lower pecuniary returns on firms whose characteristics they value more, reducing those firms' cost of capital and generating a composition effect. At the same time, the nonpecuniary benefits from their holdings provide an additional motive to accumulate wealth. These two mechanisms form the basis of the wealth–ownership feedback studied in Section 3, where saving and portfolio returns jointly determine the evolution of investors' relative financial capacity.

Firms. Firms are indexed by a characteristic $g \in [0,1]$, with density $\phi(g)$. The characteristic g is oriented so that higher values are more attractive to Advocate investors. It can, for example, represent an ESG score or a decreasing transformation of carbon emissions intensity.

All firms have the same technology, $y = f(k)$ with $f' > 0$ and $f'' < 0$, and begin with common capital k_0.[5] Capital fully depreciates after production. Let $R(g)$ denote the pecuniary return required by the marginal investor in firm g. Firm value maximization implies

$$f'(k(g)) = R(g), \tag{1}$$

and the ex-dividend equity price is

$$q(g) = \frac{f(k(g))}{R(g)}. \tag{2}$$

Differences in required returns therefore translate into differences in productive capital across otherwise identical firms.

[5]The common technology can be interpreted as the aggregate production function of a group of firms with characteristic g, each of which may differ in productivity. If the conditional productivity distribution is the same across g, efficient within-group capital allocation yields a common representative technology. Capital $k(g)$ then denotes group-average capital, so equal capital across characteristics does not require equal capital across individual firms. Finally, the aggregate productivity within group g is normalized to one because it is the same across all groups. Appendix H provides the details for this interpretation.

Households and Sustainable Wealth. There are two households, Advocate and Benchmark. Households' portfolio holdings for firm g are denoted by $z^A(g)$, $z^B(g)$, respectively. We prohibit short sales, so holdings must be nonnegative, and market clearing requires $z^A(g) + z^B(g) = 1$. We let $z(g) = z^A(g)$, so $z^B(g) = 1 - z^A(g)$. Advocate derives utility from sustainable wealth which we define as

$$G = \int_0^1 v(g)q(g)z(g)\phi(g)\,dg. \tag{3}$$

The sustainability weight $v(g)$ values individual firm characteristics; it is continuously differentiable and strictly increasing, with $v(g_n) = 0$ for a neutral characteristic $g_n \in (0, 1)$. Holdings above g_n contribute positively to sustainable wealth; holdings below it contribute negatively.[6]

Advocate preferences can then be represented by

$$U^A = u(c_0^A) + \beta u(c_1^A) + \mathcal{V}(G), \tag{4}$$

where $\beta > 0$, $u' > 0$, $u'' < 0$, $\mathcal{V}' > 0$, and $\mathcal{V}'' \leq 0$. The curvature of $\mathcal{V}$ governs diminishing marginal utility from sustainable wealth, so the marginal nonpecuniary value of an additional unit of wealth invested in firm g is $\mathcal{V}'(G)v(g)$. Benchmark values only consumption, so its preferences are simply: $U^B = u(c_0^B) + \beta u(c_1^B)$.

Let W_0 denote aggregate initial cum-dividend wealth and $w_0 \in (0, 1)$ Advocate's share, implemented through proportional initial ownership of firms. Households allocate this wealth between period-0 consumption and equity purchases, then consume portfolio dividends in period 1. The full derivation of the optimality conditions and the equilibrium are reported in Appendix A.

Portfolio Allocation and the Composition Effect. For a firm held in positive quantity, Advocate's portfolio condition is

$$\frac{u'(c_0^A)}{\beta u'(c_1^A)} = R(g) + \frac{m}{\beta} v(g), \tag{5}$$

where m is defined as

$$m \equiv \frac{\mathcal{V}'(G)}{u'(c_1^A)} > 0. \tag{6}$$

[6]The density $\phi(g)$ and loading $v(g)$ jointly accommodate different empirical measures of sustainability. If g is an ESG score, ϕ describes its distribution and v its valuation by Advocate. If $g = T(e)$ is a decreasing transformation of, say, carbon emissions, e, both v and ϕ must be expressed in the transformed characteristic. Relabeling firms leaves the economic environment unchanged when the density and weights are transformed consistently. With a continuous characteristic distribution, one can alternatively use its sustainability rank as the index so that $\phi(g) = 1$ by construction, and the empirical heterogeneity is represented by the mappings from rank to emissions and to sustainability value. Our quantitative analysis in Section 4 exploits such flexibility.

and it represents Advocate's marginal sustainable wealth valuation. Benchmark's portfolio condition is the familiar

$$\frac{u'(c_0^B)}{\beta u'(c_1^B)} = R(g). \tag{7}$$

Note that, at zero equity holdings, the marginal benefit of purchasing the asset cannot exceed its marginal cost, so the portfolio conditions hold with the inequality $\geq$.

Because $m > 0$ and v is strictly increasing, Advocate's relative willingness to hold a firm increases with g. With positive saving by both households, this single-crossing property implies a threshold allocation,

$$z(g) = \begin{cases} 0, & \text{if} \quad g < g^*, \\ 1, & \text{if} \quad g \geq g^*. \end{cases} \tag{8}$$

where the ownership at the measure-zero cutoff g^* is immaterial. Let $R^* = R(g^*)$, Benchmark pricing then gives $R(g) = R^*$ below the threshold g^*, while Advocate pricing gives

$$R(g) = R^* - \frac{m}{\beta}[v(g) - v(g^*)], \qquad g \geq g^*. \tag{9}$$

Combining this return schedule with the firm optimality condition yields the capital allocation

$$k(g) = \begin{cases} (f')^{-1}(R^*), & g < g^*, \\ (f')^{-1}\left(R^* - \dfrac{m}{\beta}[v(g) - v(g^*)]\right), & g \geq g^*. \end{cases} \tag{10}$$

Condition (10) shows that capital is flat across Benchmark-priced firms and increasing in g across Advocate-priced firms. In other words, the capital composition effect takes a "hockey-stick" shape. In summary, Advocate's nonpecuniary preferences affect its portfolio allocation, reducing the equilibrium returns and thus the cost of capital for firms in its portfolio, which ultimately leads to an impact on productive capital allocation.

Nonpecuniary Saving Motive. Together with choosing the composition of their portfolios, households determine how much to save. Let a_1^A (resp. a_1^B) denote the wealth carried into period 1 by Advocate (resp. Benchmark). The pecuniary return on Advocate's saving can be thus defined as $R_w^A \equiv c_1^A / a_1^A$. Integrating Advocate's portfolio condition over its holdings results in the Euler

condition for saving

$$\frac{u'(c_0^A)}{\beta u'(c_1^A)} = R_w^A + \frac{m}{\beta}\frac{G}{a_1^A}. \tag{11}$$

The right hand side of (11) shows that alongside the typical pecuniary return in marginal utility of future consumption, R_w^A, saving provides the nonpecuniary return proportional to $G\mathcal{V}'(G)$. This nonpecuniary return is positive when $G > 0$ and negative when $G < 0$.

How Advocate saving compares to Benchmark is determined by Advocate's net effective saving return relative to Benchmark. Integrating the return schedule (9) over Advocate's holdings allows one to express the net effective saving return as

$$\frac{m}{\beta}v(g^*) = R_w^A - R^* + \frac{m}{\beta}\frac{G}{a_1^A}$$

Recall that Advocate accepts a lower average pecuniary return, $R_w^A - R^*$, because it holds firms with $v(g) > v(g^*)$. The nonpecuniary return offsets this disadvantage, and the net effective saving return is positive, zero, or negative according to whether g^* is above, at, or below g_n. In particular, the nonpecuniary saving return can remain positive when $g^* < g_n$, provided $G > 0$, even though Advocate's effective return is then below Benchmark's. At $g^* = g_n$, the two return components exactly offset.

These return comparisons describe the incentives to carry wealth into the next period, but relative saving also depends on consumption preferences, initial wealth, and equilibrium prices. Under the log-utility, which we use in our numerical illustration below, the saving-rate formulas imply that Advocate saves a larger fraction of initial wealth than Benchmark exactly when $G > 0$. More generally, expanding Advocate's portfolio below g_n adds negatively valued holdings, so the nonpecuniary return depends on the composition and valuation of the entire portfolio and on the curvature of $\mathcal{V}$.

Wealth and the Ownership Threshold. In the equilibrium of the two-period economy, the ownership threshold g^* is jointly determined with asset prices and saving strategies. Let s^A and s^B denote the fractions of initial wealth saved by Advocate and Benchmark respectively. Asset-market

clearing requires

$$a_1^A = \int_{g^*}^1 q(g)\phi(g)\,dg = s^A w_0 W_0, \tag{12}$$

$$a_1^B = \int_0^{g^*} q(g)\phi(g)\,dg = s^B(1-w_0)W_0. \tag{13}$$

Taking the ratio one gets

$$\frac{\int_{g^*}^1 q(g)\phi(g)\,dg}{\int_0^{g^*} q(g)\phi(g)\,dg} = \frac{s^A}{s^B}\frac{w_0}{1-w_0}. \tag{14}$$

which shows how asset prices, saving, and threshold g^* affect one another in equilibrium. Suppose for the moment that saving rates are equal across households, and asset prices are equal across all firms, equation (14) would reduce to

$$\Phi(g^*) = 1 - w_0, \tag{15}$$

where Φ is the cdf of ϕ. In other words, the initial fraction of wealth of Advocate would correspond to the fraction of firms above g^*. Two forces move the threshold away from this counterfactual allocation. On the one hand, higher relative valuations of high-g firms allow a given amount of Advocate wealth to purchase fewer firms, thereby raising g^*. On the other hand, a higher Advocate saving rate increases its share of wealth to be carried into next period and thus lowers g^*. The two competing effects encapsulate the tension that shapes the dynamics of Advocate wealth and its impact on the ownership threshold g^*: higher prices (lower pecuniary returns) reduce its financial capacity while the higher saving rate increases it. Ultimately, the evolution of g^* over time depends on the combined effect of these two forces.

Numerical Illustration. We provide next a numerical illustration of the equilibrium mechanisms just described. We set $f(k) = k^\alpha$, $\phi(g) = 1$, and $u(c) = \log c$, which implies $s^A = \frac{\beta + G\mathcal{V}'(G)}{1+\beta+G\mathcal{V}'(G)}$, $s^B = \frac{\beta}{1+\beta}$. Utility from sustainable wealth is specified such that $\mathcal{V}'(G) = (1+\kappa G)^{-\sigma_G}$ with $1+\kappa G > 0$. We set $\kappa = 4.64$ for this illustration and $\sigma_G = 0.5$, as maintained in our quantitative analysis, $\alpha = 0.33$, and $\beta = 1$. We specify linear preferences over the firm characteristic g, namely

$$v(g) = \nu\frac{g - g_n}{1 - g_n}, \tag{16}$$

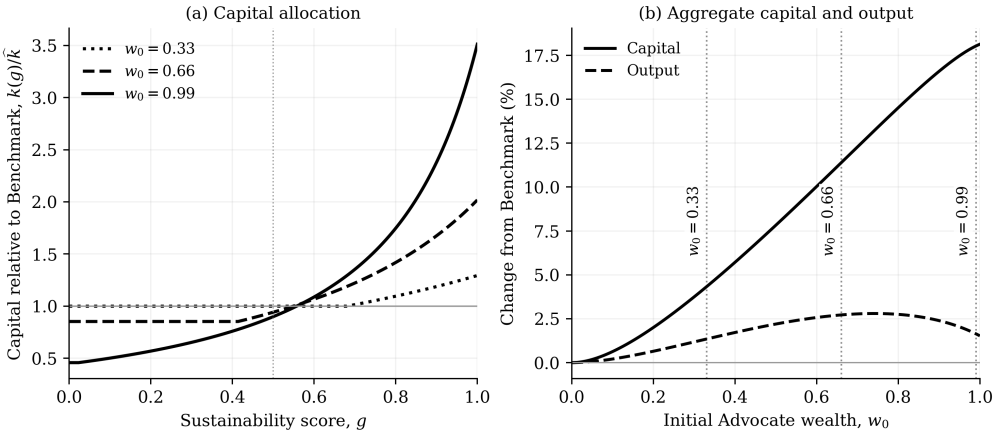


Figure 1: Capital Composition and Aggregate Impact

Notes: Panel (a) reports capital relative to the all-Benchmark allocation $\hat{k}$, shown by the horizontal line at one. The vertical dotted line marks $g_n = 0.5$. The schedules correspond to $w_0 = 0.33$, 0.66, and 0.99. Panel (b) reports percentage deviations of aggregate capital and output from the all-Benchmark economy as Advocate's initial wealth share varies. The vertical dotted lines in panel (b) mark $w_0 = 0.33$, 0.66, and 0.99, corresponding to the schedules in panel (a). Both panels use $\phi(g) = 1$ and $\nu = 3$; the remaining parameters are those of the baseline numerical illustration.

and we set $\nu = 3$ and $g_n = 0.5$. Initial capital is $k_0 = \hat{k} = (\alpha\beta/(1+\alpha\beta))^{1/(1-\alpha)}$, the capital that would result in an economy with only Benchmark households.

Composition Effect on Capital. Figure 1(a) reports capital relative to the all-Benchmark allocation. Consistently with condition (10), capital allocation follows a hockey-stick shape with a kink at the threshold g^*. As Advocate's initial wealth share increases from 0.33 to 0.66 and 0.99, the threshold falls from approximately 0.68 to 0.41 and 0.02 and Advocate prices an expanding set of firms. At $w_0 = 0.66$, capital at the lowest-g firms is about 15 percent below Benchmark, while capital at $g = 1$ is about 102 percent above it. The figure illustrates how the capital reallocation associated with heterogeneous investor preferences depends on Advocate's financial capacity. Figure 1(b) shows that aggregate capital rises by about 4.3 percent at $w_0 = 0.33$ and 11.4 percent at $w_0 = 0.66$, and approaches an increase of approximately 18.1 percent as w_0 approaches one. Output peaks earlier, at about 2.8 percent near $w_0 = 0.74$, and then declines while remaining above Benchmark as capital continues to rise. Concave production explains this divergence: an uneven capital allocation produces less output than a uniform allocation with the same aggregate capital.

Saving Motive and the Ownership Threshold. In Figure 2(a), Advocate's saving rate initially rises with its wealth share, reaching approximately 0.558 before declining as its portfolio extends further below g_n. Figure 2(b) separates the valuation and saving channels. At $w_0 = 0.75$, the mechanical benchmark gives $g^* = 0.25$, the valuation-only accounting benchmark gives approximately 0.355, and the full equilibrium gives 0.337. Higher valuations raise the threshold, while Advocate's

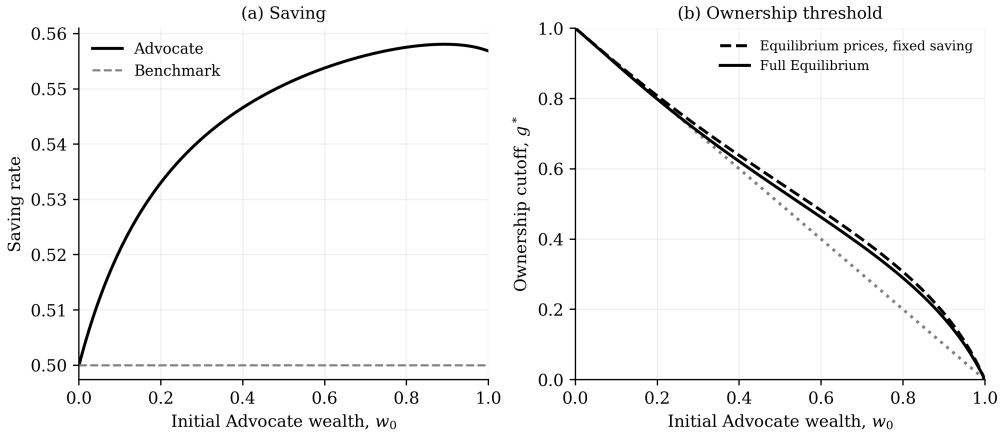


Figure 2: Saving and the Ownership Threshold

Notes: Panel (a) reports saving rates $s^A = \frac{\beta+G\mathcal{V}'(G)}{1+\beta+G\mathcal{V}'(G)}$, $s^B = \frac{\beta}{1+\beta}$. Panel (b) separates the asset-price and saving channels determining the ownership threshold. The dotted line is $g^* = 1 - w_0$, the reference with equal asset prices and equal saving rates. The dashed line retains equilibrium asset prices associated with each threshold but imposes equal saving rates as an accounting benchmark. Its horizontal coordinate is Advocate's share of aggregate equity value at those prices. The solid line plots the full-equilibrium threshold against initial Advocate wealth w_0, including equilibrium saving differences.

higher saving offsets part of that effect.

Discussion of Assumptions.

Borrowing Constraints. In the baseline model, firms finance investment through retained earnings and neither borrow nor lend. We evaluate the importance of this assumption by allowing interfirm borrowing and lending without exogenous borrowing limits, through a competitive intermediary with convex aggregate intermediation costs. Firms with low equity funding costs can thereby finance firms with higher funding costs, while the intermediation friction prevents complete arbitrage. The extension preserves threshold equity pricing but adds an upper flat segment to the capital allocation: sufficiently high-score firms lend rather than expand capital once its marginal product reaches the deposit return. Lower intermediation costs flatten the capital allocation, and aggregate capital is nonmonotonic in initial Advocate wealth in the intermediated economies. They can also raise Advocate saving by allowing greater concentration in high-score equities, even as those firms finance other firms indirectly. Thus interfirm lending moderates the capital composition effect without eliminating the nonpecuniary saving incentive. Appendix E contains the extended model and a numerical illustration. An important caveat is that Advocate investors may value high-score firms less when those firms lend to low-score firms, since their investments would indirectly finance activities they value less. Such a valuation penalty could weaken, and potentially reverse, the extent to which interfirm lending moderates the capital composition effect. Because the balance between

these forces is difficult to quantify, we retain the assumption of no interfirm borrowing or lending in our quantitative analysis.

Short Sale Constraints. The prohibition on short sales sharpens portfolio sorting but is not essential to the composition and saving mechanisms. With finite short-sale limits, Advocate shorts firms below the ownership threshold and holds larger long positions above it, while Benchmark takes the opposite positions. This preserves threshold equity pricing and the capital hockey stick composition, so short sales affect financing and saving, but need not strengthen the capital tilt. Appendix F provides the derivations and numerical illustrations which show that the impact of limited short sales on saving, aggregate capital, and output are modest at intermediate wealth shares. We do allow limited short sales in the quantitative exercise in Section 4.

Risk. We maintain a deterministic model throughout our main analysis. As outlined in our review of the literature, several prominent papers work in a context where financial markets price risk, so our choice requires justification. Our argument rests on two observations obtained by extending our baseline model in two directions. In the first extension, we consider the role of dividend risk in the presence of Advocate investors with nonpecuniary preferences. Uncertain returns add a diversification motive, so both households may hold the same firm, required returns reflect risk premia as well as sustainability preferences, and the capital schedule need not have an exactly flat segment. The composition mechanism nevertheless remains: for a given household consumption allocation and risk premium, Advocate accepts a lower expected pecuniary return on more attractive firms, lowering their cost of capital. The additional nonpecuniary saving incentive also remains operative. In our numerical illustration, baseline risk leaves Advocate saving, aggregate capital, and expected output close to their corresponding deterministic values.

In the second extension, we remove sustainability preferences and introduce differential climate-risk exposure in an all-Benchmark economy. This setting illustrates the alternative risk-based explanation for sustainable investment studied in the climate-finance literature reviewed above. Higher-score firms are less exposed to adverse climate states, which can lower their required expected returns and generate a pattern of capital allocation similar to that in the economy with Advocate investors. However, this reallocation does not generate the additional saving incentive associated with nonpecuniary preferences. In our illustration with an elasticity of intertemporal substitution of 1.5 and risk aversion of 3, climate risk strengthens precautionary demand for safe payoffs but lowers the certainty-equivalent portfolio return and thus reduces equilibrium saving and aggregate

capital. Appendix G presents the analytical results and numerical illustrations for both extensions.

3 Wealth Ownership and Capital Composition Dynamics

We now consider an infinite horizon economy. Our goal is to study how the mechanisms isolated in the two-period economy impact the dynamics of wealth ownership and, in turn, that of capital composition. We characterize the key equilibrium conditions governing such wealth-ownership feedback and the associated steady state. Household and firm problems, derivations, and technical conditions are reported in Appendix B.

Equilibrium Capital Composition. Time is continuous, $t \geq 0$, and households discount utility at rate $\rho > 0$. Advocate has flow utility, $u(c_t^A) + \mathcal{V}(G_t)$, with

$$G_t = \int_0^1 v(g) q_t(g) z_t(g) \phi(g)\, dg,$$

while Benchmark is simply $u(c_t^B)$. Borrowing and short sales are constrained as in Section 2. Let $R_t(g) = [d_t(g) + \dot{q}_t(g)]/q_t(g)$ denote the instantaneous pecuniary return of holding one unit of equity in firm with characteristics g, which includes both dividends and capital gains. Capital depreciates at the instantaneous rate $\delta > 0$, so along an interior firm optimization path[7]

$$f'(k_t(g)) = \delta + R_t(g), \tag{17}$$

Following the notation introduced in Section 2, we let

$$m_t \equiv \frac{\mathcal{V}'(G_t)}{u'(c_t^A)} > 0. \tag{18}$$

so that Advocate evaluates firm g using $R_t(g) + m_t v(g)$, while Benchmark uses $R_t(g)$. Since $m_t > 0$ for all t, the single-crossing argument we used to characterize the two-period economy equilibrium extends to the infinite horizon. The following Lemma shows that the equilibrium inherits the threshold structure of Section 2.

Lemma 1 (Threshold Equilibrium)**.** *At any date of an equilibrium in which both households hold positive financial wealth, there is an ownership threshold $g_t^* \in (0,1)$. Benchmark owns firms with*

[7]For simplicity of exposition, the model allows for instantaneous capital adjustment with no adjustment costs. We thus characterize feasible interior equilibrium paths with compatible initial capital allocations. Section 4 introduces adjustment costs to study transitions from inherited capital distributions that can be unrestricted.

$g < g_t^*$ and Advocate owns firms with $g > g_t^*$.

Letting $R_t^* = R_t(g_t^*)$, the return structure in the threshold equilibrium is $R_t(g) = R_t^*$, when $g < g_t^*$, and $R_t(g) = R_t^* - m_t[v(g) - v(g_t^*)]$, when $g \geq g_t^*$, which implies

$$k_t(g) = \begin{cases} (f')^{-1}(\delta + R_t^*), & g < g_t^*, \\ (f')^{-1}(\delta + R_t^* - m_t[v(g) - v(g_t^*)]), & g \geq g_t^*. \end{cases} \tag{19}$$

Hence, the hockey-stick shape of capital allocation characterized in Section 2 also applies at each instant in the infinite horizon economy. Crucially, its level, slope, and ownership boundary all evolve over time with R_t^*, m_t, and g_t^*.

The Wealth–Ownership Feedback. We are now ready to characterize the dynamics of wealth and its impact on the threshold g_t^* and the equilibrium capital composition. Define Advocate's share of aggregate financial wealth as

$$w_t^A \equiv \frac{a_t^A}{a_t^A + a_t^B}, \tag{20}$$

where its holdings have market value

$$a_t^A = \int_{g_t^*}^1 q_t(g)\phi(g)\, dg. \tag{21}$$

For Benchmark this is $a_t^B = \int_0^{g_t^*} q_t(g)\phi(g)\, dg$. Let $R_{w,t}^A$ denote Advocate's pecuniary portfolio return, Advocate's wealth then grows at $\dot{a}_t^A/a_t^A = R_{w,t}^A - c_t^A/a_t^A$, while Benchmark's grows at $\dot{a}_t^B/a_t^B = R_t^* - c_t^B/a_t^B$. Hence, differentiating (20) with respect to time one can show that

$$\dot{w}_t^A = w_t^A(1 - w_t^A)\left[\underbrace{R_{w,t}^A - R_t^*}_{\text{pecuniary-return effect}} \quad \underbrace{-\left(\frac{c_t^A}{a_t^A} - \frac{c_t^B}{a_t^B}\right)}_{\text{saving effect}}\right]. \tag{22}$$

Equation (22) shows that relative wealth dynamics depends on two factors. The first is a pecuniary-return effect, and it consists of the difference between the return on the portfolios of the two households. This return effect is always negative since $v(g) > v(g_t^*)$ for firms above the equilibrium threshold, which implies $R_{w,t}^A < R_t^*$. Advocate's willingness to accept lower returns therefore reduces its relative wealth growth. The second is a saving effect, which consists of the negative difference between the consumption-wealth ratios of the two households. If the ratio for Advocate

is lower than Benchmark, that is Advocate's saving is stronger, the saving effect is overall positive and contributes to Advocate's relative wealth growth.

As seen in Section 2, the relative difference in saving between Advocate and Benchmark depends on the net effective saving return, $m_t v(g_t^*)$. Letting ϵ denote a constant intertemporal elasticity of substitution, the difference in consumption growth rates can be written as

$$\frac{\dot{c}_t^A}{c_t^A} - \frac{\dot{c}_t^B}{c_t^B} = \epsilon \, m_t v(g_t^*). \tag{23}$$

Equation (23) shapes households' relative consumption paths and thereby enters the wealth-share dynamics through the difference in consumption-wealth ratios. Because the Euler equation determines consumption growth rather than its level, the comparison between Advocate's and Benchmark's consumption–wealth ratios depends on the full equilibrium path, but the presence of a nonpecuniary saving motive can sustain a positive saving effect and thus imply a net expansion of Advocate's relative wealth.

How does the wealth share dynamics impact the equilibrium ownership threshold? To answer this question we first differentiate equation (21) and use Advocate's flow budget to get:

$$\dot{g}_t^* = \frac{c_t^A - \int_{g_t^*}^1 d_t(g)\phi(g)\, dg}{q_t(g_t^*)\phi(g_t^*)}. \tag{24}$$

Equation (24) describes what measure of saving translates into ownership. If Advocate consumes less than its dividend income, it purchases firms immediately below the threshold and g_t^* falls. If it consumes more, it sells at the boundary and g_t^* rises. It is dividend income net of consumption that determines the direction and magnitude of the ownership change, the latter also depending on $q_t(g_t^*)\phi(g_t^*)$ which measures the expenditure required for a marginal expansion of its ownership range. Together with (23), this condition connects the nonpecuniary saving motive to the evolution of ownership: preferences for sustainable wealth affect consumption and hence the resources available to purchase additional equity. The resulting ownership allocation, in turn, determines Advocate's subsequent dividend income and sustainable wealth, feeding back into its consumption and portfolio decisions. Interestingly, capital gains cancel from this equation as they change the wealth embodied in existing holdings without directly changing the firms owned. At the same time, equity values still affect the speed of acquisition: for a given flow of purchases, more valuable firms at the boundary imply a smaller movement in g_t^*. Together, equations (22) and (24) describe the

dynamics of the wealth-ownership feedback that shapes the equilibrium capital composition over time.

Long Run Impact. Our previous analysis shows that Advocate sustainability preferences can impact the dynamic equilibrium allocation. We now ask whether the impact is long-lasting, and, if so, what shape it takes. To do so, we consider a steady state of the infinite horizon economy where capital $k(g)$ is constant for all g. An interior steady state has constant consumption, wealth, firm capital and prices, and ownership threshold, with both households holding positive wealth. We study the regularity conditions sufficient for the existence of a steady state in Proposition 2 of Appendix B. When those conditions are satisfied, the steady state is characterized by the following proposition.

Proposition 1 (Steady State)**.** *In any interior steady-state equilibrium, the ownership threshold is*

$$g_{ss}^* = g_n, \tag{25}$$

and capital allocation satisfies

$$k_{ss}(g) = \begin{cases} (f')^{-1}(\rho + \delta), & g < g_n, \\ (f')^{-1}(\rho + \delta - m_{ss} v(g)), & g \geq g_n, \end{cases} \tag{26}$$

where $m_{ss} = \mathcal{V}'(G_{ss})/u'(c_{ss}^A) > 0$.

Condition (26) is a direct consequence of the threshold equilibrium allocation (19). The substantive result of Proposition 1 is that the equilibrium threshold g_t^* in steady state corresponds to the neutral characteristic g_n. The stationary capital allocation is then determined jointly with the marginal sustainable wealth valuation, m_{ss}. Given the opposing forces shaping the dynamics of wealth and of the ownership threshold identified earlier, the steady-state result reveals that the Advocate saving decisions expand its share of wealth beyond the valuation effects to acquire firms' equity with lower g as long as their nonpecuniary contribution $v(g)$ is positive. Along an equilibrium path converging to an interior steady state, the wealth–ownership feedback therefore leads to complete ownership of firms with valuable characteristics by Advocate households. The stationary ownership threshold is independent of initial wealth, so under mild-regularity conditions that ensure a unique steady state, all convergent equilibrium paths also have the same limiting capital

allocation regardless of the initial wealth share. This result shows that sustainable investing can translate investors' preferences into long-lasting macroeconomic impact. A closer look at (26), however, reveals an important limitation. Relative to an all-Benchmark economy, the long-run capital allocation to firms with undesirable characteristics is not impacted. Benchmark investors' wealth share adjusts endogenously, allowing them to hold the firms that Advocate investors are unwilling to own. Therefore, when relative wealth is allowed to endogenously adjust, sustainable investing has an asymmetric long-run impact: it expands capital at firms with desirable characteristics, but does not permanently reduce capital at firms with undesirable characteristics.

The steady-state characterization identifies the lasting effects of sustainable investing, but leaves open how the economy approaches that allocation and how long the adjustment takes. We therefore turn to simulations of a quantitative version of the model to examine how the wealth–ownership feedback shapes the transition and the macroeconomic effects realized along the way.

4 Quantitative Analysis

In this section we present empirically disciplined calculations of what a macroeconomic impact driven by sustainable investing can look like. Our goal is to trace a systematic sensitivity map from sustainable-investing preferences to their impact on equilibrium wealth, capital, and output. We discipline preference parameters where empirical evidence is available and vary those that remain weakly identified, interpreting the results as conditional implications of the model rather than point predictions.

The exercise consists of introducing Advocate investors in an economy whose capital corresponds to an all-Benchmark economy steady state. Essentially, we assume that the preferences of a fraction of households corresponding to w_0 wealth share are permanently changed to include $\mathcal{V}(G)$. We are especially interested in evaluating how strongly, and how quickly, the dynamic wealth-ownership feedback can change the allocation of productive capital. We present two exercises. In the first, we assume that the characteristics Advocate value are firms' ESG scores. In the second, we assume that the valued characteristics are grounded in physical measures of environmental impact, specifically Scope 1 and 2 carbon emissions. The empirical distributions of firms' productive capacity across these two measures differ substantially, so wealth–ownership feedback and capital composition interact differently across the two exercises.

4.1 Quantitative Environment and Calibration We work with a quantitative version of the model presented in Section 3. Specifically, we extend the model by adding capital adjustment costs and bounded short sales. Investment satisfies $\dot{k}_t(g) = i_t(g) - \delta k_t(g)$, and dividends are

$$d_t(g) = f(k_t(g)) - i_t(g) - \frac{\psi}{2}\left(\frac{i_t(g)}{k_t(g)} - \delta\right)^2 k_t(g), \tag{27}$$

Each investor can short up to $\bar{s}$ of the shares outstanding of each firm, which corresponds to setting $z_t(g) \in [-\bar{s}, 1+\bar{s}]$ for Advocate, with Benchmark holding the offsetting position. Flow utility from consumption is $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$, and time is measured in years. We specify sustainable-wealth preferences as

$$v(g) = \nu\frac{g - g_n}{1 - g_n}, \qquad \mathcal{V}'(G) = (1+\kappa G)^{-\sigma_G}. \tag{28}$$

In each exercise, we jointly choose ν and κ to satisfy

$$R_0^* - R_{w,0}^A \in \{10\ bp, 20\ bp, 40\ bp\}, \qquad \frac{\mathcal{V}'(2G_0)}{\mathcal{V}'(G_0)} = .90. \tag{29}$$

The difference $R_0^* - R_{w,0}^A$ measures Advocate's initial pecuniary portfolio return shortfall relative to Benchmark's marginal return. As discussed earlier, existing estimates of similar differentials show substantial variation over time, so we take the 20 basis point estimate from Baker, Egan, and Sarkar (Baker et al., 2022) as our illustrative calibration benchmark, with 10 bp and 40 bp considered to assess the sensitivity of our results. The curvature condition disciplines the decline in marginal sustainable-wealth utility. With $\sigma_G = .5$, the maintained ratio of .90 implies $\kappa G_0 = 19/62$. We adopt this ratio as a transparent calibration assumption rather than an empirical estimate and examine sensitivity to alternative values. We set Advocate's initial wealth share to $w_0^A = .33$, a high-participation benchmark informed by broad measures of sustainable investing rather than a direct estimate of wealth held by investors with nonpecuniary preferences.[8] Finally, the choice of g_n is specific to each exercise, so it is discussed separately.

To construct an empirical counterpart to $\phi(g)$, we need a measure of economic activity across

[8]The Global Sustainable Investment Alliance (2021) reports that sustainable investment assets represented 35.9% of professionally managed assets across the markets covered in 2020, including 33.2% in the United States. Its 2022 review reports lower shares of 24.4% and 13%, respectively, under revised definitions (GSIA, 2023). These measures classify assets and investment strategies rather than investor preferences: sustainable investing can reflect financial objectives, while investors with nonpecuniary preferences can also hold conventional assets. They therefore provide reference points for our calibration, but do not identify the initial Advocate wealth share. The stationary equilibrium conditions do not depend on the initial wealth share. When the stationary allocation is unique and the equilibrium path converges to it, initial wealth affects the transition and its speed, but not the limiting allocation.

Table 1: Maintained parameters

Object	Value	Role
α	.33	Curvature of $f(k) = k^\alpha$
ρ	$-4\log(.99)$	Annual discount rate
δ	$1 - .975^4$	Annual depreciation parameter
ψ	5.48	Quadratic capital adjustment costs
σ	.5	Inverse intertemporal elasticity
σ_G	.5	Sustainable-wealth utility curvature
$\bar{s}$	.05	Per-firm short-position bound
w_0^A	.33	On-impact share of Advocate net financial wealth

firm characteristics. We use net property, plant, and equipment (PPE) to measure firms' initial productive capacity.[9] We obtain firm-level ESG scores, emissions, revenue, and PPE data for fiscal year 2023 from LSEG (formerly Refinitiv), accessed through LSEG Workspace. Normalizing each firm's PPE by total sample PPE gives its initial-capital weight, which we use to construct the fixed density of model production units, $\phi(g)$. Because initial capital per model unit is common, these weights reproduce the observed distribution of initial productive capacity.[10] Table 1 lists the maintained parameters. Appendix C reports the complete equilibrium, and Appendix I provides details of the numerical implementation.

Finally, in our two quantitative exercises we summarize the macroeconomic impact of firm-level adjustments by changes in aggregate capital K_t and output Y_t relative to their all-Benchmark levels. To assess the impact on capital composition we also report capital K_t^{25} and output Y_t^{25} in the highest-scoring group accounting for 25% of initial capital, keeping firms in that group fixed throughout each exercise.

4.2 ESG-Based Characteristics

In the first exercise, we consider preferences over firm characteristics represented by LSEG (ex-Refinitiv) ESG scores by setting g equal to the ESG score divided by 100. We construct the empirical distribution $\phi(g)$ from 2,871 consolidated issuers in the 2023 LSEG sample with matched ESG scores and positive net PPE.[11] Thus $\phi(g)$ reproduces the

[9] Simply counting firms would give equal weight to firms with very different scales of operation. Employment or market value would provide alternative measures of size, but capital is most closely aligned with our analysis of productive investment.

[10] This measurement convention does not fix the subsequent distribution of capital: as investment changes capital per unit, $k_t(g)$, the capital share associated with any range of characteristics changes with the integral of $k_t(g)\phi(g)$ over that range, relative to aggregate capital. Thus, $\phi(g)$ remains fixed while investment endogenously changes capital concentration.

[11] After removing documented overlapping subsidiaries, the sample contains \$8.263 trillion of recorded capital, or 96.24% of eligible recorded capital in the extract. These are sample coverage statistics, not shares of all US productive capital.

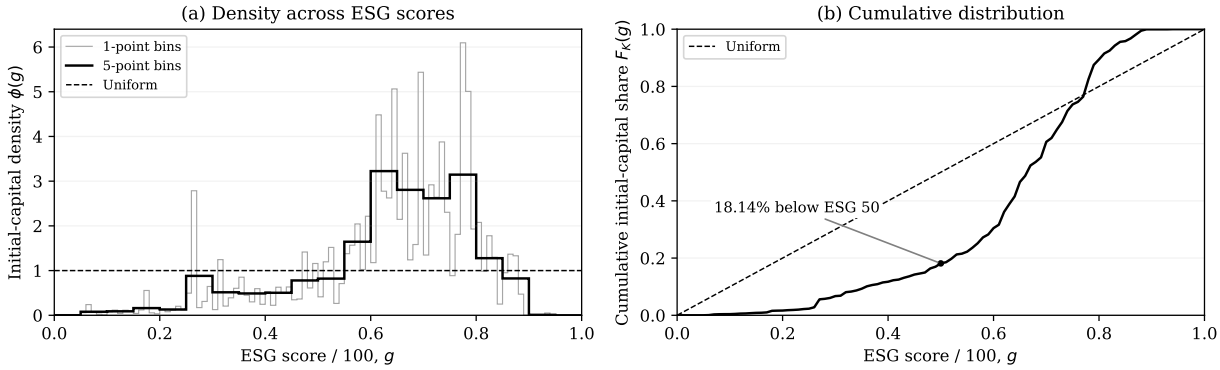


Figure 3: Initial productive capacity: ESG-based characteristics

Notes: Initial-capital weights are proportional to recorded PPE in the consolidated 2023 ESG sample. The left panel reports $\phi(g)$ per unit of normalized ESG score: one-point bins are the baseline, the thicker five-point outline shows the broader shape, and the dashed line is uniform. The right panel reports the corresponding cumulative share $\Phi(g)$ and the dashed uniform comparison, $\Phi(g) = g$.

distribution of initial productive capacity rather than an equally weighted distribution of observed firms. Figure 3 shows that 81.86% of capital lies at or above the neutral score of 50. We compare the uniform and empirical distributions to highlight how this distribution shapes the quantitative results.

We first establish the long-run effects before examining the transition and the wealth–ownership feedback that generates them. Panels A and B of Table 2 report the baseline cases under the uniform and empirical distributions, respectively. At the 20 bp calibration, long-run capital and output rise by 1.95% and .63% under the uniform distribution, compared with 9.53% and 3.01% under the empirical distribution. Advocate's wealth share increases from 33% to 52.65% and 89.69%, respectively. Consistent with Proposition 1, all of the long-run expansion occurs above the neutral score, while firms below it remain at the all-Benchmark allocation. The expansion is also disproportionately concentrated at the highest scores. Under the empirical distribution, capital and output in the fixed highest-scoring group accounting for 25% of initial capital rise by 18.51% and 5.76%, respectively, exceeding their aggregate counterparts.

The larger initial mass of positively valued productive capacity and the stronger nonpecuniary saving incentive jointly shape the empirical results. At the same 20 bp portfolio return differential, Advocate's initial net effective saving return, $m_0 v(g_0^*)$, is 65 bp under the empirical distribution, compared with 18 bp under the uniform distribution. The resulting wealth accumulation expands Advocate's ownership and the range of firms whose cost of capital reflects its preferences. A large increase in Advocate wealth nevertheless need not imply a similarly large reallocation of physical

Table 2: Long Run Macroeconomic Impact of Sustainable Investing

$R_0^* - R_{w,0}^A$ (bp)	ΔK_{ss} (%)	ΔY_{ss} (%)	ΔK_{ss}^{25} (%)	ΔY_{ss}^{25} (%)	w_{ss}^A (%)
Panel A: Uniform distribution, baseline ($g_n = 0.50$)					
10	0.92	0.30	2.77	0.91	51.23
20	1.95	0.63	5.88	1.90	52.65
40	4.49	1.43	13.69	4.32	56.40
Panel B: Empirical distribution, baseline ($g_n = 0.50$)					
10	4.15	1.34	7.91	2.54	87.18
20	9.53	3.01	18.51	5.76	89.69
40	20.32	6.11	40.85	11.94	98.36
Panel C: Empirical distribution, alternative neutral scores					
g_n					
0.33	13.39	4.19	21.15	6.53	98.59
0.66	3.62	1.16	10.88	3.46	56.11
Panel D: Empirical distribution, alternative curvature targets, 20 bp					
$\mathcal{V}'(2G_0)/\mathcal{V}'(G_0)$					
0.75	7.73	2.46	14.92	4.69	88.85
0.98	10.74	3.37	20.95	6.47	90.26

Notes: Aggregate capital and output changes are relative to the initial stationary all-Benchmark economy. K_t^{25} and Y_t^{25} are total capital and output at the highest-scoring group accounting for 25% of initial capital; group membership remains fixed throughout each exercise and across the neutral-score and curvature comparisons. The corresponding columns report $100(K_{ss}^{25}/K_0^{25} - 1)$ and $100(Y_{ss}^{25}/Y_0^{25} - 1)$. The score thresholds defining this group are approximately $g = .75$ under the uniform distribution and $g = .7623$ under the empirical distribution. Wealth shares are percentage levels, starting from $w_0^A = .33$. Panels A and B recalibrate ν and κ to each stated initial portfolio return differential and the maintained curvature ratio of .90. Panel C holds both parameters at the empirical distribution's $g_n = .50$, 20 bp baseline calibration in Panel B and re-solves the equilibrium as g_n changes. The resulting initial return differentials are 14.08 bp at $g_n = .33$ and 32.80 bp at $g_n = .66$. Holding ν fixed preserves $v(1) = \nu$; the slope $\nu/(1 - g_n)$ changes with g_n. Panel D maintains $g_n = .50$ and recalibrates ν and κ to a 20 bp initial portfolio return differential and each stated curvature ratio.

capital: much of this wealth supports the acquisition of claims on existing productive capacity. For firms near the neutral score, nonpecuniary benefits are small, so changes in ownership induce only modest reductions in the cost of capital and increases in capital per production unit.

Panels C and D assess sensitivity under the empirical distribution relative to the 20 bp baseline of Panel B. Lowering the neutral score while holding ν and κ fixed broadens the set of positively valued firms and increases the long-run expansion; raising it has the opposite effect. Stronger diminishing marginal utility of sustainable wealth reduces the expansion when preferences are recalibrated to the same portfolio return differential. Across the values considered, the quantitative outcomes are more sensitive to the neutral score than to the curvature target.

Figure 4 illustrates how the wealth–ownership feedback shapes the transition. The expansion of aggregate capital and capital at the highest-scoring firms unfolds alongside the accumulation of Advocate wealth. As the ownership threshold moves toward lower scores, Advocate's preferences

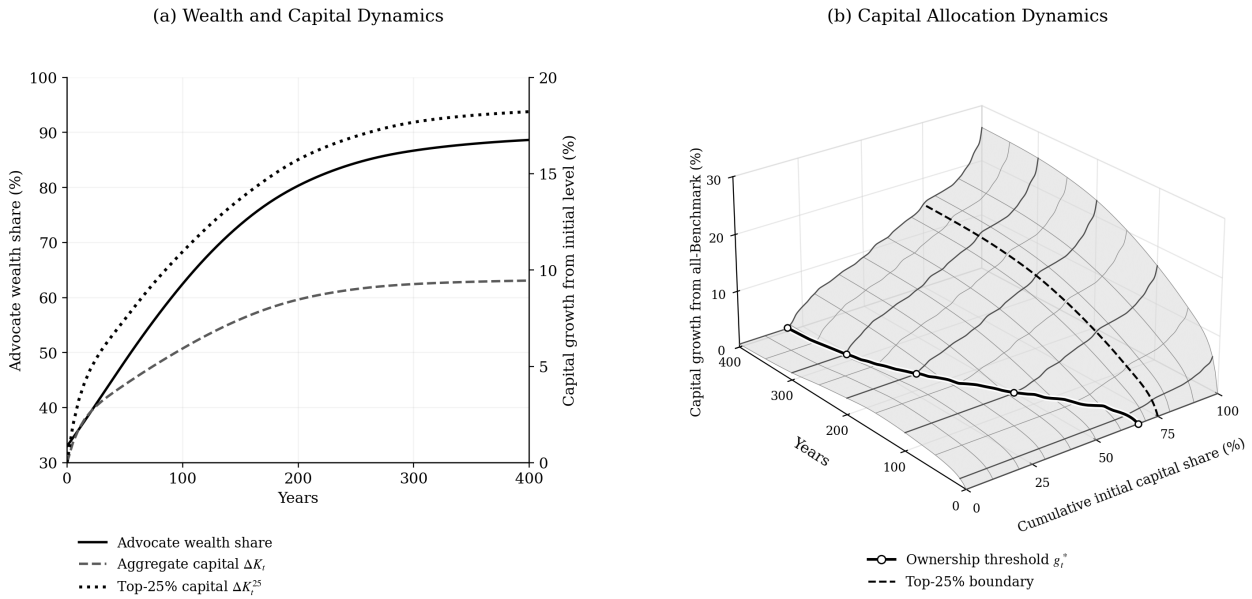


Figure 4: Wealth and Capital Transition Dynamics

Notes: Both panels use the empirical distribution and the baseline calibration targeting an initial portfolio return differential of 20 bp. Initial capital is at the all-Benchmark stationary allocation, and Advocate's initial wealth share is 33%. Panel (a) reports Advocate's wealth share on the left axis and percentage growth in aggregate capital, ΔK_t, and capital at the highest-scoring group accounting for 25% of initial capital, ΔK_t^{25}, on the right axis. Each capital series is measured relative to its own initial level, and group membership remains fixed throughout the transition. Panel (b) shows percentage capital growth per production unit relative to the all-Benchmark allocation. The horizontal axis measures cumulative initial capital share, ordered from lowest to highest ESG score. The solid marked line traces the ownership threshold g_t^*; the dashed line marks the boundary of the highest-scoring group accounting for 25% of initial capital.

influence investment at a progressively larger part of the initial capital distribution. The transition therefore combines broader Advocate ownership with an expansion of productive capacity that remains disproportionately concentrated at high scores.

The speed of wealth accumulation is a central determinant of the duration of this transition. Advocate's initial wealth limits the extent of its ownership, and the saving incentive takes time to translate into the additional wealth that supports a broader allocation. Under the empirical distribution, the time required to complete 90% of the long-run increase in Advocate's wealth share falls from approximately 485 years at 10 bp to 243 years at 20 bp and 130 years at 40 bp. Stronger calibrated preferences therefore generate both larger long-run effects and a faster approach to those outcomes. Wealth and capital nevertheless follow distinct adjustment paths: firms already held by Advocate can expand before its wealth share changes substantially, while subsequent wealth accumulation extends its investment incentives to lower-scoring firms.

Figure 5 sheds light on the saving mechanism and its reflection in equity values and productive

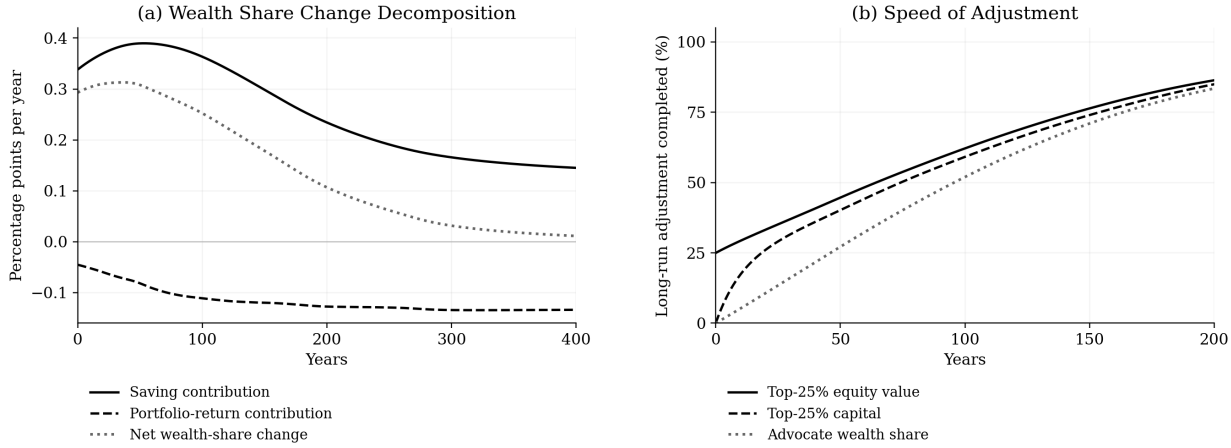


Figure 5: Wealth Share Decomposition and Speed of Adjustment

Notes: Both panels use the empirical distribution and the baseline calibration targeting an initial portfolio return differential of 20 bp. Panel (a) decomposes the change in Advocate's wealth share into the saving contribution and the portfolio-return contribution, measured in percentage points per year, over years 0–400. The dotted line is their sum. Panel (b) shows the first 200 years of adjustment in total equity value and capital at the fixed highest-scoring group accounting for 25% of initial capital, together with Advocate's aggregate wealth share. Each series reports the percentage of its own long-run change completed. Equity value is the integral of $q_t(g)$ over this group using the empirical distribution. Equity-value and capital changes are measured from their initial all-Benchmark levels, while wealth-share changes are measured from $w_0^A = .33$. The equity-value series therefore includes the initial revaluation, while capital and wealth-share adjustment start at zero.

capital. Panel (a) shows that the positive saving contribution more than offsets the negative portfolio-return contribution, allowing Advocate's wealth share to rise despite its lower pecuniary return. Panel (b) compares the adjustment of equity value and capital in the fixed top-25% group with that of Advocate's aggregate wealth share, each expressed as a fraction of its own long-run change. Equity values complete approximately one-quarter of their long-run increase on impact, anticipating subsequent capital accumulation, while capital and wealth adjust gradually. Equity values then continue to rise as productive capacity expands and Advocate accumulates wealth.

To assess the quantitative role of endogenous wealth dynamics, we use the empirical 20 bp calibration in a pooled-family counterfactual, similar to Chittaro et al. (2025), that keeps Advocate's share of aggregate net equity at 33% throughout the transition. Advocate continues to value sustainable holdings, so the portfolio allocation mechanism and a direct saving motive remain. The family nevertheless chooses saving jointly and assigns 33% of consumption to Advocate, rather than allowing the two investors to choose consumption independently. The comparison therefore includes the equilibrium consequences of this different saving arrangement. Appendix J provides the full details.

Table 3 shows that fixing the wealth share substantially reduces the long-run expansion. Ag-

Table 3: Full and Fixed Wealth Share Comparison

Year	Outcome	Full	Fixed	Difference
100	ΔK (%)	5.93	3.65	2.28
100	ΔY (%)	1.91	1.19	0.72
100	ΔK^{25} (%)	10.95	5.95	5.01
100	ΔY^{25} (%)	3.49	1.92	1.56
100	Wealth share (%)	62.48	33.00	29.48
100	Ownership threshold	0.64	0.73	-0.09
ss	ΔK (%)	9.53	3.65	5.88
ss	ΔY (%)	3.01	1.19	1.82
ss	ΔK^{25} (%)	18.51	5.95	12.57
ss	ΔY^{25} (%)	5.76	1.92	3.84
ss	Wealth share (%)	89.69	33.00	56.69
ss	Ownership threshold	0.50	0.73	-0.23

Notes: Results use the empirical distribution and the 20 bp preference calibration. Aggregate capital and output are percentage changes relative to the initial stationary all-Benchmark economy. ΔK^{25} and ΔY^{25} report percentage growth in capital and output at the fixed highest-scoring group accounting for 25% of initial capital, each relative to its own initial level. This is the same group used in Table 2. Wealth shares and ownership thresholds are levels. Difference subtracts Fixed from Full; differences between growth rates and between wealth shares are in percentage points. The pooled-family investment-account and consumption shares are both .33, with preferences held at the full economy's calibration. The comparison includes the equilibrium consequences of the family's different saving arrangement.

gregate capital rises by 3.65%, compared with 9.53% in the full economy, while capital in the fixed top-25% group rises by 5.95%, compared with 18.51%. Relative to this family benchmark, the differences amount to approximately 60% of the full economy's long-run capital and output expansion and two-thirds of the expansion within the highest-scoring group. These differences widen over the transition as Advocate wealth accumulates. Endogenous wealth dynamics are therefore quantitatively important for both the aggregate expansion of productive capacity and its concentration at high-scoring firms.

Taking stock, three insights emerge from the ESG-based characteristics exercise. First, preference-driven sustainable investing can generate a substantial expansion of productive capacity, especially in firms at the higher end of the ESG score distribution. Second, the wealth–ownership feedback sustains and broadens this expansion. The speed at which Advocate accumulates wealth governs how quickly its influence extends to firms with lower scores. In the absence of this mechanism, the impact of sustainable investing would be substantially more limited. Third, the wealth–ownership feedback makes the magnitude, timing, and concentration of the impact sensitive to the strength and shape of sustainability preferences. This underscores the importance of the empirical work aimed at quantifying the preference-driven motives for sustainable investing as emphasized by Starks (2023).

4.3 Emissions-Based Characteristics

The ESG-based exercise illustrates how investor preferences over a broad sustainability score may affect productive capital allocation. However, changes in capital across ESG scores might not directly impact outcomes investors ultimately care about. For example, Berg et al. (2022) show that ESG ratings differ substantially across providers because of differences in scope and measurement, while Christensen et al. (2022) show that greater disclosure need not resolve disagreement about firms' sustainability performance. We therefore turn to an application in which Advocate investors directly value firms with lower emissions intensity. We measure a firm's emission intensity by summing Scope 1 and Scope 2 emissions measured in tonnes of CO_2 equivalent and dividing by the firm's revenues in million US dollars. We apply the same capital-weighting principle used in the ESG exercise to firms' emission intensity. Figure 6(a) shows the resulting distribution: the capital-weighted mean is 470.18, compared with a median of 118.46, reflecting a long emissions-intensive tail.[12] For this application, we let $g \in [0,1]$ denote the initial-capital-weighted rank of production units ordered from higher to lower emissions intensity. Writing $H_K(e)$ for the capital-weighted cumulative distribution of footprint intensity, the corresponding decreasing intensity curve is $e(g) = H_K^{-1}(1-g)$. Figure 6(b) reports the monotone, mean-preserving approximation to this curve used in the simulations.[13]

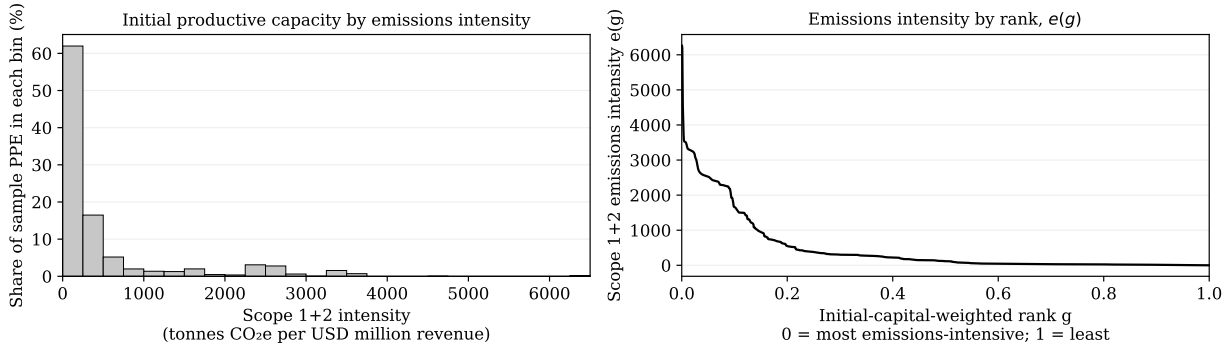


Figure 6: Initial productive capacity: emissions-based characteristics

Notes: Initial-capital weights are proportional to recorded PPE in the 2023 LSEG emissions sample. The left panel reports the share of initial productive capacity in bins of 250 tonnes CO_2e per million US dollars of revenue; the bars sum to 100%. The right panel expresses the same empirical heterogeneity through the decreasing footprint-intensity curve $e(g)$ used in the computations, with $g = 0$ denoting the dirtiest end. Initial productive capacity is uniform in rank, but intensity is not.

[12] The emissions sample contains 1,170 issuers and represents 87.87% of the original ESG sample's recorded capital. We normalize the retained PPE weights to sum to one. Appendix I.6 describes the sample, measurement, and aggregation.

[13] Equal rank intervals contain equal shares of initial productive capacity. For a continuous capital-weighted intensity distribution, the change of coordinates is $g = 1 - H_K(e)$, so higher ranks correspond to lower emissions intensity. The density in g is therefore one, while empirical heterogeneity is retained in the decreasing curve $e(g)$.

We assume that a firm with emissions intensity rank g is valued by Advocate according to

$$v(g) = \nu \frac{e(g_n) - e(g)}{e(g_n) - e(1)}, \tag{30}$$

The empirical intensity distribution has a pronounced dirty tail; consequently, preferences that are linear in intensity generate a nonlinear weight in rank. Large intensity differences among dirty firms translate into large differences in nonpecuniary valuations, whereas firms in the cleaner part of the distribution are more similar.

Table 4 reports the long-run effects of emissions-based preferences. We take median neutrality, $g_n = .5$, as the baseline and consider $g_n = .33$ and .66 while holding both ν and κ at their baseline, 20 bp calibration. The positively valued shares of initial productive capacity are 67%, 50%, and 34% at $g_n = .33, .50, .66$, respectively.

Table 4: Long Run Macroeconomic Impact of Emissions-based Sustainable Investing

g_n	ΔK_{ss} (%)	ΔY_{ss} (%)	ΔK_{ss}^{25} (%)	ΔY_{ss}^{25} (%)	w_{ss}^{A} (%)	ΔE_{ss} (%)	$\Delta(E/Y)_{ss}$ (%)
Panel A: Baseline neutral rank							
0.50	14.06	4.23	34.11	10.16	71.06	0.23	-3.84
Panel B: Alternative neutral ranks, fixed baseline preferences							
0.33	21.45	6.35	41.77	12.21	88.19	0.59	-5.42
0.66	4.55	1.42	16.73	5.18	42.21	0.04	-1.36

Notes: Aggregate capital, output, emissions, and emissions intensity are percentage changes from the stationary all-Benchmark economy. ΔK_{ss}^{25} and ΔY_{ss}^{25} report percentage growth in capital and output at the cleanest group accounting for 25% of initial capital, each relative to its own initial level. Since g is the capital-weighted dirty-to-clean emissions rank, this group is $g \in [.75, 1]$ and remains fixed across the neutral-rank comparisons. Wealth shares are percentage levels, starting from $w_0^A = .33$. Both ν and κ remain at their $g_n = .50$, 20 bp calibration. The initial return sacrifices at $g_n = .33, .50, .66$ are 6.88, 20.00, and 83.98 bp, respectively. Emissions use the Scope 1 plus Scope 2 intensity mapping.

At median neutrality, long-run capital and output increase by 14.06% and 4.23%, and Advocate's wealth share reaches 71.06%. These scale effects exceed those in the empirical ESG 20 bp case, despite a lower stationary Advocate wealth share, highlighting the importance of the preference profile alongside investment capacity. The expansion is disproportionately concentrated at the cleanest firms: capital and output in the group accounting for 25% of initial capital at the highest ranks rise by 34.11% and 10.16%, respectively. Lowering g_n to .33 strengthens wealth accumulation and capital expansion: the stationary wealth share reaches 88.19% and aggregate capital rises by 21.45%; raising g_n to .66 instead limits these increases to 42.21% and 4.55%.

These substantial changes in productive capacity translate into much smaller changes in total emissions. With fixed firm intensities, total emissions are $E_t = \int_0^1 e(g) k_t(g)^\alpha \, dg$, so production

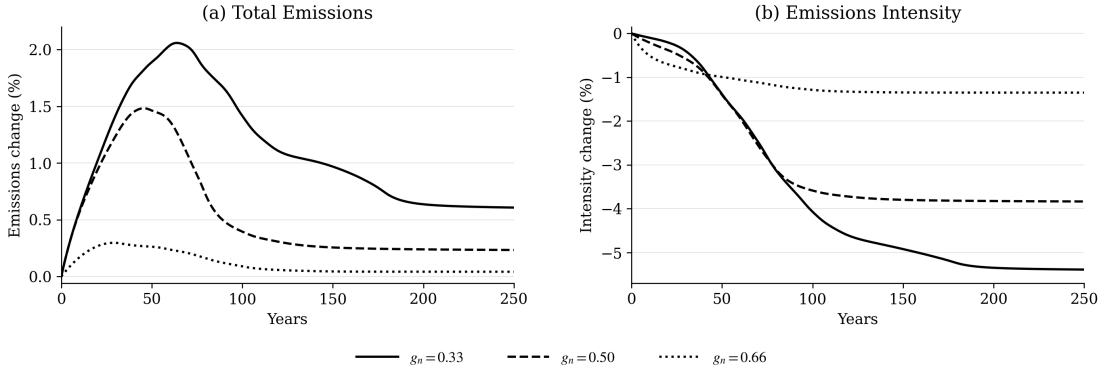


Figure 7: Emissions Dynamics under Emissions-Based Preferences

Notes: All cases start with Advocate wealth equal to 33% of aggregate net equity and with all-Benchmark capital at every rank. Both ν and κ remain at their $g_n = .50$, 20 bp calibration; EIS equals two. Panels (a) and (b) report percentage changes in total emissions, E_t, and aggregate emissions intensity, E_t/Y_t, respectively, relative to the stationary all-Benchmark economy. Total emissions denote the attributed Scope 1 plus Scope 2 footprint, which can count the same physical emissions at producers and users. Initial ranks and firm-level emissions intensities remain fixed throughout the transition.

at emissions-intensive firms contributes disproportionately to the aggregate footprint. Capital expands most at cleaner firms, where additional output generates relatively little additional emissions, whereas firms below the neutral rank retain their all-Benchmark capital in the stationary allocation. Expanding clean production therefore does not displace the production responsible for much of the initial total emissions. In the baseline, total emissions rise by only .23%, while the 4.23% increase in output lowers aggregate emissions intensity, E_t/Y_t, by 3.84%. The same distinction holds across the alternative neutral ranks: emissions intensity falls by 5.42% and 1.36% at $g_n = .33$ and .66, respectively, but total emissions remain above their all-Benchmark level in both cases.[14]

Figure 7 shows how these emissions outcomes emerge during the transition. Panel (a) reports an initial increase in total emissions followed by a decline toward the stationary level. Emissions peak approximately 2.06% above their initial level at year 64 when $g_n = .33$, 1.48% at year 46 when $g_n = .50$, and .30% at year 29 when $g_n = .66$. Early in the transition, the equilibrium return required by Benchmark falls, supporting capital accumulation even at emissions-intensive firms. Their large contribution to the initial footprint makes this temporary expansion important for total emissions. As the required return recovers toward its stationary level, their capital expansion unwinds, while cleaner firms continue to accumulate capital. Total emissions can consequently fall

[14]Firm-level emissions intensities are fixed in this exercise, so changes in aggregate emissions intensity arise entirely from changes in firms' shares of aggregate output. Cleaner production can therefore lower aggregate intensity without a contraction of emissions-intensive production. Allowing firms to invest in abatement would introduce an additional margin, which we leave to future work.

while aggregate capital continues to rise, although they eventually settle above their initial level.

Panel (b) shows the corresponding decline in aggregate emissions intensity. The paths are initially similar and separate as the differences in wealth accumulation and capital expansion become more pronounced. Because firm-level intensities remain fixed, these reductions arise entirely from a shift in the composition of output toward cleaner firms, so they reflect a cleaner composition of expanding production, rather than a reduction in the total footprint.

The exercise illustrates both the quantitative importance of the wealth–ownership feedback and the distinction between its effects on capital and on emissions. Advocate's wealth accumulation supports a substantial expansion of cleaner productive capacity, but Benchmark's wealth and portfolio holdings also adjust in equilibrium. A declining Benchmark wealth share does not imply a corresponding contraction of emissions-intensive production: Benchmark continues to hold claims on these firms and its required return supports their all-Benchmark capital in the long run. Preference-driven sustainable investing can therefore substantially change ownership and productive capacity without reducing total emissions. This distinction motivates the next section, where we ask whether sustainable investing can implement an allocation designed to achieve an explicit emissions-reduction target.

5 Policy Implementation via Sustainable Investing

In this section, we ask whether preference-driven sustainable investing can implement the allocation generated by an emissions-reduction policy which combines an emission tax with an output subsidy, as studied in Fischer (2001).[15] We choose the tax and subsidy to deliver a ten percent long-run reduction in total emissions relative to the initial all-Benchmark allocation, with a balanced policy budget in steady state.[16]

We work with the quantitative version of the model of Section 4, including the same initial-capital ranking procedure, but we focus on Scope 1 emission for the policy exercise.[17] Thus $g \in [0, 1]$

[15]Such instruments have an established role in environmental policy, including Canada's output-based pricing system and China's output-based allocation of emissions allowances, where they seek to preserve incentives for cleaner production while limiting the burden on regulated producers. Their effects on total emissions nevertheless depend on the interaction between production composition and scale: as Holland et al. (2009) demonstrate for low-carbon fuel standards, encouraging cleaner production can lower emissions intensity without necessarily reducing total emissions.

[16]The magnitude has a historical precedent in the original Regional Greenhouse Gas Initiative agreement, which specified a ten percent reduction in participating states' emissions budgets by 2018 relative to their initial budgets. We adopt this percentage as an illustrative implementation target.

[17]We use Scope 1 emissions in the policy exercise because the emissions target and the tax apply to emissions directly generated by production. Including Scope 2 would also attribute emissions from purchased electricity to the firms using it, potentially counting the same emissions at both the electricity producer and the purchaser. Scope 1

orders production units from dirtier to cleaner, and $e(g)$ denotes their fixed Scope 1 emissions intensity. Recall that output and total emission are $Y_t = \int_0^1 k_t(g)^\alpha \, dg$ and $E_t = \int_0^1 e(g) k_t(g)^\alpha \, dg$, respectively, while aggregate emission intensity is simply E_t/Y_t. Since firm-level intensities are fixed, changes in aggregate intensity arise from changes in the composition of production.

The exercise proceeds in three steps. First, we construct the policy allocation in a version of our quantitative model economy populated entirely by Benchmark households. Second, we consider an economy with all Advocate households and no policy, and we set Advocate's preferences over emissions intensity so to implement the same stationary allocation as under policy. Third, we consider an economy with both Advocate and Benchmark investors while holding Advocate's preferences as specified in the second step, and consider initial Advocate wealth shares of 33, 66, and 99 percent and ask how close the equilibrium allocation is to the policy allocation. Our exercise is thus similar in spirit to Chittaro et al. (2025), with the important difference that we allow for the wealth share to endogenously adjust over time.

Emissions Policy in the All-Benchmark Economy We study the emissions policy in the All-Benchmark economy first. The policy combines a permanent tax τ per unit of emissions with an output subsidy τe_n per unit of production. A firm with rank g thus makes a net payment to the government of $\tau[e(g) - e_n]k_t(g)^\alpha$. Firms with intensity above e_n pay a net tax, while those below e_n receive a net subsidy. The threshold e_n determines policy neutrality, and τ determines the strength of incentives on either side of that threshold.

Let E_P denote total emissions, $k_P(g)$ capital in firm with rank g, and Y_P total output, in the steady state under policy. We jointly choose (τ, e_n) to satisfy the stationary balanced budget condition

$$\int_0^1 \tau[e(g) - e_n]k_P(g)^\alpha \, dg = \tau(E_P - e_n Y_P) = 0.$$

with $E_P = 0.9E_0$. The condition implies $e_n = E_P/Y_P$, so the neutral intensity equals the output-weighted average intensity in the steady state under policy.[18]

The policy changes a firm's effective output price to $1 + \tau(e_n - e(g))$. This results in a wedge between the capital a firm would choose in the absence of policy, $\hat{k}$, and the capital under policy,

therefore provides a consistent basis for aggregating emissions and applying the policy at their source. In Section 4.3, Scope 1 and 2 instead provide a broader measure of firms' emissions footprints over which investors may have preferences.

[18] During adjustment, net revenues are returned to households as lump-sum transfers. The tax and subsidy rates remain fixed; they neither impose an immediate emissions cap nor balance the policy budget period by period.

$k_P(g)$, corresponding to

$$k_P(g) = \hat{k}[1 + \tau(e_n - e(g))]^{1/(1-\alpha)}. \tag{31}$$

Under the policy, firms with emissions intensity above e_n reduce their capital, while firms with intensity below e_n increase it. Under our calibration, to achieve 10% emissions reduction, the emission tax is approximately \$116.92 per tonne, and the neutral emissions level is $e_n = 386.33$ tonnes CO_2 per million dollars of revenue, corresponding to $g_n \simeq .23$. Under policy, aggregate emissions intensity falls by 9.66 percent and output and capital both fall by 0.37 percent, resulting in the ten percent reduction in total emissions. Most of the emissions reduction therefore comes from reallocating production away from emissions-intensive firms.

Implementation in the All-Advocate Economy We next remove the policy and consider an economy with all Advocate households. We construct preferences that support the entire stationary capital allocation in (31), and hence the same stationary output, emissions intensity, and total emissions. The preferences that implement the policy allocation take the form

$$v(g) = \nu(\rho + \delta)\frac{\tau[e_n - e(g)]}{1 + \tau[e_n - e(g)]}, \tag{32}$$

where the parameters τ and e_n are those obtained in the first step. We keep the sustainable-wealth utility so that $\mathcal{V}'(G) = (1 + \kappa G)^{-\sigma_G}$, holding κ and σ_G fixed, and choose ν jointly with the implied marginal sustainable wealth valuation m to implement the desired stationary allocation.[19]

To implement the policy steady state allocation, a firm's capital choice must satisfy $R(g) = \alpha k_P(g)^{\alpha-1} - \delta = \rho - v(g)/\nu$. Since Advocate's stationary Euler condition requires $R(g) = \rho - mv(g)$, for every firm, choosing $\nu = 1/m$ with m evaluated at the policy allocation makes the Euler condition consistent with the firm's required capital choice. These preferences therefore support both the expansion of clean production and the contraction of dirty production required by the policy allocation.

The condition $\nu = 1/m$ applies only at the steady state under policy allocation; $m_t\nu$ need not equal one along the transition, so matching the stationary allocation does not guarantee that the transition is identical. Figure 8 shows that the policy and all-Advocate paths are nevertheless close. A five percent reduction in total emissions takes 6.6 years under policy and 6.5 years under all-Advocate implementation; a nine percent reduction takes 21.5 and 21.1 years, respectively. Thus,

[19] Differently from Section 4's linear-intensity valuation, the functional form for $v(g)$ is chosen for policy implementation and is generally nonlinear in physical intensity as well as in rank.

the fixed preferences implement the stationary objective and approximately reproduce the policy's adjustment speed, without being calibrated to match that speed.

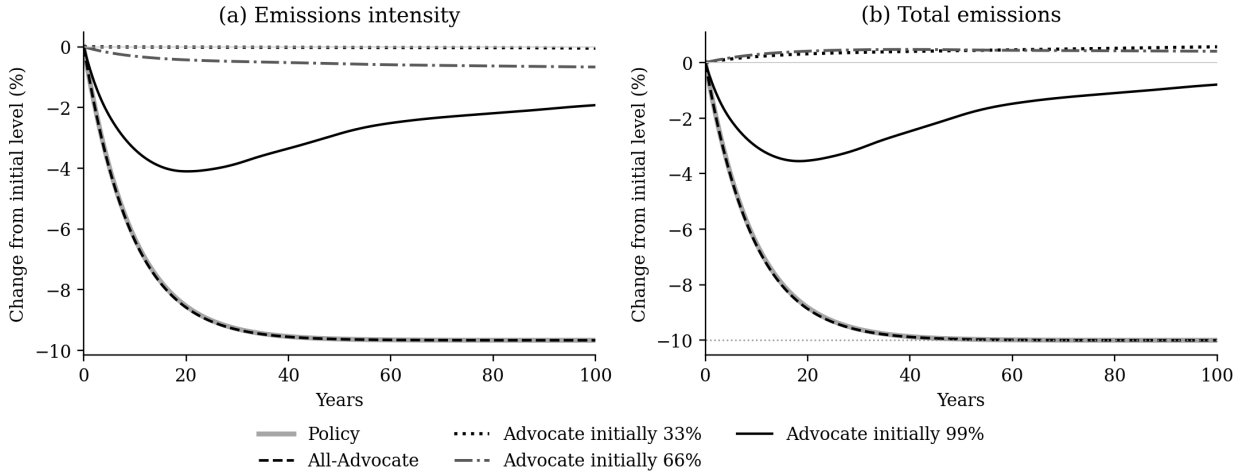


Figure 8: Emissions intensity and total emissions during adjustment

Notes: Percentage changes relative to the initial all-Benchmark allocation over the first 100 years. All economies start from the same capital allocation. Policy operates in an all-Benchmark economy. The all-Advocate and Advocate+Benchmark economies have no policy and share the same fixed Advocate preferences. Emissions intensity is E_t/Y_t. The policy and all-Advocate paths nearly coincide. The horizontal dotted line in panel (b) marks the ten percent total-emissions reduction target.

Implementation in the Advocate+Benchmark Economy We now consider an economy with both Advocate and Benchmark households as in Section 4, but with the Advocate preferences specified above. We consider three experiments where initial Advocate wealth shares are 33, 66, and 99 percent and wealth shares subsequently evolve endogenously.

As expected, all three economies converge to the same interior stationary equilibrium, where Advocate holds 81.58 percent of financial wealth. Table 5 compares this outcome with the policy and all-Advocate allocations. In the Advocate+Benchmark economy, emissions intensity falls by 1.06 percent, but capital expands by 3.65 percent and output expands by 1.18 percent, so total emissions rise by 0.11 percent. The improvement in composition is therefore more than offset by the scale effect, and even 99 percent initial Advocate wealth does not deliver lasting implementation.

The source of implementation failure is Benchmark's eventual ownership of firms with emission intensity above the neutral level e_n. To implement the policy allocation, firms with $e(g) > e_n$ should have a cost of capital above ρ. The all-Advocate economy allows the higher return because Advocate is the marginal investor and requires compensation for the nonpecuniary penalty $v(g) < 0$. In the economy with both households, however, Benchmark is the steady-state marginal investor

Table 5: Long-run policy implementation

Outcome	Policy	All-Advocate	Advocate+Benchmark
Total emissions	-10.000	-10.000	0.113
Emissions intensity	-9.665	-9.665	-1.057
Aggregate output	-0.371	-0.371	1.183
Aggregate capital	-0.371	-0.371	3.647
Advocate wealth share	—	100.00	81.58

Notes: Emissions, output, and capital entries are percentage changes relative to the initial all-Benchmark allocation. Wealth shares are percentage levels. Advocate + Benchmark economies starting from 33, 66, and 99 percent Advocate wealth converge to the same stationary allocation. Advocate preferences are identical across All-Advocate and Advocate+Benchmark.

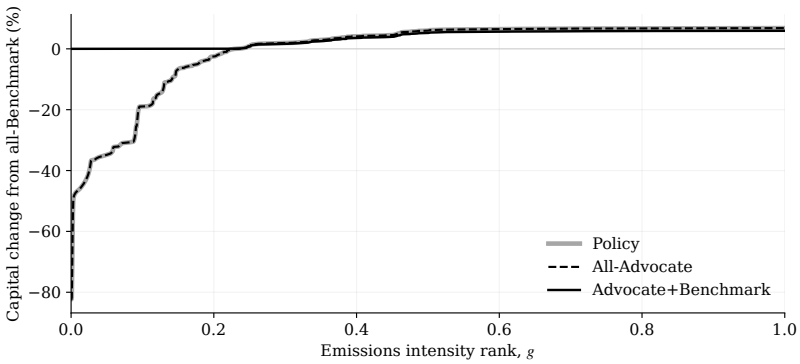


Figure 9: Stationary capital allocations across emissions intensity

Notes: Capital per model production unit is expressed as a percentage change from the all-Benchmark allocation. Higher emissions intensity ranks correspond to lower emissions intensity. Policy and All-Advocate coincide in the stationary allocation. Advocate+Benchmark uses the same preferences as All-Advocate and allows wealth shares to adjust endogenously.

in firms with $e(g) > e_n$, which means that for firms with emission intensity above e_n, $R(g) = \rho$ and capital remains at $\hat{k}$. Figure 9 illustrates the resulting difference in capital allocations. For Policy and All-Advocate capital contracts at low ranks, where emissions intensity is high, and expands at high ranks, where emissions intensity is low. Advocate+Benchmark retains the expansion at high ranks, although at a slightly smaller magnitude, but does not generate any contraction at low ranks.

Figure 8 shows that initial wealth matters for the transition dynamics. Panel (a) shows emissions intensity declining for all three levels of initial wealth considered: as more capital is accumulated by low-intensity firms, emissions per unit of output decline. The decline is minimal for the 33 and 66 percent cases, while it is more pronounced for the 99 percent case. The stronger decline is, however, temporary since Advocate is initially forced to hold firms with emissions intensity above the neutral level e_n, which pushes their capital accumulation down. But over time, Advocate reallocates its wealth to firms with emissions intensity below e_n, while Benchmark takes ownership

of firms above e_n and their capital expands again, reducing the initial drop in average intensity. Panel (b) shows that the impact on total emissions for the Advocate+Benchmark economy is small and of the opposite sign from the one under policy. The wealth-ownership feedback changes the strength and reach of Advocate's influence along the transition, even though preferences remain fixed. A large initial Advocate wealth share consequently need not sustain an allocation close to the all-Advocate economy.

The exercise shows that preferences capable of implementing the policy allocation under sole Advocate ownership need not deliver its emissions objective when investors with different objectives coexist, and wealth shares are allowed to endogenously adjust. The same preferences improve the composition of production, but Benchmark's endogenous ownership of high emissions intensity firms prevents the capital contraction needed for long-run implementation.

6 Conclusion

In this paper we study how preference-driven sustainable investing translates investors' values into lasting changes in productive capital allocation. The central mechanism we identify is a wealth–ownership feedback whereby nonpecuniary preferences affect both portfolio choices and saving incentives, while the resulting distribution of investor wealth determines the range of firms whose cost of capital reflects those preferences. Our quantitative exercises show that this feedback can generate a substantial expansion of productive capacity, disproportionately concentrated at firms with highly valued characteristics, although its reach expands gradually as Advocate investors accumulate wealth. The long-run effect is nevertheless asymmetric. In the long run, capital expands at firms with positively valued characteristics, but Benchmark investors' endogenous wealth adjustment sustains the capital allocation to firms Advocate prefer to avoid. Consequently, a substantial impact on capital need not translate into a substantial impact on what investors value: in our emissions exercise, cleaner production can expand without displacing emissions-intensive production. The policy exercise makes this distinction concrete. Preferences that implement an emissions-reduction allocation in the All-Advocate economy fail to deliver the same outcome in the Advocate+Benchmark economy, even when Advocates initially hold almost all financial wealth.

These findings provide a benchmark for studying sustainable investing when firms can also change how they produce. Our analysis holds technologies and firm characteristics fixed, abstracting from technological innovation and from endogenous technological choice, such as investment in

emissions abatement. Such endogenous choices could alter both the magnitude and the direction of the macroeconomic impact. The literature on induced innovation and directed technical change shows how economic incentives can influence the development and adoption of cleaner technologies, see for example Popp (2002) and Acemoglu et al. (2012, 2016). Extending our framework to incorporate these choices would allow investor preferences to affect not only capital allocation across existing technologies, but also the incentives to develop cleaner technologies and reduce emissions within firms. Changes in technology would, in turn, affect financial returns, the characteristics investors value, and the evolution of ownership. Our framework provides a basis for studying these interactions while retaining the endogenous wealth distribution that determines investors' capacity to influence production. We leave the analysis of these additional margins, and their implications for the macroeconomic impact of sustainable investing, to future work.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT Astra-6 (OpenAI) for grammatical editing and proofreading assistance as well as support in coding for data and numerical analysis. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

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Appendix

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Appendix I: Analytical Derivations

A Two-Period Model: Derivations

This appendix provides the derivations underlying Section 2.

A.1 Firm and Household Conditions Firm mass is normalized to one, $\int_0^1 \phi(g)\,dg = 1$. Taking $R(g)$ as given, firm g chooses $k(g)$ to maximize $d_0(g) + q(g) = f(k_0) - k(g) + f(k(g))/R(g)$. The interior condition is $f'(k(g)) = R(g)$ and $q(g) = f(k(g))/R(g)$, establishing equations (1)–(2). In the baseline version of the model, firms are borrowing constrained, so it must be that $d_0(g) \geq 0$.

Write $K = \int k(g)\phi(g)\,dg$ and $Q = \int q(g)\phi(g)\,dg$. Aggregate initial wealth is the cum-dividend value of initial ownership,

$$W_0 = \int_0^1 [d_0(g) + q(g)]\phi(g)\,dg = f(k_0) - K + Q. \tag{33}$$

An initial ownership share w_0 in each firm gives Advocate wealth $w_0 W_0$ and Benchmark wealth $(1 - w_0)W_0$. Thus w_0 is exogenous, while the level W_0 adjusts with equilibrium investment and equity values.

Let a_1^A denote Advocate saving. With $z(g)$ denoting Advocate's share of firm g, one has

$$a_1^A = \int_0^1 z(g)q(g)\phi(g)dg, \qquad c_0^A = w_0 W_0 - a_1^A,$$
$$c_1^A = \int_0^1 z(g)f(k(g))\phi(g)dg, \quad G = \int_0^1 v(g)z(g)q(g)\phi(g)dg.$$

The optimal portfolio allocation condition for Advocate is obtained by differentiating $u(c_0^A) + \beta u(c_1^A) + \mathcal{V}(G)$ with respect to $z(g)$ under the budget constraints above to get

$$u'(c_0^A)q(g) \geq \beta u'(c_1^A)f(k(g)) + \mathcal{V}'(G)v(g)q(g),$$

with equality when holdings $z(g)$ are strictly positive. Dividing both sides by $q(g)$ and using $f(k(g))/q(g) = R(g)$ gives equation (5). For Benchmark, $a_1^B = \int z^B(g)q(g)\phi(g)\,dg$, $c_0^B = (1 -$

$w_0)W_0 - a_1^B$, and $c_1^B = \int z^B(g) f(k(g))\phi(g)\, dg$ with optimality condition $u'(c_0^B) \geq \beta u'(c_1^B)R(g)$. Market clearing, $z^A(g) + z^B(g) = 1$, implies $a_1^A + a_1^B = Q$. Using the budget constraints and equation (33) gives

$$c_0^A + c_0^B + K = f(k_0), \qquad c_1^A + c_1^B = \int_0^1 f(k(g))\phi(g)\, dg.$$

A.2 Ownership Threshold and Capital Allocation Define $m = \mathcal{V}'(G)/u'(c_1^A) > 0$ as in equation (6). Advocate's effective valuation of firm g is then proportional to $R(g) + (m/\beta)v(g)$, whereas Benchmark values only $R(g)$. Because $v(g)$ is strictly increasing, relative willingness to hold a firm is strictly increasing in g, implying a unique cutoff whenever the two types coexist. Thus Advocate owns firms above g^* and Benchmark those below it, as stated in equation (8).

Let $R^* = R(g^*)$. Benchmark pricing gives $R(g) = R^*$ below the cutoff. For any Advocate-priced firm, subtracting Advocate's portfolio condition at g^* from that at g gives

$$R(g) - R^* = -\frac{m}{\beta}\left[v(g) - v(g^*)\right],$$

which is equation (9). Substitution into $f'(k(g)) = R(g)$ immediately gives the capital schedule in equation (10). Since f' is decreasing and v increasing, capital is flat below g^* and increasing in g above it.

The cutoff is jointly determined by saving and equity values. Asset-market clearing gives $a_1^A = \int_{g^*}^1 q(g)\phi(g)dg = s^A w_0 W_0$ and $a_1^B = \int_0^{g^*} q(g)\phi(g)dg = s^B(1 - w_0)W_0$. Taking their ratio yields equation (14). If $q(g)$ is constant and $s^A = s^B$, it reduces to $\Phi(g^*) = 1 - w_0$, equation (15).

A.3 Nonpecuniary Saving Motive Integrating Advocate's portfolio condition over its holdings gives

$$u'(c_0^A)a_1^A = \beta u'(c_1^A)c_1^A + \mathcal{V}'(G)G. \tag{34}$$

The first term is the usual pecuniary payoff to saving and the second is the nonpecuniary saving motive. Dividing by $\beta u'(c_1^A)a_1^A$ and writing the gross portfolio return for Advocate as $R_w^A = c_1^A/a_1^A$ gives equation (11). Integrating the return schedule over Advocate's portfolio also gives

$$R_w^A = R^* - \frac{m}{\beta}\left[\frac{G}{a_1^A} - v(g^*)\right].$$

Thus the effective saving return relative to Benchmark is $(m/\beta)v(g^*)$, whereas the nonpecuniary contribution from the entire portfolio is $(m/\beta)G/a_1^A$. These are distinct objects. To see this, note that under log consumption utility, equation (34) becomes $a_1^A/c_0^A = \beta + G\mathcal{V}'(G)$, so

$$s^A = \frac{\beta + G\mathcal{V}'(G)}{1 + \beta + G\mathcal{V}'(G)}, \qquad s^B = \frac{\beta}{1+\beta}.$$

An interior solution requires $\beta + G\mathcal{V}'(G) > 0$. On this domain, $G > 0$ raises Advocate saving relative to Benchmark and $G < 0$ lowers it.

If $g^* > g_n$, every firm held by Advocate has $v(g) > 0$, so $G > 0$. If $g^* < g_n$, firms in $[g^*, g_n)$ contribute negatively to sustainable wealth, while firms in $(g_n, 1]$ contribute positively. The sign of G, and hence of the saving differential, therefore depends on the market-value-weighted balance of these two regions; crossing g_n does not by itself imply that G becomes negative.

Concavity of $\mathcal{V}$ alone does not make the saving wedge increasing in G: its derivative is $\mathcal{V}'(G) + G\mathcal{V}''(G)$. A decline in G weakens the wedge when this expression is positive. For $\mathcal{V}'(G) = (1 + \kappa G)^{-\sigma_G}$, the derivative equals $(1+\kappa G)^{-\sigma_G - 1}[1+(1-\sigma_G)\kappa G]$, which is positive throughout $1+\kappa G > 0$ when $\sigma_G = 1/2$. This condition governs changes in the wedge conditional on G; equilibrium portfolio expansion also changes prices and need not reduce G.

Finally, aggregate capital and output are $K = \int k(g)\phi(g)dg$ and $Y = \int f(k(g))\phi(g)dg$. Because f is concave, a more dispersed capital allocation can reduce output for a given K; equivalently, $\int f(k(g))\phi(g)dg \leq f(\int k(g)\phi(g)dg)$. This is why aggregate output can peak before aggregate capital in the two-period illustration.

A.4 Numerical Implementation

For $f(k) = k^\alpha$, the interior firm condition gives $q(g) = k(g)/\alpha$. In the all-Benchmark economy, log utility implies $c_0 = K/(\alpha\beta)$; goods-market clearing therefore gives $\hat{k} = [\alpha\beta/(1 + \alpha\beta)]f(k_0)$. The initial stock k_0 in the numerical illustration is set equal to $\hat{k}$. The replication files for the numerical illustration are available upon request.

B Dynamic Model: Derivations

This appendix contains the derivations for the results of Section 3. Because there are no adjustment costs in the model for illustration purposes, initial capital and ownership must be compatible with all equilibrium, feasibility, and transversality conditions. We introduce adjustment costs in Section 4, so the initial condition requirements above are not needed in the quantitative results.

B.1 Households, Firms, and Portfolio Choice Let $z_t^i(g)$ denote the share of firm g held by household $i \in \{A, B\}$. Financial wealth is $a_t^i = \int_0^1 q_t(g) z_t^i(g) \phi(g) dg$, and market clearing requires $z_t^A(g) + z_t^B(g) = 1$, with $z_t^i(g) \geq 0$. With the instantaneous pecuniary return

$$R_t(g) = \frac{d_t(g) + \dot{q}_t(g)}{q_t(g)},$$

household wealth evolves according to $\dot{a}_t^i = \int_0^1 R_t(g) q_t(g) z_t^i(g) \phi(g) dg - c_t^i$. The "dot" notation is the standard notation for time derivative in continuous time.

Advocate maximizes $\int_0^\infty e^{-\rho t}[u(c_t^A) + \mathcal{V}(G_t)]dt$, where $G_t = \int v(g) q_t(g) z_t^A(g) \phi(g) dg$. Let R_t^i denote the shadow return on household $i \in \{A, B\}$ wealth. The consumption first-order condition gives the costate $\lambda_t^A = u'(c_t^A)$; the interior portfolio condition and costate equation are, respectively,

$$R_t(g) + m_t v(g) = R_t^A, \qquad \frac{\dot{\lambda}_t^A}{\lambda_t^A} = \rho - R_t^A, \qquad m_t \equiv \frac{\mathcal{V}'(G_t)}{u'(c_t^A)}.$$

Benchmark satisfies $R_t(g) = R_t^B$ on assets it prices and $\dot{\lambda}_t^B / \lambda_t^B = \rho - R_t^B$. Defining the local EIS $\epsilon(c) \equiv -u'(c)/[c\, u''(c)]$, the consumption Euler equations are

$$\frac{\dot{c}_t^A}{c_t^A} = \epsilon(c_t^A)(R_t^A - \rho), \qquad \frac{\dot{c}_t^B}{c_t^B} = \epsilon(c_t^B)(R_t^B - \rho). \tag{35}$$

At zero holdings, $R_t(g) + m_t v(g) \leq R_t^A$ for Advocate and $R_t(g) \leq R_t^B$ for Benchmark. We impose the transversality condition $\lim_{t\to\infty} e^{-\rho t} u'(c_t^i) a_t^i = 0$, along the paths considered.

Firm g finances investment through retained earnings, so $d_t(g) = f(k_t(g)) - \dot{k}_t(g) - \delta k_t(g) \geq 0$. Taking $R_t(g)$ as given, it maximizes discounted dividends. To make the interior qualification explicit, let $\mathcal{Q}_t^K(g)$ be the shadow value of installed capital in dividend units, distinct from equity value $q_t(g)$. The Hamiltonian conditions are $\mathcal{Q}_t^K \geq 1$, $(\mathcal{Q}_t^K - 1) d_t = 0$, and $\dot{\mathcal{Q}}_t^K = [R_t + \delta - f'(k_t)] \mathcal{Q}_t^K$. On an interval with $d_t(g) > 0$, $\mathcal{Q}_t^K(g) = 1$, implying

$$f'(k_t(g)) = \delta + R_t(g), \tag{36}$$

which is equation (17). If $d_t(g) = 0$ binds over an interval, the shadow-value conditions replace this equality. Section 3 restricts attention to feasible paths in the interior regime by assuming the appropriate initial conditions, as remarked above. Imposing the no-bubble condition, $\lim_{T\to\infty} e^{-\int_t^T R_s(g) ds} q_T(g) = 0$ the asset price is determined by $q_t(g) = \int_t^\infty e^{-\int_t^s R_\tau(g) d\tau} d_s(g)\, ds$.

B.2 Threshold Equilibrium Under the no-short-sale restrictions, the relevant comparison between the two investors is $m_t v(g) - (R_t^A - R_t^B)$. It is strictly increasing in g, so it crosses zero at most once. Define g_t^* by $m_t v(g_t^*) = R_t^A - R_t^B$. Then Benchmark owns firms below g_t^* and Advocate those above it, establishing Lemma 1.

Let $R_t^* = R_t(g_t^*)$. At the threshold, Benchmark's condition gives $R_t^B = R_t^*$ and Advocate's gives $R_t^A = R_t^* + m_t v(g_t^*)$. Hence

$$R_t(g) = R_t^* \quad (g < g_t^*), \qquad R_t(g) = R_t^* - m_t[v(g) - v(g_t^*)] \quad (g \geq g_t^*),$$

which give the return schedule underlying Lemma 1. Combining these expressions with equation (36) gives the capital schedule (19).

Finally, substitution into equation (35) gives the Euler equations at the threshold,

$$\frac{\dot{c}_t^A}{c_t^A} = \epsilon(c_t^A)\left[R_t^* + m_t v(g_t^*) - \rho\right], \qquad \frac{\dot{c}_t^B}{c_t^B} = \epsilon(c_t^B)(R_t^* - \rho),$$

For a common constant EIS ϵ, a condition that holds under our utility specification in Section 3, subtracting these equations gives equation (23).

B.3 Relative Wealth and the Ownership Threshold Under the threshold allocation, $a_t^A = \int_{g_t^*}^1 q_t(g)\phi(g)dg$ and $a_t^B = \int_0^{g_t^*} q_t(g)\phi(g)dg$. Define Advocate's average pecuniary portfolio return by

$$R_{w,t}^A \equiv \frac{\int_{g_t^*}^1 R_t(g) q_t(g)\phi(g)dg}{a_t^A}.$$

Using the Advocate return schedule and the definition of G_t,

$$R_{w,t}^A = R_t^* - m_t\left[\frac{G_t}{a_t^A} - v(g_t^*)\right]. \tag{37}$$

Because v is strictly increasing, $G_t/a_t^A > v(g_t^*)$, so $R_{w,t}^A < R_t^*$.

The wealth laws are therefore $\dot{a}_t^A/a_t^A = R_{w,t}^A - c_t^A/a_t^A$ and $\dot{a}_t^B/a_t^B = R_t^* - c_t^B/a_t^B$. Differentiating $w_t^A = a_t^A/(a_t^A + a_t^B)$ gives

$$\dot{w}_t^A = w_t^A(1 - w_t^A)\left[R_{w,t}^A - R_t^* - \left(\frac{c_t^A}{a_t^A} - \frac{c_t^B}{a_t^B}\right)\right],$$

which is equation (22). The two terms are the pecuniary-return and saving effects emphasized in

the main text.

At dates where the threshold is differentiable and ϕ is continuous with $q_t(g_t^*)\phi(g_t^*) > 0$, the threshold law follows by differentiating $a_t^A = \int_{g_t^*}^1 q_t(g)\phi(g)dg$ with respect to time:

$$\dot{a}_t^A = \int_{g_t^*}^1 \dot{q}_t(g)\phi(g)dg - q_t(g_t^*)\phi(g_t^*)\dot{g}_t^*.$$

The flow budget gives $\dot{a}_t^A = \int_{g_t^*}^1 [d_t(g) + \dot{q}_t(g)]\phi(g)dg - c_t^A$, so capital gains cancel, leaving

$$\dot{g}_t^* = \frac{c_t^A - \int_{g_t^*}^1 d_t(g)\phi(g)dg}{q_t(g_t^*)\phi(g_t^*)}, \tag{38}$$

which is equation (24).

For completeness, integrating the dividend identity over firms and using aggregate consumption equal to aggregate dividends yields the resource constraint $\dot{K}_t = Y_t - c_t^A - c_t^B - \delta K_t$, where $K_t = \int k_t(g)\phi(g)dg$ and $Y_t = \int f(k_t(g))\phi(g)dg$.

B.4 Steady State: Characterization, Existence, and Uniqueness In an interior steady state, equation (35) implies $R_{ss}^* = \rho$ for Benchmark and $R_{ss}^* + m_{ss}v(g_{ss}^*) = \rho$ for Advocate. Since $m_{ss} > 0$ and v is strictly increasing with $v(g_n) = 0$, it follows that $g_{ss}^* = g_n$. The firm condition then gives exactly the capital allocation in Proposition 1: $f'(k_{ss}(g)) = \rho + \delta$ for $g < g_n$ and $f'(k_{ss}(g)) = \rho + \delta - m_{ss}v(g)$ for $g \geq g_n$.

We next provide sufficient conditions for existence.

Proposition 2 (Existence of an Interior Steady State)**.** *Suppose $\rho > 0$, $\delta > 0$, and f is continuous at zero with $f(0) = 0$, twice continuously differentiable on $(0, \infty)$, strictly increasing and strictly concave, with the usual Inada conditions $f'(0^+) = \infty$ and $f'(\infty) = 0$; $u \in C^2((0, \infty))$ has $u' > 0$ and $u'' < 0$; $\mathcal{V}$ is twice continuously differentiable on a domain containing $[0, \infty)$, with $\mathcal{V}' > 0$, $\mathcal{V}'' \leq 0$, and $\lim_{G\to\infty} \mathcal{V}'(G) = 0$; v is continuously differentiable and strictly increasing with $v(g_n) = 0$ and $v(1) > 0$; and ϕ places positive mass on both sides of g_n and is continuous and bounded away from zero near $g = 1$. Then at least one interior steady state with positive wealth for both investor types exists.*

Proof. Let $\bar{v} = v(1)$ and $\bar{m} = \rho/\bar{v}$. For any candidate $m \in [0, \bar{m})$, set $R_m(g) = \rho - mv(g)$ for $g \geq g_n$ and define $k_m(g)$ by $f'(k_m(g)) = \delta + R_m(g)$. Stationary dividends are $d_m(g) = f(k_m(g)) - \delta k_m(g)$, Advocate consumption is $c^A(m) = \int_{g_n}^1 d_m(g)\phi(g)dg$, and equity values are $q_m(g) = d_m(g)/R_m(g)$.

Hence

$$G(m) = \int_{g_n}^{1} v(g) \frac{d_m(g)}{\rho - mv(g)} \phi(g) dg, \qquad \Psi(m) \equiv \frac{\mathcal{V}'(G(m))}{u'(c^A(m))}.$$

These objects are continuous for $m < \bar{m}$, and $\Psi(0) > 0$. As $m \uparrow \bar{m}$, $R_m(1) \downarrow 0$ while $k_m(g)$ and $d_m(g)$ remain bounded and strictly positive near $g = 1$. Indeed, $k_m(g) \leq (f')^{-1}(\delta) < \infty$, and strict concavity with $f(0) = 0$ gives positive net output there. Let $\varepsilon = \rho - m\bar{v}$. Bounded v' implies $\rho - mv(g) \leq \varepsilon + C(1-g)$ near one for some $C > 0$, while the numerator $v(g)d_m(g)\phi(g)$ is bounded below by a positive constant. The integral defining $G(m)$ therefore diverges at least logarithmically. Thus $G(m) \to \infty$, while $c^A(m)$ remains bounded away from zero and infinity. The assumption $\mathcal{V}'(G) \to 0$ therefore implies $\Psi(m) \to 0$ as $m \uparrow \bar{m}$.

Consequently $\Psi(m) - m$ is positive at zero and negative for m sufficiently close to $\bar{m}$. Continuity gives a fixed point $m_{ss} = \Psi(m_{ss}) \in (0, \bar{m})$. Setting $g^*_{ss} = g_n$ and using the stationary return, capital, dividend, and price schedules constructed above yields an interior steady state. Positive mass on both sides of g_n gives positive wealth for both households. □

Existence does not by itself imply uniqueness. The steady state is pinned down by the scalar fixed point $m = \Psi(m)$. A sufficient condition for a unique crossing is $\Psi'(m) < 1$ throughout $[0, \bar{m})$; the stronger condition $\Psi'(m) < 0$ is especially transparent. Where differentiable,

$$\frac{\Psi'(m)}{\Psi(m)} = \frac{\mathcal{V}''(G(m))}{\mathcal{V}'(G(m))} G'(m) - \frac{u''(c^A(m))}{u'(c^A(m))} c^{A\prime}(m).$$

Thus uniqueness obtains, for example, when the decline in marginal sustainable-wealth utility induced by higher $G(m)$ dominates the effect of higher Advocate consumption on $1/u'(c^A(m))$. Numerically, boundary signs and a scan for crossings provide diagnostics; a uniqueness claim requires ruling out additional crossings over the full admissible interval.

Every interior steady state has $g^*_{ss} = g_n$. With a unique fixed point, the stationary capital allocation is also unique, so any convergent equilibrium path has the same limit regardless of its initial wealth distribution.

B.5 Transitional Capital Relative to Benchmark

Let $f'(\hat{k}) = \rho + \delta$. On an interior path, Benchmark-priced capital satisfies $f'(k^B_t) - f'(\widehat{k}) = R^*_t - \rho = \dot{c}^B_t / [\epsilon(c^B_t) c^B_t]$. For an Advocate-priced firm,

$$f'(k_t(g)) - f'(\hat{k}) = (R^*_t - \rho) - m_t[v(g) - v(g^*_t)].$$

Thus $k_t(g) < \hat{k}$ exactly when the right-hand side is positive; $v(g) < 0$ alone does not imply capital below Benchmark. In particular, if $R_t^* = \rho$, Benchmark-priced firms remain at $\hat{k}$ and Advocate-priced firms above the cutoff exceed it. This comparison characterizes allocations on an interior path; it does not establish a transition from an arbitrary inherited capital stock.

C Quantitative Model

This appendix specifies the equilibrium and reporting conventions underlying Section 4. Both quantitative exercises use CES consumption utility with $\sigma = .5$, so the intertemporal elasticity of substitution is $\epsilon = 1/\sigma = 2$.

C.1 Preferences, Portfolios, and Equilibrium

Write $I[x] = \int_0^1 x(g)\phi(g)\,dg$ and suppress time subscripts when convenient. Both households have $u'(c) = c^{-\sigma}$. Advocate additionally values sustainable wealth, with

$$\mathcal{V}(G) = \frac{(1+\kappa G)^{1-\sigma_G} - 1}{\kappa(1-\sigma_G)}, \qquad \sigma_G = .5, \qquad 1 + \kappa G > 0. \tag{39}$$

Let $z(g) = z^A(g)$ and $z^B(g) = 1 - z(g)$. The per-firm short-sale bounds imply the sharp ownership allocation

$$z(g) = -\bar{s} + (1 + 2\bar{s})\mathbf{1}\{g > g^*\}, \qquad \bar{s} = .05. \tag{40}$$

Financial wealth, pecuniary portfolio returns, and household budgets satisfy

$$a^j = I[qz^j], \qquad R_w^j = \frac{I[Rqz^j]}{a^j}, \qquad \dot{a}^j = I[Rqz^j] - c^j, \qquad G = I[vqz]. \tag{41}$$

Under limited short sale, sustainable wealth includes "signed" positions: a short position in a firm with negative $v(g)$ contributes positively to G.

Portfolio optimality and the Euler equations give

$$R(g) = R^* - m\max\{v(g) - v(g^*), 0\}, \qquad m = (c^A)^\sigma \mathcal{V}'(G), \tag{42}$$

$$\frac{\dot{c}^A}{c^A} = \frac{R^* + mv(g^*) - \rho}{\sigma}, \qquad \frac{\dot{c}^B}{c^B} = \frac{R^* - \rho}{\sigma}. \tag{43}$$

Here R^* is Benchmark's marginal return; with short positions, it need not equal Benchmark's

portfolio return R_w^B. Differentiating $a^A = I[qz]$ and using the household budget gives

$$\dot{g}^* = \frac{c^A - I[zd]}{(1+2\bar{s})q(g^*)\phi(g^*)}, \tag{44}$$

where the density at the ownership threshold is positive.

Distinguish the marginal value of installed capital, $\mathcal{Q}(g)$, from equity value, $q(g)$. The investment and price equations are

$$\dot{k}(g) = \frac{k(g)}{\psi}[\mathcal{Q}(g) - 1], \tag{45}$$

$$\dot{\mathcal{Q}}(g) = R(g)\mathcal{Q}(g) + \delta - f'(k(g)) - \frac{[\mathcal{Q}(g)-1]^2}{2\psi}, \tag{46}$$

$$\dot{q}(g) = R(g)q(g) - d(g). \tag{47}$$

Dividends follow equation (27), and output market clearing requires $c^A + c^B = I[d]$.

Initial capital is $k_0(g) = \hat{k} = [\alpha/(\rho+\delta)]^{1/(1-\alpha)}$, and $I[q_0 z_0]/I[q_0] = .33$. This wealth share is evaluated at on-impact equilibrium prices. Initial equity values, the ownership threshold, and consumption are endogenous. We select paths approaching a bounded stationary allocation and satisfying household transversality and firm no-bubble conditions. At an interior stationary allocation, $R^* = \rho$, $g^* = g_n$, and

$$k_{ss}(g) = \left[\frac{\alpha}{\delta + R_{ss}(g)}\right]^{1/(1-\alpha)}, \quad \mathcal{Q}_{ss}(g) = 1, \quad q_{ss}(g) = \frac{f(k_{ss}(g)) - \delta k_{ss}(g)}{R_{ss}(g)}. \tag{48}$$

C.2 Reporting Capital and Output Let $K_t = I[k_t]$ and $Y_t = I[k_t^\alpha]$. The highest-score group contains 25% of initial capital and has a fixed lower boundary g_{25}, defined by

$$\int_{g_{25}}^1 k_0(g)\phi(g)\,dg = .25K_0, \qquad K_t^{25} = \int_{g_{25}}^1 k_t(g)\phi(g)\,dg, \qquad Y_t^{25} = \int_{g_{25}}^1 k_t(g)^\alpha \phi(g)\,dg. \tag{49}$$

Membership remains fixed throughout the transition and the preference comparisons. Each reported change is relative to the corresponding initial level: $\Delta X_t = 100(X_t/X_0 - 1)$ for $X \in \{K, Y, K^{25}, Y^{25}\}$. Thus ΔK_t^{25} and ΔY_t^{25} measure growth within the group, not changes in its share of aggregate capital or output.

C.3 Wealth Share Counterfactual The counterfactual pools resources in a price-taking family that maintains Advocate's investment account at the fraction $\bar{w} = .33$ of aggregate net equity,

$I[qz] = \bar{w}I[q]$. The family maximizes discounted $u(c^A) + \chi u(c^B) + \mathcal{V}(G)$, where $\chi = [(1-\eta)/\eta]^\sigma$ and $\eta = .33$, giving $c^A = \eta C$ and $c^B = (1-\eta)C$. This consumption-sharing rule is an additional assumption, not a consequence of fixing the wealth share. Preferences, technology, and short-sale bounds retain their full-economy calibration.

Portfolio sorting and pricing satisfy equations (40) and (42), with $m = (\eta C)^\sigma \mathcal{V}'(G)$. The family chooses saving jointly, replacing the two household Euler equations with

$$\frac{\dot{C}}{C} = \frac{R^* + \bar{w}mv(g^*) - \rho}{\sigma}, \qquad C = I[d]. \tag{50}$$

The direct nonpecuniary saving motive remains, but the comparison with the full economy includes the consequences of this different saving and consumption-sharing arrangement. Appendix J gives the family problem and its equilibrium conditions; Appendix I provides calibration, measurement, and numerical details.

D Policy Construction and Implementation

The policy exercise constructs a policy in the all-Benchmark economy and asks whether sustainable-investing preferences can implement its stationary allocation, first in the All-Advocate economy and then in the Advocate+Benchmark economy. The target is a 10% reduction in total emissions relative to the initial all-Benchmark allocation.

D.1 The Policy Allocation

The policy combines an emissions tax τ with an output subsidy τe_n. A firm's net payment is $\tau[e(g) - e_n]k(g)^\alpha$. In steady state its capital is

$$k_P(g) = \hat{k}\{1 + \tau[e_n - e(g)]\}^{1/(1-\alpha)}, \qquad \hat{k} = \left(\frac{\alpha}{\rho + \delta}\right)^{1/(1-\alpha)}. \tag{51}$$

We choose τ and e_n jointly to satisfy

$$\int_0^1 e(g)k_P(g)^\alpha \, dg = .9E_0, \qquad \tau \int_0^1 [e(g) - e_n]k_P(g)^\alpha \, dg = 0. \tag{52}$$

The second condition balances taxes and subsidies at the stationary allocation, implying $e_n = E_P/Y_P$. During the transition, policy parameters remain fixed and net fiscal revenue is returned to households as lump-sum transfers; budget balance is imposed only in the steady state.

D.2 All-Advocate Implementation Remove the policy and specify preferences as

$$v(g) = \nu(\rho + \delta)\frac{\tau[e_n - e(g)]}{1 + \tau[e_n - e(g)]}. \tag{53}$$

At the policy capital allocation, the required untaxed financial return is

$$R_{\text{req}}(g) = \alpha k_P(g)^{\alpha-1} - \delta = \rho - \frac{v(g)}{\nu}. \tag{54}$$

The All-Advocate stationary Euler condition, $R_{\text{req}}(g) + mv(g) = \rho$, therefore holds for all firms when $m\nu = 1$. To choose the scale consistently, define untaxed dividends $d_P(g) = k_P(g)^\alpha - \delta k_P(g)$, equity values $q_P(g) = d_P(g)/R_{\text{req}}(g)$, and consumption $C_P = \int_0^1 d_P(g)\,dg$. With the maintained sustainable-wealth curvature, ν solves

$$C_P^{2\sigma}\nu^2 - \kappa\nu \int_0^1 [\rho - R_{\text{req}}(g)]q_P(g)\,dg - 1 = 0. \tag{55}$$

We select the unique positive root and verify the utility domain and the unsquared condition $m\nu = 1$. This construction implements stationary capital, output, and emissions. It does not impose identical transition paths: m_t changes during adjustment, while $v(g)$ remains fixed.

D.3 Implementation in the Advocate+Benchmark Economy The Advocate+Benchmark economy satisfies the equilibrium conditions in Appendix C, with initial Advocate wealth shares of .33, .66, and .99 and sustainability preferences specified as above. At an interior stationary allocation, the two Euler equations imply $R^* = \rho$ and $g^* = g_n$, where $e(g_n) = e_n$. Consequently,

$$R_M(g) = \rho - m_M \max\{v(g), 0\}, \qquad k_M(g) = \left[\frac{\alpha}{\delta + R_M(g)}\right]^{1/(1-\alpha)}. \tag{56}$$

For firms below the neutral score, Benchmark financing leaves $R_M(g) = \rho$ and $k_M(g) = \hat{k}$, whereas the emissions reduction policy requires a contraction. Even setting $m_M\nu = 1$ would not remove this difference. A small initial net Benchmark wealth share is also distinct from the absence of Benchmark investors: short positions can finance larger gross holdings, and relative wealth adjusts along the transition. Appendix K provides the full policy and implementation equations, transition conditions, and numerical checks.

Appendix II: Model Extensions

E Costly Firm Borrowing Extension

This appendix extends the two-period model to allow firms to borrow and lend through a representative competitive intermediary. The purpose is to examine to what extent interfirm borrowing and lending affect the capital composition and saving incentive of sustainable investing. Firms valued by Advocate have a lower required pecuniary return on equity and can channel retained resources to firms with a higher required return. We introduce a representative intermediary that faces a monitoring cost which is convex in the level of equilibrium intermediation.

E.1 Interfirm Financing

The production technology, household preferences, and firm characteristics are those of the two-period model. Initial capital is common across firms, initial output is $Y_0 = f(k_0)$, and $\int_0^1 \phi(g)dg = 1$. Let $b(g) \geq 0$ denote a firm's borrowing from the intermediary and $x(g) \geq 0$ its deposits with it. The corresponding gross returns are denoted by R_b and R_x respectively. Firm dividends and equity prices satisfy

$$\begin{aligned} d_0(g) &= Y_0 - k(g) + b(g) - x(g) \geq 0, \\ d_1(g) &= f(k(g)) + R_x\, x(g) - R_b\, b(g) \geq 0, \qquad\qquad q(g) = \frac{d_1(g)}{R_s(g)}. \end{aligned} \tag{57}$$

where $R_s(g)$ is the return on equity. Taking returns as given, firms choose $k(g), b(g), x(g) \geq 0$ to maximize $d_0(g) + q(g)$. There is no exogenous upper bound on $b(g)$, so borrowing is not directly constrained, but we assume full repayment of debt. Borrowing can finance investment or an initial dividend; new equity issuance, household borrowing, and short sales remain prohibited. Extending the borrowing possibility to households does not significantly change the results because households do not face differential cost of funds.

Intermediation is exclusively *interfirm*: households hold equity, while firms hold deposits and issue loans. We assume that lending passes through a representative financial intermediary, whose problem is described below. We assume that firm scores remain fixed, and neither loans nor deposits enter household sustainable wealth separately. Thus the low-cost funding of high-score firms reflects shareholders' willingness to retain resources at a low pecuniary return on equity; it does not require a score-dependent rate on otherwise identical debt. When a high-score firm places a deposit, the deposit is reflected in its equity value and retains the sustainability valuation

associated with that firm's score. This is admittedly a strong assumption as Advocate households might value differently a firm with a high score that lends to firms with a lower score. The results of this section should thus be considered as illustrating the most favorable scenario for borrowing to impact capital composition via arbitrage.

E.2 Representative financial intermediary Let L denote aggregate lending. Borrowing/lending markets clear when

$$L = \int_0^1 x(g)\phi(g)dg = \int_0^1 b(g)\phi(g)dg. \tag{58}$$

The representative intermediary takes the borrowing return R_b, the deposit return R_x, and the monitoring fee per unit of principal p_M as given. The intermediary chooses its volume $\ell \geq 0$ to maximize profits

$$\Pi(\ell) = (R_b - R_x - p_M)\ell. \tag{59}$$

Profit maximization and free entry imply $R_b - R_x = p_M$, so the interest margin covers monitoring expenses and intermediary profits, and equity values are zero.

We assume a reduced-form monitoring-price schedule that increases with aggregate intermediation, specifically

$$p_M = \frac{\chi}{2}\frac{L}{k_0}, \qquad R_b - R_x = p_M, \qquad M \equiv p_M L = \frac{\chi}{2}\frac{L^2}{k_0}, \qquad \chi > 0. \tag{60}$$

In equilibrium, $\ell = L$, and the intermediary takes p_M as given in its volume choice; the dependence of this price on aggregate L is an equilibrium relationship. Monitoring expenses are therefore linear in the intermediary's own volume at given prices, while aggregate monitoring payments are quadratic in equilibrium volume. This determines the level of borrowing and lending in the economy.

Finally, we assume that monitoring services are produced using labor from households. Households supply labor inelastically and receive monitoring income in proportion to their initial financial wealth shares:

$$y_M^A = w_0 M, \qquad y_M^B = (1 - w_0)M. \tag{61}$$

Households take their monitoring income as given when choosing consumption and portfolios.

Write $a_0^A = w_0 W_0$, $a_0^B = (1 - w_0)W_0$, and $Q^i = \int_0^1 q(g)z^i(g)\phi(g)dg$. The budget constraints for

households are

$$c_0^i + Q^i = a_0^i, \qquad c_1^i = \int_0^1 d_1(g) z^i(g) \phi(g) dg + y_M^i, \qquad i \in \{A, B\}. \tag{62}$$

Equity clears at $z^A(g) + z^B(g) = 1$, with $z^i(g) \geq 0$.

E.3 Equilibrium Define household shadow returns and the marginal valuation of sustainable wealth by

$$R^i = \frac{u'(c_0^i)}{\beta u'(c_1^i)}, \qquad m = \frac{\mathcal{V}'(G)}{u'(c_1^A)}. \tag{63}$$

For increasing $v(g)$, the ownership threshold g^* and equity returns satisfy

$$R_s(g) = \begin{cases} R^B, & g < g^*, \\ R^B - \frac{m}{\beta}[v(g) - v(g^*)], & g \geq g^*, \end{cases} \qquad R^A - R^B = \frac{m}{\beta} v(g^*). \tag{64}$$

Benchmark holds equity below g^* and Advocate above it. Monitoring income affects consumption and therefore the equilibrium shadow returns, but does not change the portfolio-sorting condition. Here R^B is Benchmark's household shadow return, while R_b is the corporate borrowing return.

The firm conditions clarify how intermediation changes investment. Let $\lambda(g) \geq 0$ be the multiplier on $d_0(g) \geq 0$, and define $\widetilde{R}(g) = R_s(g)[1 + \lambda(g)]$. With positive capital and residual equity, optimality implies

$$\begin{aligned} f'(k(g)) &= \widetilde{R}(g), & \lambda(g) d_0(g) &= 0, \\ R_x \leq \widetilde{R}(g) &\leq R_b, & x(g)[\widetilde{R}(g) - R_x] &= 0, \\ & & b(g)[R_b - \widetilde{R}(g)] &= 0. \end{aligned} \tag{65}$$

Consider the active-intermediation branch with both household types holding positive equity. A depositing firm invests until $f'(k) = R_x$ and deposits its remaining internal resources. Positive deposits require $(f')^{-1}(R_x) < Y_0$. Since $R_b > R_x$, a borrower likewise chooses capital below Y_0 and has a positive initial dividend. Its condition is therefore $R_s(g) = R_b$. Combining this with (64) yields $R_b = R^B$: the Benchmark-priced segment is indifferent about debt issuance, and its aggregate issuance adjusts to absorb deposits.

Let g_L denote the deposit threshold, defined by $R_s(g_L) = R_x$. When $g^* < g_L < 1$, the capital

allocation is

$$k(g) = \begin{cases} (f')^{-1}(R^B), & g < g^*, \\ (f')^{-1}\Big(R^B - \frac{m}{\beta}[v(g) - v(g^*)]\Big), & g^* \le g < g_L, \\ (f')^{-1}(R_x), & g \ge g_L. \end{cases} \tag{66}$$

The *lower flat segment* has the same origin as in the main model: Benchmark prices these firms at a common return, so they choose the same capital. The middle segment is increasing in g because Advocate accepts progressively lower equity returns at higher scores. The *upper flat segment* arises from the option to deposit: once a firm's equity return falls below R_x, additional retained resources earn more in deposits than in direct production investment. Its investment stops at the common level $(f')^{-1}(R_x)$, while additional resources are deposited. Thus intermediation truncates the rising part of the hockey stick without eliminating preference-driven equity sorting.

Writing $\Phi(g) = \int_0^g \phi(s)ds$, the active branch satisfies

$$L = [1 - \Phi(g_L)]\big[Y_0 - (f')^{-1}(R_x)\big], \qquad \frac{m}{\beta}[v(g_L) - v(g^*)] = R_b - R_x = \frac{\chi L}{2k_0}. \tag{67}$$

In the numerical implementation, debt is distributed equally across the indifferent Benchmark-priced firms: $b(g) = L/\Phi(g^*)$ for $g < g^*$ and zero above.

Let $K = \int_0^1 k(g)\phi(g)dg$, $Y = \int_0^1 f(k(g))\phi(g)dg$, and $Q = Q^A + Q^B$. Aggregating firm and household budgets gives

$$W_0 = Y_0 - K + Q, \qquad c_0^A + c_0^B + K = Y_0, \qquad c_1^A + c_1^B = Y. \tag{68}$$

Corporate dividends at date one sum to $Y - M$, but households also receive $y_M^A + y_M^B = M$, so total output to households is still Y. However, the distribution of monitoring payment affects households' saving decision. We demonstrate next that the nonpecuniary saving incentive survives the introduction of monitoring income in the saving decision. For log consumption utility, define $s^i = Q^i/a_0^i$, $\Gamma = G\mathcal{V}'(G)$, and $\Omega^i = \beta y_M^i/c_1^i$. Integrating household portfolio optimality over equity holdings and using (62) yields

$$\frac{Q^A}{c_0^A} = \beta + G\mathcal{V}'(G) - \Omega^A, \qquad \frac{Q^B}{c_0^B} = \beta - \Omega^B, \tag{69}$$

so that

$$s^A = \frac{\beta + G\mathcal{V}'(G) - \Omega^A}{1 + \beta + \Gamma - \Omega^A}, \qquad s^B = \frac{\beta - \Omega^B}{1 + \beta - \Omega^B}. \tag{70}$$

The nonpecuniary saving incentive thus still increases the incentive to save for Advocate, while anticipated monitoring income works in the opposite direction: it finances future consumption and reduces the need to acquire equity. In particular, Benchmark saving is no longer fixed at $\beta/(1+\beta)$. Nor is financial saving proportional to productive capital, since the equity of depositing firms incorporates financial claims on other firms. Numerically, the impact of monitoring income on saving is limited.

An equilibrium consists of household consumption and equity holdings, firm investment and financing choices, equity prices, the ownership and deposit thresholds, and intermediary returns and volume satisfying household and firm optimality, intermediary profit maximization, and equity, credit, and goods-market clearing. The characterization above is conditional on active intermediation and positive residual equity; it does not assert global uniqueness or rule out other financing corners.

E.4 Numerical illustration We retain the Section 2 specification: $f(k) = k^{0.33}$, $u(c) = \log c$, $\beta = 1$, $\phi(g) = 1$, $v(g) = 3(g - 0.5)/(1 - 0.5)$, and $\mathcal{V}'(G) = (1 + 4.64G)^{-0.5}$. Initial capital is $k_0 = \hat{k} = [\alpha\beta/(1 + \alpha\beta)]^{1/(1-\alpha)} = 0.125$, and $Y_0 = \widehat{Y} = 0.503$. We compare the illustrative friction coefficients $\chi = 0.25, 1, 4$, without imposing borrowing limits in any case.

Figure 10(a) holds initial Advocate wealth at $w_0 = 0.66$. Lower intermediation frictions compress the difference between borrowers' and depositors' returns. As χ falls from 4 to 0.25, the borrowing return falls and the deposit return rises, raising capital at lower-score firms and lowering the upper plateau. Depositing begins at a lower score, so the rising middle segment also becomes shorter. Intermediated principal increases from 23.05 to 81.79 percent of initial capital. The resulting capital allocation is substantially flatter, although the equity portfolios remain sorted by score.

Panel (b) shows that aggregate capital is nonmonotonic in initial Advocate wealth for each friction coefficient. Capital peaks at approximately 6.39, 7.79, and 10.92 percent above all-Benchmark for $\chi = 0.25, 1, 4$, respectively. Greater Advocate wealth initially supports capital accumulation, but an increasing share of its financial claims can be held through the equity of depositing firms. These firms stop expanding their own productive capital once its marginal product reaches the deposit return. Changes in the deposit return and in the masses of borrowers and depositors then

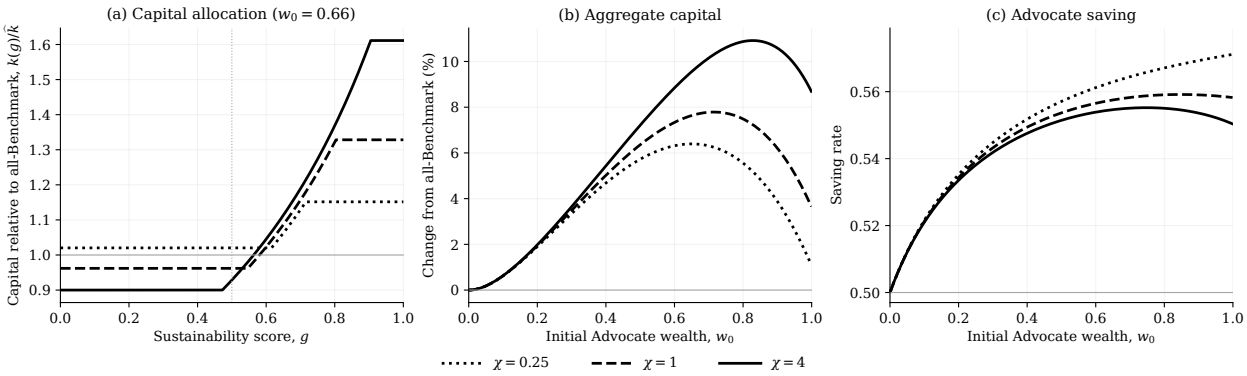


Figure 10: Capital Allocation, Aggregate Capital, and Advocate Saving with Intermediation

Notes: Line styles represent $\chi = 0.25, 1, 4$ in (60). There are no exogenous borrowing limits. Monitoring income is distributed according to (61). Panel (a) fixes $w_0 = 0.66$; the horizontal line marks all-Benchmark capital and the vertical dotted line marks $g_n = 0.5$. Panels (b) and (c) vary w_0 over $[0.001, 0.999]$. Aggregate capital is measured relative to all-Benchmark, and Advocate saving is defined in (70). The horizontal saving reference is 0.5. The exact All-Advocate endpoint is not included.

allow aggregate capital to decline even as Advocate's financial capacity increases. This aggregate decline is an equilibrium outcome of the illustration, not a direct implication of the individual-firm plateau alone.

Panel (c) shows that weaker frictions can nevertheless raise Advocate saving. At a given initial wealth share, more intermediation allows Advocate to hold a narrower set of higher-score equities while those firms finance other firms indirectly. At $w_0 = 0.66$, reducing χ from 4 to 0.25 raises the ownership threshold from 0.472 to 0.616 and increases sustainable wealth from 0.496 to 0.606. Although $\mathcal{V}'(G)$ falls, the product $G\mathcal{V}'(G)$ rises. The stronger nonpecuniary saving incentive reflects both the value of Advocate's equity holdings and their concentration in higher-score firms, including the deposits held within those firms.

Monitoring income partly offsets this incentive, rather than causing the increase in saving. At $w_0 = 0.66$, $G\mathcal{V}'(G) - \Omega^A$ equals approximately 0.290, 0.261, and 0.246 for $\chi = 0.25, 1, 4$, yielding Advocate saving rates of 56.32, 55.77, and 55.47 percent. Thus, the equilibrium increase in the net nonpecuniary saving incentive explains why weaker frictions generate higher saving in the figure.

The comparison illustrates why financial saving and productive capital must be distinguished. Interfirm lending reduces the dispersion of capital across scores and permits additional financial wealth to be held through depositing firms, while the sustainability valuation continues to depend on the score of the equity issuer. Consequently, higher Advocate saving can coexist with a smaller aggregate capital stock.

F Short Sales Extension

This appendix relaxes the short sales constraint while retaining the technology, sustainable-wealth preferences, and retained-earnings financing of Section 2. While finite short positions preserve the composition effect and the nonpecuniary saving motive, they change the mapping from wealth to portfolio positions and introduce a constraint term into the saving equations.

F.1 Environment and portfolio conditions We assume that each household may short at most a fraction $\bar{s} \geq 0$ of the outstanding shares of each firm:

$$z^i(g) \geq -\bar{s}, \qquad i \in \{A, B\}, \qquad z^A(g) + z^B(g) = 1. \tag{71}$$

Consequently, $-\bar{s} \leq z^A(g) \leq 1 + \bar{s}$ in equilibrium. The upper bound follows from market clearing and the other household's short-sale limit. Short-sale proceeds enter the seller's initial budget, the seller pays the terminal dividend on the borrowed shares and there are no stock-loan fees.

Write $Y_0 = f(k_0)$, $K = \int k(g)\phi(g)dg$, and $Q = \int q(g)\phi(g)dg$. Firm dividends remain $d_0(g) = Y_0 - k(g) \geq 0$ and $d_1(g) = f(k(g))$. Thus

$$k(g) = \min\{(f')^{-1}(R(g)), Y_0\}, \qquad q(g) = \frac{f(k(g))}{R(g)}. \tag{72}$$

For household i, net financial saving is $a_i = \int q z^i \phi \, dg$, terminal consumption is $c_1^i = \int f(k) z^i \phi \, dg$, and $c_0^i + a_i = w_i W_0$, where $w_A = w_0$, $w_B = 1 - w_0$, and $W_0 = Y_0 - K + Q$. Short positions cancel in the aggregate: $a_A + a_B = Q$ and $c_0^A + c_0^B + K = Y_0$.

Sustainable wealth remains $G = \int v(g)q(g)z^A(g)\phi(g)dg$, now evaluated at signed positions. In particular, shorting a firm with $v(g) < 0$ contributes positively to G. Advocate's utility remains $u(c_0^A) + \beta u(c_1^A) + \mathcal{V}(G)$. Define the nonnegative marginal-utility gaps

$$\begin{aligned} \lambda_A(g) &= u'(c_0^A) - \beta u'(c_1^A) R(g) - \mathcal{V}'(G) v(g), \\ \lambda_B(g) &= u'(c_0^B) - \beta u'(c_1^B) R(g). \end{aligned} \tag{73}$$

Portfolio optimality requires $\lambda_i(g) \geq 0$ and $\lambda_i(g)[z^i(g) + \bar{s}] = 0$. A household prices every firm at which its position is strictly above its lower bound, including a negative position above $-\bar{s}$.

F.2 Sorting, investment, and the wealth threshold Let $r_i = u'(c_0^i)/[\beta u'(c_1^i)]$ and $m = \mathcal{V}'(G)/u'(c_1^A)$. The portfolio inequalities imply $R(g) \leq \min\{r_B, r_A - \frac{m}{\beta} v(g)\}$. At least one household is above its lower bound at every firm, so one inequality must be an equality. With an interior crossing $r_A - r_B = \frac{m}{\beta} v(g^*)$, strict monotonicity of v gives

$$z^A(g) = \begin{cases} -\bar{s}, & g < g^*, \\ 1 + \bar{s}, & g > g^*, \end{cases} \qquad R(g) = \begin{cases} R^*, & g < g^*, \\ R^* - \frac{m}{\beta}[v(g) - v(g^*)], & g > g^*, \end{cases} \tag{74}$$

where $R^* = r_B$. Benchmark is the marginal investor below the cutoff and Advocate above it. The threshold now separates the households' long and short positions, so it no longer describes exclusive positive ownership.

Equation (74) retains the capital hockey stick shape of the baseline model in the main text. Below g^*, capital is constant; above it, capital increases with g wherever the retained-earnings ceiling is slack. A larger short-sale limit does not necessarily steepen this schedule as R^*, $\frac{m}{\beta}$, and g^* all adjust in equilibrium.

Let $Q_H = \int_{g^*}^1 q\phi \, dg$ and $Q_L = Q - Q_H$. Market clearing implies

$$a_A = (1 + \bar{s})Q_H - \bar{s}Q_L, \qquad a_B = (1 + \bar{s})Q_L - \bar{s}Q_H, \quad \frac{Q_H}{Q} = \frac{a_A/Q + \bar{s}}{1 + 2\bar{s}}. \tag{75}$$

This is the short-sale counterpart of the wealth–ownership relation in Section 2.

F.3 Saving with binding short-sale limits Multiplying equation (73) by $q(g)z^i(g)$ and integrating yields

$$u'(c_0^i)a_i = \beta u'(c_1^i)c_1^i + \mathbf{1}_{\{i=A\}} G\mathcal{V}'(G) - \bar{s}L_i, \qquad L_i = \int_0^1 \lambda_i(g)q(g)\phi(g)dg \geq 0. \tag{76}$$

The last term follows from complementary slackness so that a nonzero multiplier applies at $z^i = -\bar{s}$. Under log consumption utility, the saving rates are

$$s^A = \frac{\beta + G\mathcal{V}'(G) - \bar{s}L_A}{1 + \beta + G\mathcal{V}'(G) - \bar{s}L_A}, \qquad s^B = \frac{\beta - \bar{s}L_B}{1 + \beta - \bar{s}L_B}. \tag{77}$$

Note that the nonpecuniary saving motive $G\mathcal{V}'(G)$ remains, but short sales work in the opposite direction. In addition, Benchmark's saving rate is no longer fixed at $\beta/(1 + \beta)$. The equilibrium

impact of short sales on saving rates as well as capital composition is evaluated in the following numerical illustration.

F.4 Numerical illustration We use the Section 2 primitives: $f(k) = k^{0.33}$, $\beta = 1$, $\phi(g) = 1$, $g_n = 0.5$, $k_0 = \hat{k} = [\alpha\beta/(1+\alpha\beta)]^{1/(1-\alpha)}$, and

$$v(g) = 3\frac{g - 0.5}{1 - 0.5}, \qquad \mathcal{V}'(G) = (1 + 4.64G)^{-1/2}, \tag{78}$$

The all-Benchmark allocation has $\hat{K} = 0.125$ and $\hat{Y} = 0.503$. We compare $\bar{s} = 0, 0.1, 0.2, 0.5$.

Table 6: Limited short sales: saving and aggregate outcomes

w_0	$\bar{s}$	g^*	s^A (%)	s^B (%)	ΔK (%)	ΔY (%)
0.33	0.0	0.680	54.29	50.00	4.32	1.34
0.33	0.1	0.655	54.27	49.88	4.05	1.26
0.33	0.2	0.637	54.32	49.76	3.84	1.19
0.33	0.5	0.602	54.61	49.38	3.37	1.04
0.66	0.0	0.414	55.52	50.00	11.38	2.72
0.66	0.1	0.434	55.73	48.80	10.50	2.59
0.66	0.2	0.447	55.90	47.83	9.79	2.48
0.66	0.5	0.471	56.29	45.67	8.29	2.21
0.99	0.0	0.023	55.71	50.00	18.03	1.64
0.99	0.1	0.153	56.21	-60.91	15.96	1.85
0.99	0.2	0.226	56.57	-138.79	14.56	2.01
0.99	0.5	0.334	57.26	-283.88	12.05	2.23

Notes: Capital and output are measured relative to the all-Benchmark economy. Saving is the market value of the household's *net* equity position divided by its initial wealth. The position limit is a fraction of each firm's outstanding shares.

Table 6 and Figures 11–13 show that finite short sales preserve the main mechanisms. At $w_0 = 0.33$, raising $\bar{s}$ from zero to 0.1 lowers the cutoff from 0.680 to 0.655; at $w_0 = 0.66$, it raises it from 0.414 to 0.434. The capital hockey stick remains in both cases, while shorting allows stronger financial positions without a proportionate increase in productive capital.

At $w_0 = 0.66$ and $\bar{s} = 0.1$, Advocate's saving rate is 55.73%, compared with 55.52% without short sales, while Benchmark saves 48.80% rather than 50%. Aggregate capital rises by 10.50% rather than 11.38%, and output by 2.59% rather than 2.72%. Thus greater scope for portfolio tilts need not amplify the aggregate capital response. Finally, the output response remains nonmonotonic in Advocate wealth as the gains from additional capital eventually face a larger cost from its unequal allocation across otherwise identical technologies.

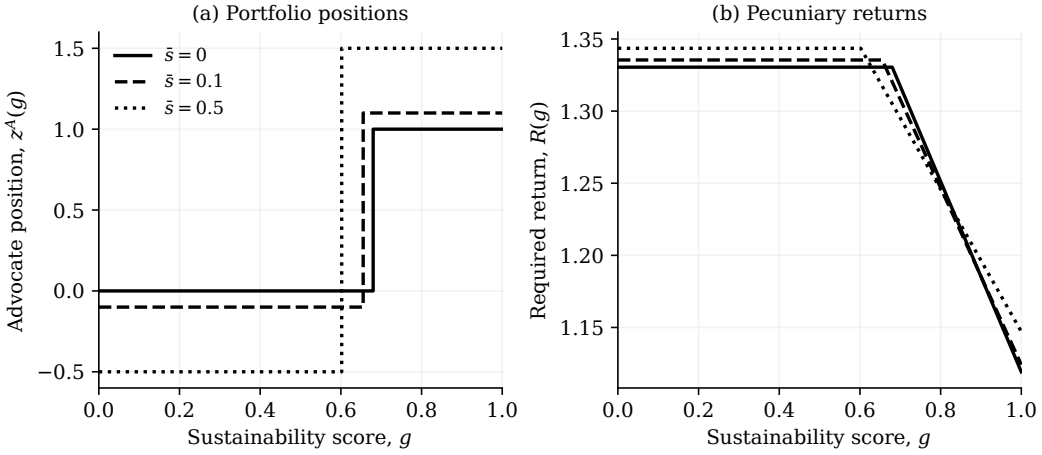


Figure 11: Short sales, portfolio positions, and required returns

Notes: $w_0 = 0.33$. Positions are fractions of outstanding shares. Required returns include the endogenous adjustment of prices, saving, and the pricing cutoff.

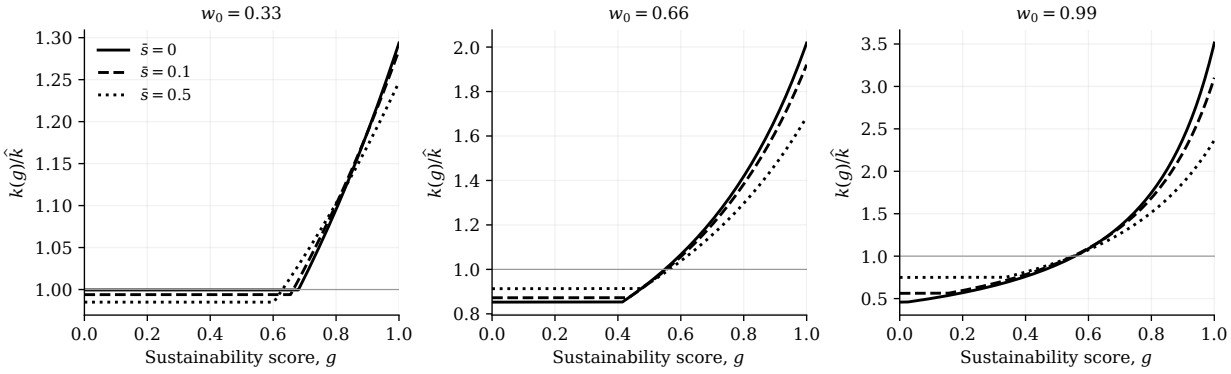


Figure 12: The capital hockey stick with limited short sales

Notes: Each panel fixes initial Advocate wealth. Capital is normalized by the common all-Benchmark capital stock. The loading is linear with $\nu = 3$ in every panel.

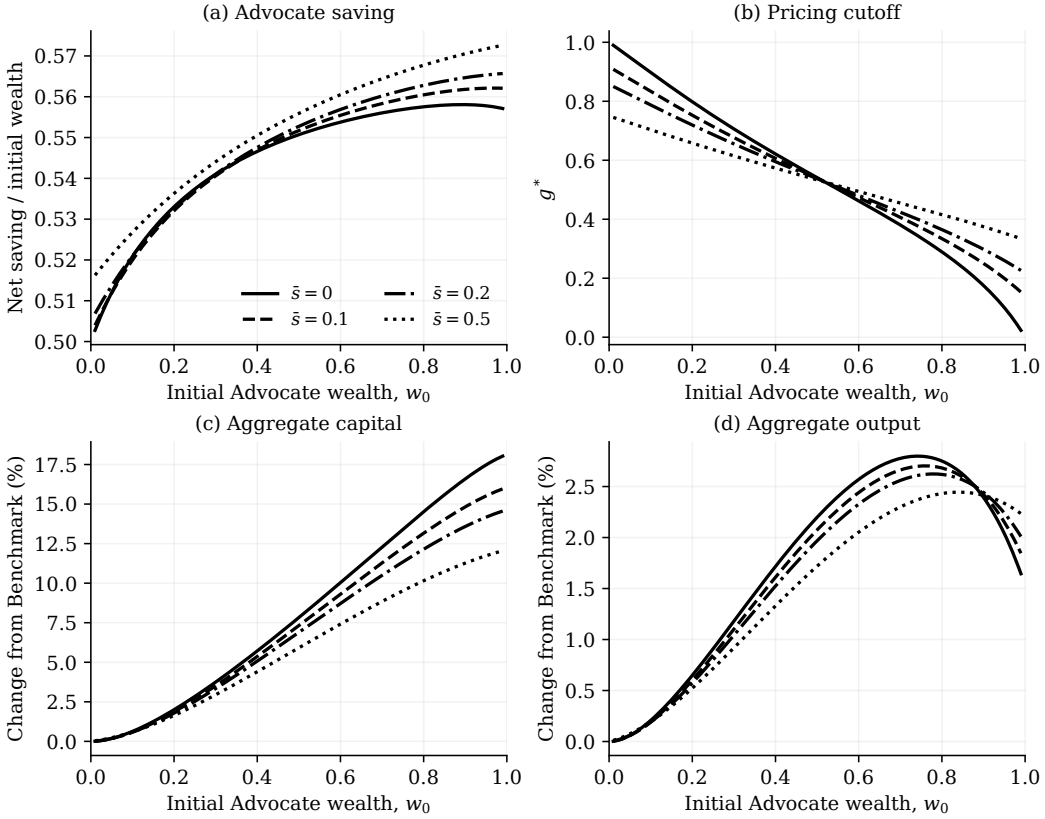


Figure 13: Short-sale limits, the wealth threshold, and the scale effect

Notes: Each curve is a sequence of equilibria indexed by $w_0 \in [0.01, 0.99]$. The cutoff separates long and short positions. Capital and output are relative to the all-Benchmark economy.

G Return Risk and Portfolio Diversification

This appendix introduces uncertain returns while retaining the sustainable-wealth preferences and retained-earnings financing of Section 2. To evaluate the impact of risk, we study two extensions of the baseline model. In the first extension, we consider the role of dividend risk in the presence of Advocate investors with nonpecuniary preferences. Uncertain returns add a diversification motive, so both households may hold the same firm, required returns reflect risk premia as well as sustainability preferences, and the capital schedule need not have an exactly flat segment. The composition mechanism nevertheless remains: for a given household consumption allocation and risk premium, Advocate accepts a lower expected pecuniary return on more attractive firms, lowering their cost of capital. The additional nonpecuniary saving incentive also remains operative. In our numerical illustration, baseline risk leaves Advocate saving, aggregate capital, and expected output close to their corresponding deterministic values.

In the second extension, we remove sustainability preferences and introduce differential climate-risk exposure in an all-Benchmark economy. This setting illustrates the alternative risk-based explanation for sustainable investment studied in the climate-finance literature reviewed above. Higher-score firms are less exposed to adverse climate states, which can lower their required expected returns and generate a pattern of capital allocation similar to that in the economy with Advocate investors. However, this reallocation does not generate the additional saving incentive associated with nonpecuniary preferences. In our illustration with Epstein-Zin preferences with an elasticity of intertemporal substitution of 1.5 and risk aversion of 3, climate risk strengthens precautionary demand for safe payoffs but lowers the certainty-equivalent portfolio return and thus reduces equilibrium saving and aggregate capital.

G.1 Output risk and firm investment

We assume that output in period 1 is given by $f(k(g))Z(g,\omega)$, where $Z > 0$, $\mathbb{E}[Z(g,\omega)] = 1$, and the distribution of Z is independent of the firm's capital choice. It is useful to define $D(g) > 0$ to be the market valuation of one unit of the payoff $Z(g,\omega)$. An atomistic firm takes this valuation schedule as given and maximizes

$$Y_0 - k(g) + D(g)f(k(g)), \qquad 0 \le k(g) \le Y_0. \tag{79}$$

Its equity value, realized gross return, and expected gross return are

$$q(g) = D(g)f(k(g)), \qquad \widetilde{R}(g,\omega) = \frac{Z(g,\omega)}{D(g)}, \qquad R(g) = \mathbb{E}[\widetilde{R}(g,\omega)] = \frac{1}{D(g)}. \tag{80}$$

With $f(k) = k^\alpha$, optimal investment is $k(g) = \left[\frac{\alpha}{R(g)}\right]^{\frac{1}{1-\alpha}}$, and equity is valued at $q(g) = \frac{k(g)^\alpha}{R(g)}$, so multiplicative output risk is priced through $D(g)$.

Household $i \in \{A, B\}$ chooses initial consumption and its portfolio before uncertainty resolves. Terminal consumption is

$$C_1^i(\omega) = \int_0^1 z^i(g) f(k(g)) Z(g,\omega)\phi(g)\, dg.$$

where we use the uppercase notation for consumption in period 1 to denote that it is now a random variable. Advocate maximizes $u(c_0^A) + \beta\mathbb{E}[u(C_1^A)] + \mathcal{V}(G)$, where sustainable wealth is measured at date 0, $G = \int_0^1 v(g)q(g)z^A(g)\phi(g)\, dg$. Define $a_i = \int qz^i\phi\, dg$, $K = \int k\phi\, dg$, and $Q = \int q\phi\, dg$. Initial wealth is $W_0 = Y_0 - K + Q$, and budgets are $c_0^A + a_A = w_0 W_0$ and $c_0^B + a_B = (1 - w_0)W_0$. Market clearing for equity, $z^A + z^B = 1$, implies market clearing for goods in period 0 and in period 1 contingent on any realized state.

G.2 Risk premia and the composition effect

The optimal portfolio allocation conditions under risk are

$$\begin{aligned} \lambda_i(g) &= u'(c_0^i) - \beta\mathbb{E}[u'(C_1^i)\widetilde{R}(g)] - \mathbf{1}_{\{i=A\}}\mathcal{V}'(G)v(g) \geq 0, \\ 0 &= \lambda_i(g) z^i(g). \end{aligned} \tag{81}$$

To separate tastes from risk, let $\overline{u}_i' = \mathbb{E}[u'(C_1^i)]$ and define

$$r_i = \frac{u'(c_0^i)}{\beta\overline{u}_i'}, \qquad \pi_i(g) = -\frac{\mathrm{Cov}(u'(C_1^i), \widetilde{R}(g))}{\overline{u}_i'}, \qquad m^r = \frac{\mathcal{V}'(G)}{\overline{u}_A'}. \tag{82}$$

where r^i corresponds to the household-specific return that would be required on a risk-free asset, while $\pi^i(g)$ is the risk premium required by household i on firm g. It follows that

$$R(g) = \begin{cases} r_A + \pi_A(g) - \frac{m^r}{\beta}v(g), & \text{if Advocate is marginal,} \\ r_B + \pi_B(g), & \text{if Benchmark is marginal.} \end{cases} \tag{83}$$

As in the deterministic model, a more attractive score lowers Advocate's required expected return, holding its risk premium and household-specific risk-free rate fixed. The investment rule translates this return wedge into a capital-allocation wedge. Note that, differently from Section 2, both households can be marginal at the same firm because its payoff covaries differently with their consumption. For a firm held by both households,

$$\frac{m^r}{\beta} v(g) = (r_A - r_B) + [\pi_A(g) - \pi_B(g)]. \tag{84}$$

Risk can consequently replace a sharp ownership cutoff by overlap across scores. Whether broad sharing emerges depends on the strength and correlation of risk relative to the taste wedge.

G.3 The saving motive under uncertainty Multiply equation (81) by qz^i, integrate, and use the state-contingent budget and complementary slackness to get

$$u'(c_0^i)a_i = \beta \mathbb{E}[u'(C_1^i)C_1^i] + \mathbf{1}_{\{i=A\}} G\mathcal{V}'(G). \tag{85}$$

For $u(c) = \log c$, $u'(C_1^i)C_1^i = 1$ in every state. With no short sales, saving as a share of initial household wealth satisfies

$$s^A = \frac{\beta + G\mathcal{V}'(G)}{1 + \beta + G\mathcal{V}'(G)}, \qquad s^B = \frac{\beta}{1+\beta}. \tag{86}$$

The result exactly replicates the saving rate in the deterministic model. Advocate saves more than Benchmark precisely when $G > 0$, given $\mathcal{V}'(G) > 0$. Risk affects G through portfolios and prices but adds no separate term to these equilibrium saving formulas under log utility. This invariance does not eliminate precautionary valuation of safe payoffs, as the all-Benchmark exercise below shows. The result need not hold under a general consumption utility because the expectation in equation (85) need not be constant.

G.4 Numerical specification We assume that the score distribution is uniform on $[0, 1]$, and the technology and preference parameters are

$$\alpha = 0.33, \quad \beta = 1, \quad g_n = 0.5, \quad \nu = 3, \quad v(g) = \nu \frac{g - g_n}{1 - g_n}, \quad \mathcal{V}'(G) = (1 + 4.64G)^{-1/2}.$$

and we set $k_0 = \hat{k}$. The production computations use $N = 321$ equal-mass score blocks with $g_j = (j - 1/2)/N$ and $p_j = 1/N$. Firms within a block share its score-specific shock. The output multiplier is

$$Z(g, \omega) = 1 + \sigma_f (g - g_n) F(\omega) + \sigma_\varepsilon \varepsilon(g, \omega), \qquad \sigma_f = 0.05. \tag{87}$$

The common factor F takes values -1 and 1 with equal probability. Independently of F, $\varepsilon(g)$ is a stationary, symmetric two-state process along the score axis. Its initial sign is equally likely to be -1 or 1, and its sign flips at the arrival times of a Poisson process with rate $1/(2\ell)$. We fix $\ell = 0.15$, so

$$\mathbb{E}[\varepsilon(g)] = 0, \qquad \mathrm{Corr}(\varepsilon(g), \varepsilon(h)) = \exp(-|g - h|/\ell).$$

This specifies the full joint distribution needed for expected utility, rather than only its covariance matrix. Note that ℓ is not time (as in the typical Poisson process), but it describes dependence across firms with different sustainability scores. In particular,

$$\mathrm{Cov}(Z(g), Z(h)) = \sigma_f^2 (g - g_n)(h - g_n) + \sigma_\varepsilon^2 \exp(-|g - h|/\ell).$$

The exponential kernel is the correlation of ε, not of total output shocks, which also contain the common factor.

Baseline risk sets $\sigma_\varepsilon = 0.40$. Sensitivity cases set $\sigma_\varepsilon = 0.60$ or 0.80, keeping σ_f and ℓ fixed. Realized equity returns equal $R(g)Z(g, \omega)$; because $Z \geq 1 - 0.025 - \sigma_\varepsilon$, output is positive in every state, including the highest-risk case. The amplitudes are illustrative output-risk parameters, not empirically calibrated or annualized return volatilities.

Let $e_{ij} = p_j z_j^i f(k_j)$ denote expected dividend exposure. Then

$$\begin{aligned} \mathbb{E}[C_1^i] &= \sum_j e_{ij}, \\ \mathrm{Var}(C_1^i) &= \sigma_f^2 \left[\sum_j (g_j - g_n) e_{ij} \right]^2 + \sigma_\varepsilon^2 \sum_{j,k} e_{ij} e_{ik} \exp(-|g_j - g_k|/\ell). \end{aligned} \tag{88}$$

Diversifying across distant scores reduces exposure to the score-specific component. Increasing σ_ε strengthens that incentive; increasing the common factor alone primarily changes the incentive to hedge its loading.

For each specification, we report the changes with respect to a baseline all-Benchmark economy

with the same shocks and portfolio constraints. Write the allocation of the baseline economy as $\hat{k}^r(g)$ and its aggregates as $\hat{K}^r$ and $\hat{Y}^r$. Reported changes are $\Delta K = 100(K/\hat{K}^r - 1)$ and $\Delta\mathbb{E}[Y] = 100(\mathbb{E}[Y]/\hat{Y}^r - 1)$.

Table 7: Risk, saving, and the scale effect: $\nu = 3$

w_0	Economy	s^A (%)	ΔK (%)	$\Delta\mathbb{E}[Y]$ (%)
0.33	Certainty	55.02	5.07	1.56
0.33	$\sigma_\varepsilon = 0.40$	55.00	5.05	1.56
0.33	$\sigma_\varepsilon = 0.60$	54.60	4.63	1.42
0.33	$\sigma_\varepsilon = 0.80$	53.82	3.84	1.14
0.66	Certainty	56.52	13.53	3.10
0.66	$\sigma_\varepsilon = 0.40$	56.51	13.51	3.11
0.66	$\sigma_\varepsilon = 0.60$	56.39	13.25	3.04
0.66	$\sigma_\varepsilon = 0.80$	55.96	12.31	2.75
0.99	Certainty	57.00	22.38	1.90
0.99	$\sigma_\varepsilon = 0.40$	56.98	22.29	1.93
0.99	$\sigma_\varepsilon = 0.60$	56.95	22.12	1.94
0.99	$\sigma_\varepsilon = 0.80$	56.90	21.75	1.97

Notes: All cases use the same 321-point score grid and prohibit short sales. Risk cases hold $\sigma_f = 0.05$ and $\ell = 0.15$ fixed; certainty removes both shocks. Capital and expected output are compared with the all-Benchmark economy under the corresponding shocks and constraints. Benchmark saves 50% throughout.

G.5 Numerical results Table 7 and Figures 14–16 report the results. With baseline risk and $w_0 = 0.66$, Advocate saves 56.51% of initial wealth, capital exceeds its benchmark by 13.51%, and expected output exceeds its benchmark by 3.11%. The corresponding certainty values are 56.52%, 13.53%, and 3.10%. Baseline risk thus leaves the aggregate effects close to certainty while changing portfolios at some wealth levels.

Figure 14 shows Advocate's ownership share $z^A(g)$ and expected pecuniary returns over the full score range. Under baseline risk, both households hold 27 blocks at $w_0 = 0.33$, with scores from 0.634 to 0.715. Their total score mass is 8.41%. At $w_0 = 0.66$, only one block, at $g = 0.422$, is shared, with mass 0.31%. At $w_0 = 0.99$, 25 blocks are shared, from 0.020 to 0.095, with mass 7.79%. These are rounded grid locations, not estimates of exact continuous overlap endpoints.

Larger score-specific shocks produce substantially broader sharing. At $w_0 = 0.66$, the shared mass rises to 27.41% for $\sigma_\varepsilon = 0.60$ and 62.93% for $\sigma_\varepsilon = 0.80$. The corresponding shared scores run from 0.285 to 0.556 and from 0.120 to 0.746. Portfolio risk remains material despite diversification: at this wealth share, the consumption coefficients of variation are 24.93% for Advocate and 27.53% for Benchmark under baseline risk, and 45.14% and 47.47% in the 0.80 case.

Figure 15 reports $k(g)/\hat{k}^r(g)$ for the three wealth cases. The capital tilt toward higher scores

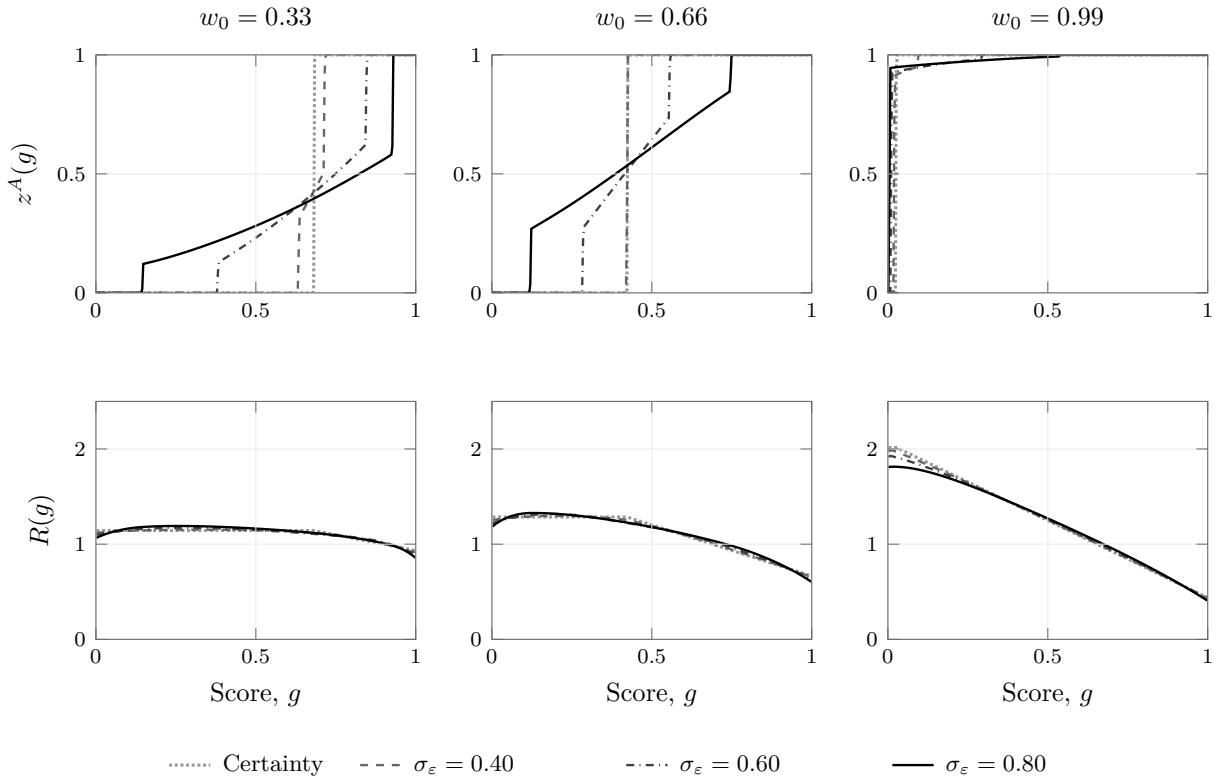


Figure 14: Risk, shared ownership, and expected returns

Notes: The upper row shows Advocate's share of each firm; Benchmark holds $1 - z^A(g)$. The lower row shows expected gross pecuniary returns. Each row uses common vertical limits across wealth cases. All panels cover the full score range. Short sales are prohibited.

remains visible under risk, although its shape reflects household-specific risk premia as well as tastes. The denominator is the benchmark allocation at the same score under the same risk specification.

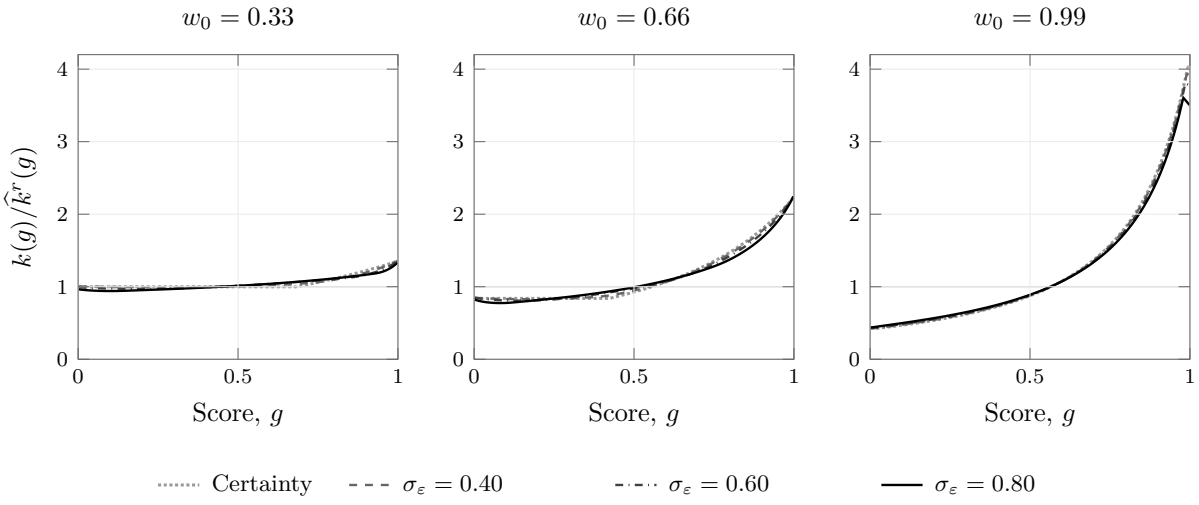


Figure 15: Capital allocation under certainty and correlated risk

Notes: Capital is divided by the all-Benchmark allocation at the same score under the corresponding risk specification. The horizontal reference line is one. Identical vertical limits, from 0 to 4.2, are used in all three panels. Short sales are prohibited.

Figure 16 varies Advocate's initial wealth share. Aggregate capital rises with w_0 over the computed range, while the expected-output response is hump-shaped: the scale gain must be weighed against the output cost of a more unequal capital allocation. Higher score-specific risk reduces Advocate's saving and the capital response at each of the three reported wealth shares. At $w_0 = 0.66$ and $\sigma_\varepsilon = 0.80$, Advocate saves 55.96%, capital rises by 12.31%, and expected output rises by 2.75%. Benchmark continues to save 50% in every no-short-sales case, as required by equation (86).

On the wealth grid, the expected-output gain peaks at 3.20% at $w_0 = 0.74$ under certainty, 3.20% at 0.74 under baseline risk, and 2.90% at 0.765 for $\sigma_\varepsilon = 0.80$. Higher risk need not lower the expected-output gain at every wealth share. At $w_0 = 0.99$, that gain rises slightly from 1.93% under baseline risk to 1.97% in the 0.80 case, even as the capital gain falls from 22.29% to 21.75%. The accompanying change in capital composition can offset part of the loss in scale.

Overall, the exercise shows that the introduction of risk in the economy with Advocate investors creates a portfolio diversification incentive that changes the stark portfolio allocation in the deterministic economy. This impacts the exact shape of both the return and capital composition schedule, but it maintains the differential returns, the favored capital composition towards higher

score firms, and the nonpecuniary saving incentive for Advocate.

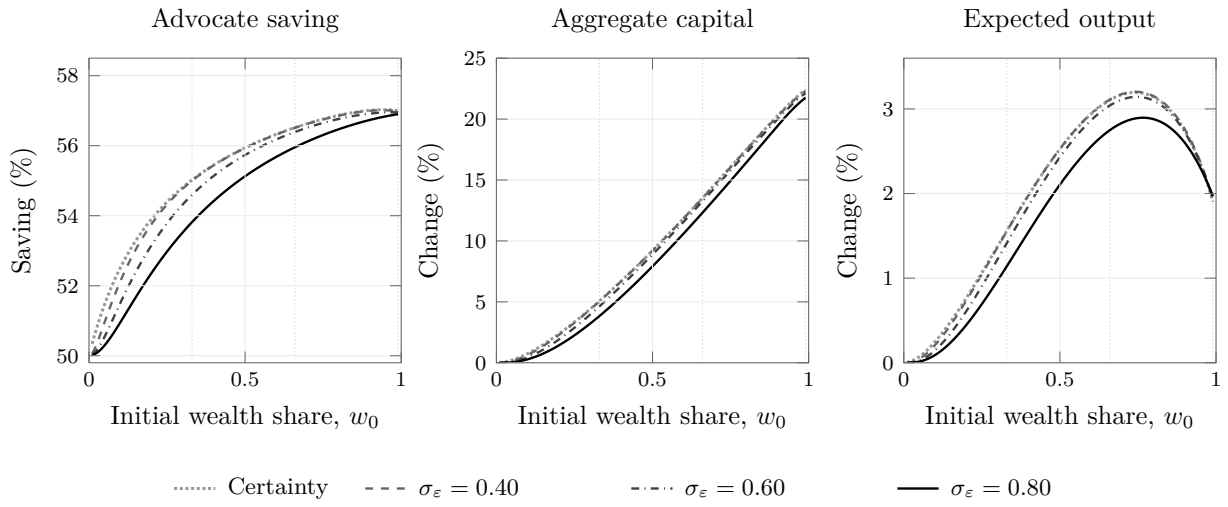


Figure 16: Risk and the aggregate effects of sustainable investing

Notes: The left panel reports Advocate's savings as a percentage of initial household wealth. The other panels show percentage changes relative to the corresponding all-Benchmark economy. Benchmark saves 50% in all cases. Vertical dotted lines mark $w_0 = 0.33, 0.66, 0.99$. Curves use 197 wealth points and prohibit short sales.

G.6 Climate risk exposure in an all-Benchmark economy We now consider an all-Benchmark economy where we allow high-score firms to have lower exposure to a common adverse climate shock. This exercise allows us to distinguish the capital reallocation induced by climate-risk pricing from the additional saving motive generated by nonpecuniary preferences when Advocate investors are present. We assume that the uncertainty term in output takes the form

$$Z^c(g,\omega) = 1 - \sigma_c(1-g)X(\omega) + 0.40\,\varepsilon(g,\omega), \qquad \sigma_c \in \{0, 0.05, 0.20, 0.40\}. \tag{89}$$

The term X is a "climate factor" that takes values -1 and 1 with equal probability, independently of the same score-specific process $\varepsilon(g)$ with $\ell = 0.15$. The climate state $X = 1$ is adverse, with larger output losses at lower scores, which is captured by the term $1 - g$. Expected output remains $k(g)^\alpha$ at every score, and $Z^c \geq 1 - \sigma_c - 0.40 \geq 0.20$. The shock amplitudes are illustrative, rather than calibrated to a particular climate scenario.

A representative Benchmark investor owns every firm, so $z^B(g) = 1$, initial consumption is $C_0 = Y_0 - K$, and terminal consumption C_1 equals aggregate output. Let M be the stochastic discount factor, then $D^c(g) = \mathbb{E}[MZ^c(g)]$ and $R^c(g) = 1/D^c(g)$, so firms that are relatively resilient when consumption is scarce command higher prices and lower required expected returns, which in

turn translate into higher productive capital investment. Note that score-specific shocks also enter valuation, so the total return schedule need not be monotone in g.

The analytical log case. To isolate the key mechanisms, let utility be logarithmic so that, $M = \beta C_0/C_1$. Summing equity values across firms gives $Q = \mathbb{E}[MC_1] = \beta C_0$. Hence

$$s^B = \frac{Q}{C_0 + Q} = \frac{\beta}{1+\beta}, \qquad K = \frac{\alpha\beta}{1+\alpha\beta} Y_0, \tag{90}$$

where we have used $Q = K/\alpha$. Climate risk can therefore change capital composition and expected returns without changing aggregate saving or capital. With our parameters, $s^B = 50\%$ and $K = 0.157$. By contrast, equation (86) retains the additional term $G\mathcal{V}'(G)$ for Advocate under the same consumption preferences, so Advocate saves more when $G > 0$.

This saving invariance does not imply the absence of precautionary demand for safe payoffs. Define the gross return on a hypothetical risk-free claim as $R_f = 1/\mathbb{E}[M]$ and let $\mu = \mathbb{E}[C_1]$. Under log utility,

$$R_f = \frac{\mu}{\beta C_0 \mathcal{P}_{\log}}, \qquad \mathcal{P}_{\log} = \mu\, \mathbb{E}\left[\frac{1}{C_1}\right] \geq 1. \tag{91}$$

For fixed C_0 and μ, a mean-preserving increase in consumption risk raises $\mathcal{P}_{\log}$ and lowers R_f. The inequality follows from convexity of $1/C_1$ and is strict under nondegenerate risk. Thus precautionary demand for safe payoffs can strengthen even while the equilibrium saving rate remains constant. Note that no traded risk-free asset is required for this pricing comparison, as R_f represents a shadow return which is well defined even in the absence of a safe saving technology.

In our numerical illustration, we consider Epstein– Zin preferences which allow us to separate risk aversion from intertemporal elasticity of substitution, but it should be clear from the results above that the introduction of differential exposure to climate risk is not per se a driver of changes in the saving rate in the economy.

Numerical Illustration In our numerical illustration, we use Epstein–Zin preferences (Epstein and Zin, 1991). Let γ denote relative risk aversion and ψ the elasticity of intertemporal substitution, and define period 1 consumption's certainty equivalent by

$$\mathcal{C}_1 = \begin{cases} \left(\mathbb{E}[C_1^{1-\gamma}]\right)^{1/(1-\gamma)}, & \gamma \neq 1, \\ \exp\{\mathbb{E}[\log C_1]\}, & \gamma = 1. \end{cases} \tag{92}$$

The intertemporal aggregator is

$$U_0 = \left[\frac{C_0^\rho + \beta \mathcal{C}_1^\rho}{1+\beta}\right]^{1/\rho}, \qquad \rho = 1 - \frac{1}{\psi}. \tag{93}$$

The continuous limit at $\psi = 1$, together with $\gamma = 1$, is ordinally equivalent to $\log C_0 + \beta \mathbb{E}[\log C_1]$. We retain the equal date weights $\beta = 1$ and the same k_0 rather than renormalizing initial resources when preferences change. Here β is a relative weight; the normalized weight on future consumption is $\beta/(1+\beta) = 1/2$. The numerical illustration sets $\psi = 1.5$ and $\gamma = 3$. This matches the EIS and approximates the risk aversion of 2.98 in the calibration of Hambel and van der Ploeg (2025).

The stochastic discount factor and aggregate equity value are

$$M = \beta C_0^{1/\psi} \mathcal{C}_1^{\gamma - 1/\psi} C_1^{-\gamma}, \qquad Q = \mathbb{E}[MC_1] = \beta C_0^{1/\psi} \mathcal{C}_1^{1-1/\psi}. \tag{94}$$

Let $\widetilde{R}_p = C_1/Q$ be the realized gross return on the market portfolio and $\mathcal{R}_p = \mathcal{C}_1/Q$ its certainty-equivalent return.

To distinguish precautionary demand for safe assets from changes in expected consumption growth, write $H = C_1/\mu$, with $\mathbb{E}[H] = 1$, and $a = \mathcal{C}_1/\mu$. Then

$$\mathbb{E}[M] = \beta \left(\frac{C_0}{\mu}\right)^{1/\psi} \mathcal{P}, \qquad \mathcal{P} = a^{\gamma - 1/\psi} \mathbb{E}[H^{-\gamma}], \qquad R_f = \frac{(\mu/C_0)^{1/\psi}}{\beta \mathcal{P}}. \tag{95}$$

At given C_0 and μ, $\mathcal{P}$ measures the price of a sure payoff relative to its value under certainty and so it isolates the precautionary component of the pricing of safe assets.

Table 8 reports the equilibrium numerical simulations. Increasing σ_c from zero to 0.40 lowers the certainty-equivalent portfolio return from 1.054 to 0.986, while the expected gross return rises from 1.128 to 1.144. Saving falls from 50.66% to 49.83%, a decline of 0.83 percentage points, while aggregate capital falls by 2.46% and expected output by 1.04%. At the same time, $\mathcal{P}$ rises from 1.123 to 1.315, and the implied risk-free gross return falls from 0.981 to 0.828. The precautionary component itself therefore strengthens, rather than the interest-rate decline merely reflecting lower expected consumption growth. Stronger precautionary demand coexists with lower net equilibrium saving because the risk-adjusted reward to investing declines and the substitution effect dominates the income effect under the chosen parameterization.

Figure 17 shows that the differential exposure to climate risk generates a capital composition effect as investment shifts toward firms with lower climate exposure and expected returns reflect

Table 8: Climate risk, precautionary demand, and equilibrium saving

Panel A. Saving and aggregate investment

σ_c	s^B (%)	Δs^B (pp)	ΔK (%)	$\Delta\mathbb{E}[Y]$ (%)
0.00	50.66	0.00	0.00	0.00
0.05	50.65	-0.01	-0.04	-0.01
0.20	50.45	-0.21	-0.62	-0.22
0.40	49.83	-0.83	-2.46	-1.04

Panel B. Portfolio returns and safe-claim valuation

σ_c	$\mathbb{E}[\widetilde{R}_p]$	$\mathcal{R}_p$	R_f	$\mathcal{P}$
0.00	1.128	1.054	0.981	1.123
0.05	1.128	1.053	0.979	1.126
0.20	1.132	1.037	0.939	1.171
0.40	1.144	0.986	0.828	1.315

Notes: All investors are Benchmark, with EIS $\psi = 1.5$, risk aversion $\gamma = 3$, and $\beta = 1$. Score-specific risk remains fixed at $\sigma_\varepsilon = 0.40$ and $\ell = 0.15$ on $N = 321$ blocks. Changes compare with $\sigma_c = 0$ at the same preferences; this control has $K = 0.160$ and $\mathbb{E}[Y] = 0.546$. Saving is a share of initial household wealth. $\mathcal{R}_p$ is the certainty-equivalent gross portfolio return; $R_f = 1/\mathbb{E}[M]$ is the implied gross return on a hypothetical safe claim. $\mathcal{P}$ isolates uncertainty's effect on the safe-claim price at matched current and mean terminal consumption, as in equation (95). Returns are for the model period and are not annualized. Short sales are prohibited and capital ceilings are slack.

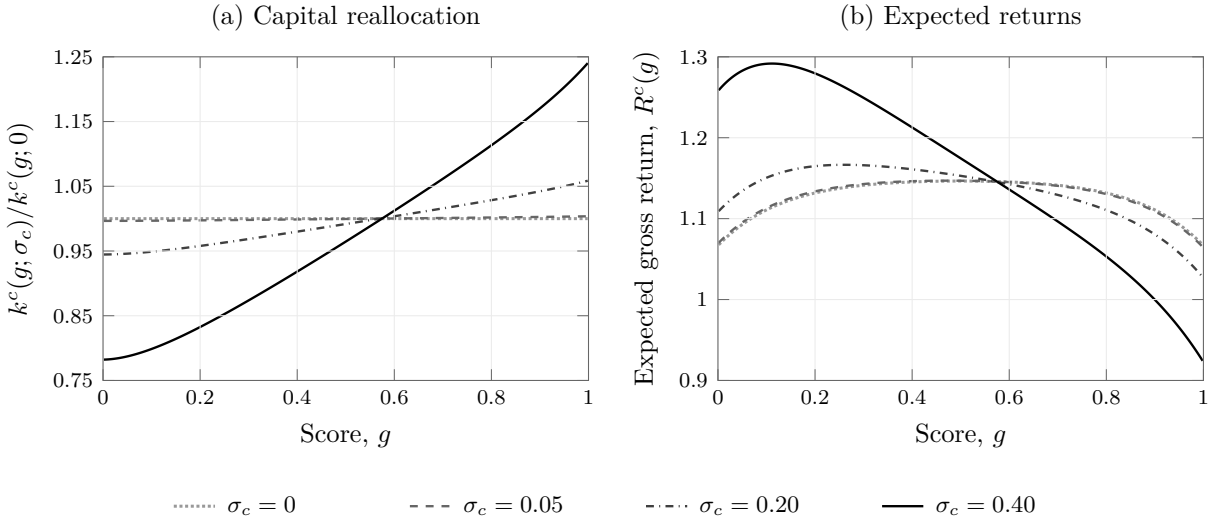


Figure 17: Climate-risk pricing and capital reallocation: EIS 1.5, risk aversion 3

Notes: All investors are Benchmark. The climate loading is $-\sigma_c(1-g)$, while correlated score-specific shocks retain $\sigma_\varepsilon = 0.40$ and $\ell = 0.15$. Capital is normalized by the allocation at $\sigma_c = 0$ with the same preferences and score-specific risk. Expected returns are gross model-period returns. Each outcome uses one common vertical range for all climate amplitudes.

differences in risk. For $\sigma_c = 0.40$, capital in the highest-score quarter changes by 15.93% relative to the no-climate-risk control, whereas capital in the lowest-score quarter changes by −19.09%. This reallocation occurs despite the decline in aggregate capital.

The exercise illustrates that introducing a risk exposure that varies depending on the characteristic g in an all-Benchmark economy can generate an equilibrium capital composition similar to the one emerging in a deterministic model with Advocate investors. However, the same economy does not necessarily generate the changes in saving rates that characterize Advocate households and that are crucial in shaping the differential wealth dynamics and the wealth-ownership feedback.

H Productivity Heterogeneity and Representative Firms

The common-technology specification in Section 2 admits an interpretation in which firms differ in productivity within each sustainability characteristic. Under the assumption that the conditional productivity distribution is the same across characteristics g, a common production function represents each group, subject to the allocation and feasibility conditions below. Equal capital across Benchmark-priced characteristics then refers to average capital within each group; it does not require identical capital stocks across the underlying firms.

For each $g \in [0, 1]$, let a continuum of firms produce $Af(k(g, A))$. Productivity is fixed over time and has conditional probability distribution μ, independent of g. This assumption requires the same conditional distribution, not merely zero correlation between productivity and the sustainability characteristic. With characteristic density $\phi(g)$, total firm mass is normalized to one and integration over firms uses $\phi(g)\, dg\, d\mu(A)$. No uniformity restriction is imposed on ϕ. Define

$$\bar{k}(g) = \int k(g, A)\, d\mu(A), \qquad \bar{y}(g) = \int Af(k(g, A))\, d\mu(A).$$

We maintain the neoclassical production assumptions and consider feasible interior firm allocations.[20]

H.1 Aggregation Within a Characteristic

The sustainability loading depends on g alone. The portfolio conditions therefore imply a common pecuniary return $R(g)$ across productivity types at a given characteristic: Advocate evaluates each firm using $R(g, A) + (m/\beta)v(g)$ in two periods

[20] For the general aggregation argument, a productivity distribution supported on a compact subset of $(0, \infty)$ is sufficient for the required integrability and regularity. The Cobb–Douglas formulas below require the stated productivity moment to be finite.

and $R_t(g, A) + m_t v(g)$ in continuous time, while Benchmark values only the pecuniary return. Under no short sales and coexistence, all productivity types above the ownership cutoff are held by Advocate and those below it by Benchmark, apart from the measure-zero cutoff. The interior firm condition is consequently

$$Af'(k(g, A)) = P(g), \qquad P(g) = R(g) \text{ or } \delta + R_t(g), \tag{96}$$

for the two-period and continuous-time models, respectively. Time subscripts on capital are suppressed when the argument applies to both.

Define the representative production function by

$$F(\kappa) = \max_{\substack{x(A)\geq 0 \\ \int x(A)\, d\mu(A) = \kappa}} \int Af(x(A))\, d\mu(A). \tag{97}$$

The first-order condition is $Af'(x(A)) = \lambda$, and the envelope theorem gives $F'(\kappa) = \lambda$. Equation (96) implements this allocation within each group, so

$$\bar{y}(g) = F(\bar{k}(g)), \qquad F'(\bar{k}(g)) = P(g). \tag{98}$$

Because μ is common across characteristics, so is F. The main text's firm conditions are recovered by interpreting its production function as F and its capital variable as $\bar{k}$. In general, $F(\kappa)$ differs from $\mathbb{E}[A]f(\kappa)$ because more productive firms receive more capital.

H.2 Financial Wealth and Ownership Let $\bar{q}(g) = \int q(g, A)\, d\mu(A)$. In two periods, common returns within a group imply $\bar{q}(g) = F(\bar{k}(g))/R(g)$. In continuous time, aggregation of dividends and the return identity gives

$$\bar{d}_t(g) = F(\bar{k}_t(g)) - \dot{\bar{k}}_t(g) - \delta \bar{k}_t(g), \qquad R_t(g) = \frac{\bar{d}_t(g) + \dot{\bar{q}}_t(g)}{\bar{q}_t(g)}. \tag{99}$$

Under cutoff ownership, Advocate wealth and sustainable wealth are therefore

$$a_t^A = \int_{g_t^*}^1 \bar{q}_t(g)\phi(g)\, dg, \qquad G_t = \int_{g_t^*}^1 v(g)\bar{q}_t(g)\phi(g)\, dg, \tag{100}$$

with the corresponding two-period definitions obtained by dropping time subscripts and using a_1^A. Thus the portfolio and saving conditions, relative-wealth equation, and ownership-boundary

equation retain their form. Within-group heterogeneity is reflected in the representative technology and asset values.

Since F is strictly concave, constant returns below the cutoff imply equal average capital across those groups, while declining returns above it imply increasing average capital. The hockey-stick allocation thus survives at the group level. Under the maintained conditions for an interior stationary equilibrium with equal patience and both households active, the stationary Euler equations still select $g^*_{ss} = g_n$. Comparing with an all-Benchmark steady state using the same F, depreciation rate, and discount rate gives $\bar{k}_{ss}(g) = \widehat{k}$ below g_n and $\bar{k}_{ss}(g) > \widehat{k}$ above it wherever the return wedge is strictly positive. Aggregate capital and output are therefore higher when the favored region has positive mass. These stationary comparisons do not establish convergence to the steady state or determine the signs of transitional effects.

H.3 Cobb–Douglas Production For $f(k) = k^\alpha$, $0 < \alpha < 1$, let $\eta = 1/(1-\alpha)$ and $M = \int A^\eta \, d\mu(A) < \infty$. The micro and group capital allocations satisfy

$$k(g, A) = \left(\frac{\alpha A}{P(g)}\right)^\eta, \qquad \bar{k}(g) = M\left(\frac{\alpha}{P(g)}\right)^\eta, \qquad k(g, A) = \frac{A^\eta}{M}\bar{k}(g). \tag{101}$$

Substitution into the output yields

$$F(\bar{k}) = A_{\text{eff}}\bar{k}^\alpha, \qquad A_{\text{eff}} = M^{1-\alpha} = \left(\int A^{1/(1-\alpha)} \, d\mu(A)\right)^{1-\alpha}. \tag{102}$$

The representative technology therefore has the same capital exponent and a common productivity coefficient. Writing it as $\bar{k}^\alpha$ corresponds to normalizing $A_{\text{eff}} = 1$, or equivalently $M = 1$. The normalization $\mathbb{E}[A] = 1$ alone is insufficient: with nondegenerate productivity dispersion, it implies $A_{\text{eff}} > 1$. Conditional on matching this coefficient and initial resources, productivity heterogeneity does not introduce an additional characteristic-dependent force into the aggregate equilibrium.

H.4 Initial Conditions and Scope The representation requires initial resources and firm-level feasibility to be matched. An efficient initial allocation within each group gives $\bar{y}_0(g) = F(\bar{k}_0(g))$; under Cobb–Douglas it takes the form $k_0(g, A) = (A^\eta/M)\bar{k}_0(g)$. By contrast, equal initial capital k_0 across heterogeneous firms gives output $\mathbb{E}[A]f(k_0)$, generally below $F(k_0)$. In the two-period model, this inherited output can instead be treated as the initial resource endowment, with F describing subsequent production. The initial capital stock of the representative firm must not then be identified mechanically with the average inherited stock.

The reduction applies to the feasible interior allocations and paths characterized in Sections 2 and 3. If financing constraints bind for some productivity types, the within-group allocation and distribution of capital can matter beyond their mean. Under Cobb–Douglas, fixed proportional capital allocations imply proportional output and dividends. Specifically, writing $h(A) = A^{\eta}/M$ gives $Af(h(A)\bar{k}) = h(A)F(\bar{k})$; with fixed productivity, investment and depreciation scale by the same factor. Firm-level dividend feasibility can therefore be checked directly against group dividends along these proportional allocations. For general technology, a proposed representative path must also admit a feasible disaggregation. Subject to these conditions, the common-technology and equal-capital results can be interpreted as statements about representative groups while allowing substantial productivity and capital dispersion among their constituent firms.

Appendix III: Computational Details

I Details on Quantitative Analysis

I.1 Calibration and the Saving Incentive The maintained parameters are reported in Table 1. For the ESG baseline, $v(g) = \nu(g - g_n)/(1 - g_n)$ with $g_n = .5$. We jointly choose ν and κ to match the initial return differential $R_0^* - R_{w,0}^A$ and the marginal-utility ratio. With $\sigma_G = .5$, the baseline curvature condition is

$$\left(\frac{1 + 2\kappa G_0}{1 + \kappa G_0}\right)^{-1/2} = .90 \quad \Longleftrightarrow \quad \kappa G_0 = \frac{19}{62}. \tag{103}$$

Thus the calibration fixes the product κG_0, not a constant elasticity of marginal sustainable-wealth utility. Table 9 records the six baseline calibrations. Parameters are expressed in the model's normalized consumption and capital units.

Table 9: Equilibrium calibration and initial marginal saving incentives

Target	Distribution	ν	κ	g_0^*	G_0	$m_0 v(g_0^*)$ (bp)
10	Uniform	0.004975	8.1963	0.6601	0.0374	8.51
10	Empirical	0.013835	3.5356	0.7241	0.0867	31.61
20	Uniform	0.010328	3.8404	0.6664	0.0798	17.86
20	Empirical	0.030778	1.5101	0.7279	0.2029	64.99
40	Uniform	0.022870	1.6218	0.6831	0.1890	40.54
40	Empirical	0.076103	0.4609	0.7608	0.6649	123.59

Notes: Targets are in basis points per year. All rows use CES consumption utility with $\sigma = .5$. The net effective saving return $m_0 v(g_0^*)$ is an outcome, not an additional calibration target. Empirical cases use one-point ESG bins. The two calibrating parameters are reported in the model's normalized consumption and capital units.

To distinguish the return target from the net effective saving return, write $v(g) = \nu \bar{v}(g)$. Under sharp ownership,

$$R_0^* - R_{w,0}^A = m_0 \nu \frac{(1 + \bar{s}) \int_{g_0^*}^1 [\bar{v}(g) - \bar{v}(g_0^*)] q_0(g) \phi(g)\, dg}{a_0^A}. \tag{104}$$

This expression averages preference differences relative to the marginal firm, whereas $m_0 v(g_0^*)$ depends on its preference weight. Matching the return target therefore need not give the same saving incentive across distributions.

Panel C of Table 2 holds $\nu = .0308$ and $\kappa = 1.510$ at the empirical 20 bp calibration and changes g_n to .33 and .66. These cases do not reimpose the initial return differential. Panel D instead maintains $g_n = .50$ and recalibrates both parameters to the 20 bp target and marginal-

utility ratios of .75 and .98. For a ratio ζ, the curvature equation becomes $\kappa G_0 = (1-\zeta^2)/(2\zeta^2-1)$.

I.2 Data Construction and Interpretation The LSEG data was obtained in September 2026 for US-headquartered public companies, including active and inactive instruments. The analysis uses annual ESG and capital observations for the calendar year 2023, with exactly matching dates. Missing ESG scores, nonpositive capital, and ambiguous multiple annual observations are excluded. Capital is net PPE, field `TR.F.PPENetTot`, expressed in million US dollars.

We exclude fund vehicles and duplicate instrument representations, then apply documented consolidation decisions. When a sampled parent has usable consolidated PPE and an ESG score on the same annual date, its reported capital and score are retained and overlapping subsidiaries are removed. Subsidiary capital is not added to the parent, and scores are not averaged. The consolidation removes 21 overlapping subsidiary rows from the eligible capital denominator, 20 of which had matched scores. The resulting sample has 2,871 matched issuers with \$8.263 trillion in PPE, representing 96.24% of recorded capital among the 4,518 eligible positive-capital issuers in the extract. The replication files report the identifier repairs, inclusion rules, and consolidation audit.

For issuer i, let P_i denote PPE and g_i its ESG score divided by 100. Define $\omega_i = P_i / \sum_j P_j$. In a bin B_ℓ of width h_ℓ in normalized-score units,

$$\phi(g) = \frac{\sum_{i:g_i \in B_\ell} \omega_i}{h_\ell}, \qquad g \in B_\ell.$$

The baseline uses one-score-point bins, so $h_\ell = .01$. Bins are left-closed and right-open, with score 100 included in the final bin. No tail mass is imputed. The histogram integrates to one and assigns 81.86% of initial capital to scores at or above 50. Because initial capital per model unit is common, PPE heterogeneity enters through ϕ once; it is not also assigned to $k_0(g)$. The uniform distribution is $\phi(g) = 1$, not the equally weighted distribution of observed issuers.

These weights describe worldwide consolidated book assets of the screened companies, not capital geographically located in the United States or the complete 2023 investable universe. PPE is an accounting proxy whose coverage varies across industries. Consolidation is targeted rather than a complete historical reconstruction of ultimate ownership. The replication files document these measurement qualifications and the associated sample and bin-width checks.

I.3 Solution Method and Accuracy We solve a boundary-value problem for nodal k, $\mathcal{Q}$, and q, together with c^A and g^*. Benchmark consumption follows from goods clearing. Let $x = (\mathcal{Q}-1)/\psi$, $D = I[d]$, $W(g) = m[v(g) - v(g^*)]_+$, and

$$d_k = f'(k) - \delta - x - \frac{\psi x^2}{2}, \qquad d_{\mathcal{Q}} = -\frac{k\mathcal{Q}}{\psi}, \qquad B_{\mathcal{Q}} = -W\mathcal{Q} + \delta - f'(k) - \frac{\psi x^2}{2}.$$

Differentiating goods clearing and using both Euler equations eliminates the common return:

$$R^* = \frac{\sigma I[d_k \dot{k} + d_{\mathcal{Q}} B_{\mathcal{Q}}] - c^A m v(g^*) + \rho D}{D + (\sigma/\psi) I[k\mathcal{Q}^2]}. \tag{105}$$

The baseline calculations use 81 spatial nodes, an adaptive time mesh, a 12,000-year terminal horizon, and a 10^{-5} RMS collocation tolerance. The horizon approximates the infinite-horizon boundary. For the empirical ESG distribution, the highest solved score is .94, the upper edge of the highest occupied bin; the normalization $v(1) = \nu$ is retained.

Ownership is regularized using the logistic function Λ:

$$z_\epsilon(g) = -\bar{s} + (1 + 2\bar{s})\Lambda\left(\frac{g - g^*}{\epsilon}\right), \quad \epsilon = .0045, \qquad \dot{g}^* = \frac{I[z_\epsilon d] - c^A}{I[(\partial z_\epsilon/\partial g^*) q]}. \tag{106}$$

This equation preserves the discretized household budget and accommodates empty histogram bins. Pricing retains the sharp positive part in equation (42); finite ϵ is therefore an approximation to the sharp ownership equilibrium.

We integrate piecewise-linear model basis functions against the histogram density, splitting at both density-bin edges and model nodes. This preserves empirical bin masses. There are $3n + 2$ state functions. Initial conditions fix n capital stocks and Advocate's wealth share. Terminal conditions impose $\mathcal{Q} = 1$ and $Rq = d$ at every node, together with $c^A = I[zd]$. When preferences are calibrated, the two calibration moments provide two additional boundary conditions for ν and κ. Parameter and distribution continuation supply initial guesses; difficult cases use $\log q$ as the equity state. Stationary allocations are also solved independently.

For the six baseline ESG paths, independent checks at saved knots, interval midpoints, and a dense early-time grid give maximum normalized pointwise differential residuals between 1.35×10^{-5} and 2.31×10^{-5}, with boundary residuals below 1.7×10^{-12}. This pointwise diagnostic differs from the solver's RMS collocation criterion. The implemented return targets match the stated 10, 20, and 40 bp values. Grid and ownership-width checks are retained in the replication files. In the

empirical CES 40 bp case, moving to 161 nodes and halving ϵ changes the wealth path by at most .0274 percentage points and the capital-growth path by .0150 percentage points; the initial return differential changes by about 1.22 bp at fixed preferences. These checks distinguish accuracy of the numerical solution from sensitivity to the ownership approximation.

I.4 Aggregate and Fixed-Group Measures Aggregate capital and output are $K_t = I[k_t]$ and $Y_t = I[k_t^\alpha]$. Define the fixed highest-scoring group by its lower boundary g_{25}, satisfying

$$\frac{\int_{g_{25}}^1 k_0(g)\phi(g)\,dg}{K_0} = .25.$$

Since $k_0(g) = \hat{k}$, this is equivalent to $\int_{g_{25}}^1 \phi(g)\,dg = .25$. The boundary is .75 under the uniform distribution, approximately .76223 under the empirical ESG distribution, and .75 under the emissions rank. Define

$$K_t^{25} = \int_{g_{25}}^1 k_t(g)\phi(g)\,dg, \qquad Y_t^{25} = \int_{g_{25}}^1 k_t(g)^\alpha \phi(g)\,dg.$$

The reported growth rates are $\Delta X_t = 100(X_t/X_0 - 1)$ for $X \in \{K, Y, K^{25}, Y^{25}\}$. Each group measure uses its own initial level. Membership is fixed over time and across the neutral-score and curvature comparisons; these measures are not changes in the group's share of current capital or output. Numerical integration restricts the same basis weights to this group, including any partially intersected bin.

For an outcome X_t with a nonzero stationary change, define

$$T_p^X = \inf\left\{t \geq 0 : \frac{X_t - X_0}{X_{ss} - X_0} \geq p\right\}, \qquad p \in \{.5, .9\}. \tag{107}$$

We evaluate the saved splines annually and interpolate the first crossing. Wealth-share adjustment times refer to the wealth share's own long-run change; they are not adjustment times for capital composition.

Figure 4 uses the cumulative initial-capital coordinate $F_K(g) = \int_0^g \phi(x)\,dx$ for the capital surface. Its display smoothing does not enter the aggregate calculations. For Figure 5, equity value in the fixed group is $\int_{g_{25}}^1 q_t(g)\phi(g)\,dg$. Equity and capital adjustment are measured relative to their initial all-Benchmark levels, while wealth-share adjustment starts from $w_0^A = .33$. Each series is divided by its own long-run change. The equity series includes the on-impact price adjustment and is not a total-return index.

I.5 Wealth-Share Decomposition

With signed portfolios, the exact wealth-share identity is

$$\dot{w}_t^A = w_t^A(1 - w_t^A)\left[R_{w,t}^A - R_{w,t}^B + \frac{c_t^B}{a_t^B} - \frac{c_t^A}{a_t^A}\right]. \tag{108}$$

The portfolio-return and saving contributions shown in Figure 5 can equivalently be calculated as

$$C_{R,t} = \frac{I[z_t q_t R_t] - w_t^A I[q_t R_t]}{I[q_t]}, \qquad C_{S,t} = \frac{w_t^A I[d_t] - c_t^A}{I[q_t]}, \qquad \dot{w}_t^A = C_{R,t} + C_{S,t}. \tag{109}$$

Multiplication by 100 expresses these contributions in percentage points per year. They include capital gains through $R_t(g) = [d_t(g) + \dot{q}_t(g)]/q_t(g)$ and do not divide by individual net wealth. Benchmark's portfolio return generally differs from its marginal return because of short positions.

I.6 Emissions Measurement, Ranking, and Calibration

Let $M_i = M_i^{(1)} + M_i^{(2)}$ be Scope 1 plus Scope 2 emissions, Y_i^{obs} revenue in million US dollars, and P_i PPE. On the retained sample $\mathcal{I}$, define

$$e_i = \frac{M_i}{Y_i^{\text{obs}}}, \qquad \omega_i = \frac{P_i}{\sum_{j\in\mathcal{I}} P_j}, \qquad H_K(e) = \sum_{i\in\mathcal{I}} \omega_i \mathbf{1}\{e_i \le e\}. \tag{110}$$

Intensity is measured in tonnes of CO_2 equivalent per million US dollars of revenue. The common sample contains 1,170 issuers and 87.87% of the original ESG sample's recorded capital. It requires positive revenue, nonnegative observations for both scopes, and verified fiscal-year 2023 metadata for revenue and emissions. Missing scopes are not replaced with zero or estimated totals. RMR is excluded because of an unresolved reporting-boundary concern. Seven retained issuers have a difference exceeding one percent between the reported combined total and the separate-scope sum; the Scope 1 plus Scope 2 measure consistently uses the latter. The replication files preserve these flags and the source observations.

Section 5 uses Scope 1 on the same sample and with the same renormalized PPE weights. Scope 1 plus Scope 2 is an attributed footprint: emissions recorded at energy producers may also enter purchasers' Scope 2. It is therefore not an unduplicated inventory of physical emissions. The downloaded Scope 2 accounting basis has not been independently reconciled.

The model index $g \in [0, 1]$ is the initial-capital-weighted rank from higher to lower intensity. The exact empirical representation is $e(g) = H_K^{-1}(1 - g)$. Equal rank intervals contain equal initial capital, so $\phi(g) = 1$; this change of coordinate does not make intensity uniform. Tied intensity observations are grouped and assigned their combined capital mass.

For the numerical approximation, order the grouped intensities e_j from highest to lowest, with capital masses μ_j, and place them at midpoint ranks $x_j = \sum_{\ell \le j} \mu_\ell - \mu_j/2$. Add endpoints at zero and one carrying the highest and lowest intensities. Let $p(g)$ be the shape-preserving piecewise cubic interpolant through these points. We use

$$e(g) = a_e p(g), \qquad a_e = \frac{\sum_j \mu_j e_j}{\int_0^1 p(g)\, dg}.$$

This procedure preserves monotonicity and the capital-weighted mean, while approximating the discrete distribution. The means are 470.1751 for Scope 1 plus Scope 2 and 427.6644 for Scope 1; the rescaling factors are approximately .99898 and 1.00134, respectively. Ranks and weights remain fixed as capital evolves.

The preference function is equation (30). At $g_n = .50$, the two baseline moments are

$$R_0^* - R_{w,0}^A = .002, \qquad \frac{\mathcal{V}'(2G_0)}{\mathcal{V}'(G_0)} = .90, \tag{111}$$

which give $\nu = .0459$ and $\kappa = .421$. Both parameters are held fixed at $g_n = .33$ and .66; neither calibration moment is reimposed in those cases. The net effective saving return remains $m_t v(g_t^*)$, with

$$\frac{\dot{c}_t^A}{c_t^A} - \frac{\dot{c}_t^B}{c_t^B} = \frac{m_t v(g_t^*)}{\sigma}. \tag{112}$$

Total emissions and emissions intensity are

$$E_t = \int_0^1 e(g) k_t(g)^\alpha \, dg, \qquad \mathcal{E}_t = E_t / Y_t. \tag{113}$$

We integrate the cubic intensity curve against piecewise-linear nodal output, splitting at every intensity-curve knot and model node. Gaussian quadrature on these intervals integrates the resulting polynomial exactly. Firm intensities are fixed, so adjustment occurs through production. This accounting does not model the energy supply chain, innovation, or within-firm abatement. Percentage changes in emissions intensity are computed from levels:

$$100 \left(\frac{\mathcal{E}_t}{\mathcal{E}_0} - 1 \right) = 100 \left[\frac{1 + \Delta E_t/100}{1 + \Delta Y_t/100} - 1 \right].$$

Subtracting the output growth rate from the emissions growth rate is only an approximation.

The emissions transitions use initial-condition homotopy. For each neutral rank, we first solve

the stationary equilibrium and then gradually move initial capital and Advocate wealth toward $k_0(g) = \hat{k}$ and $w_0^A = .33$. The calculation retains the original 81-rank quadrature, $\epsilon = .0045$, and 12,000-year horizon. Identical state paths are represented once during the solve, then expanded to the full 245-state system for validation. Checks cover both household budgets and Euler equations, capital accumulation, installed-capital and equity pricing, goods and wealth clearing, the initial and terminal conditions, and financial bounds. Across the three saved paths, maximum absolute and normalized differential residuals are below 7.4×10^{-6} and 6.6×10^{-6}, respectively, and boundary residuals are below 1.6×10^{-15}. These checks validate the implemented regularized system, not an unregularized ownership problem or an empirical estimate of transition speed.

J Fixed Wealth Share: Pooled-Family Counterfactual

J.1 Fixed Wealth Shares and Saving Incentives

Write $I[x] = \int x(g)\phi(g)\,dg$, $A = I[q]$, and $a^A = I[qz]$. The relevant investment capacity is the market value of the *net* equity portfolio, $w^A = a^A/A$. It is neither installed capital, gross long holdings, sustainable wealth G, nor accumulated dividends. As in the full economy, the initial share is evaluated at on-impact equilibrium prices.

Suppose transfers $T_A = -T_B$ enter the original financial budgets,

$$\dot{a}^A = I[Rqz] - c^A + T_A, \qquad \dot{a}^B = I[Rq(1-z)] - c^B - T_A.$$

Keeping $w^A = \bar{w}$ requires

$$T_A = \bar{w}\{I[Rq] - C\} - I[Rqz] + c^A, \qquad C = c^A + c^B. \tag{114}$$

This flow maintains the wealth share at current prices, but a complete transfer rule must also specify its response to individual deviations and hence its effect on saving incentives.

If each atomistic household takes its transfer flow as exogenous, the two original Euler equations survive. An interior bounded stationary allocation then requires $R^* = \rho$ and $mv(g^*) = 0$, hence $g^* = g_n$. All firms below g_n have the Benchmark equity value; all above have weakly larger stationary equity values. Let $P_+ = \int_{g_n}^1 \phi(g)dg$. With the maintained short bound $\bar{s} = .05$, this implies the necessary bound

$$w_{ss}^A \geq (1 + 2\bar{s})P_+ - \bar{s}. \tag{115}$$

The bound is 50% for uniform ESG weights and 85.04% for empirical ESG weights. Thus the desired 33% stationary restriction is incompatible with an interior stationary equilibrium retaining both undistorted Euler equations. This argument does not rule out every nonstationary or boundary allocation; it rules out the intended interior stationary comparison. When individual households internalize a compensating transfer rule, its derivatives instead change the saving problem. We therefore do not present equation (114) as a nondistortionary implementation of the proposed experiment.

J.2 Family Problem We use an explicit alternative with pooled resources. A price-taking family has total financial wealth a and two portfolio accounts $a^A(g)$ and $a^B(g)$, denominated in current market value. At each instant it chooses consumption and portfolios subject to

$$\dot{a} = I[R(a^A(g) + a^B(g))] - c^A - c^B, \tag{116}$$

$$I[a^A(g)] = \bar{w}a, \qquad I[a^B(g)] = (1-\bar{w})a, \tag{117}$$

$$a^j(g) \geq -\bar{s}q(g), \qquad j \in \{A, B\}. \tag{118}$$

The share constraints are continuous investment-account restrictions, anticipated in the family's optimization. Dividends and capital gains accrue to these accounts, consumption is financed from pooled resources, and costless internal transfers and trading maintain their assigned market values. No resources are created. The family takes prices and firm-share short limits as given; in equilibrium $a = A$, $a^A(g) = q(g)z(g)$, and $a^B(g) = q(g)(1 - z(g))$.

The family's objective is

$$\int_0^\infty e^{-\rho t}\left\{u(c^A) + \chi u(c^B) + \mathcal{V}(G)\right\} dt, \quad G = I[v(g)a^A(g)], \quad \chi = \left(\frac{1-\eta}{\eta}\right)^\sigma. \tag{119}$$

The original u, v, $\mathcal{V}$, ρ, technologies and financial constraints are unchanged. Consumption sharing is $c^A = \eta C$ and $c^B = (1-\eta)C$. The counterfactual in Section 4 sets $\eta = \bar{w} = .33$. This consumption-sharing rule is an additional assumption, not a consequence of fixing the wealth share. Both ν and κ remain at the empirical 20 bp calibration of the full economy.

Let $\lambda = u'(c^A)$ be the current-value multiplier on the family budget. The sustainable-wealth valuation retains its original form,

$$m = \frac{\mathcal{V}'(G)}{\lambda} = (\eta C)^\sigma (1 + \kappa G)^{-1/2}.$$

The multipliers on the two account restrictions, in consumption-value units, are r^A and r^B. Portfolio optimality implies

$$R(g) + mv(g) \leq r^A, \qquad R(g) \leq r^B, \tag{120}$$

$$[r^A - R(g) - mv(g)]\,[a^A(g) + \bar{s}q(g)] = 0, \qquad [r^B - R(g)]\,[a^B(g) + \bar{s}q(g)] = 0. \tag{121}$$

For an interior ownership threshold g^*, $r^B = R^*$ and $r^A = R^* + mv(g^*)$, giving the same sharp sorting and pricing equations as in the full economy:

$$z(g) = -\bar{s} + (1 + 2\bar{s})\mathbf{1}\{g > g^*\}, \qquad R(g) = R^* - m[v(g) - v(g^*)]_+. \tag{122}$$

The family costate equation, however, is

$$\frac{\dot{C}}{C} = \frac{\bar{w}r^A + (1 - \bar{w})r^B - \rho}{\sigma} = \frac{R^* + \bar{w}mv(g^*) - \rho}{\sigma}. \tag{123}$$

The direct nonpecuniary saving motive is present, but a single weighted marginal return replaces two independent saving decisions. With binding short positions these marginal account returns are not the average pecuniary returns on the signed portfolios. The internal transfers recovered from equation (114) implement the family allocation; they do not make its accounts independently optimizing original households. At given price paths, the family's objective is concave on the utility domain and its budget, account restrictions, and position bounds are linear. Thus the stated optimality conditions, feasibility and the transversality condition characterize a household optimum, subject to the usual admissibility conditions.

J.3 Equilibrium and Numerical Solution The original firm equations and goods clearing are retained:

$$x(g) = (\mathcal{Q}(g) - 1)/\psi, \qquad \dot{k}(g) = k(g)x(g), \tag{124}$$

$$d(g) = k(g)^\alpha - \delta k(g) - k(g)x(g) - \tfrac{1}{2}\psi k(g)x(g)^2, \qquad C = I[d], \tag{125}$$

$$\dot{\mathcal{Q}}(g) = R(g)\mathcal{Q}(g) + \delta - \alpha k(g)^{\alpha-1} - \tfrac{1}{2}\psi x(g)^2, \qquad \dot{q}(g) = R(g)q(g) - d(g). \tag{126}$$

Differentiating $I[zq] = \bar{w}I[q]$ gives

$$\dot{g}^* = \frac{\bar{w}I[\dot{q}] - I[z\dot{q}]}{I[(\partial z/\partial g^*)q]}. \tag{127}$$

For the sharp cutoff, $I[(\partial z/\partial g^*)q] = -(1+2\bar{s})q(g^*)\phi(g^*)$; numerical calculations use the original logistic regularization and its derivative. This equation enforces the wealth restriction jointly with endogenous prices, rather than resetting a previously solved trajectory.

Define $d_k = \alpha k^{\alpha-1} - \delta - x - \psi x^2/2$, $d_{\mathcal{Q}} = -k\mathcal{Q}/\psi$, $W = m[v - v(g^*)]_+$, and $B_{\mathcal{Q}} = -W\mathcal{Q} + \delta - \alpha k^{\alpha-1} - \psi x^2/2$. Differentiating goods clearing and using equation (123) gives

$$R^* = \frac{\sigma I[d_k \dot{k} + d_{\mathcal{Q}} B_{\mathcal{Q}}] - C\bar{w}mv(g^*) + \rho C}{C + (\sigma/\psi)I[k\mathcal{Q}^2]}. \tag{128}$$

There are $3n+1$ differential unknowns: nodal $k, \mathcal{Q}, \log q$ and g^*. Initial conditions fix $k = \widehat{k}$ and the value-weighted account share. At the terminal horizon, $\mathcal{Q} = 1$ and $Rq = d$. Stationary reference allocations are separately solved from the fixed-share condition and $m = (\eta I[d])^{\sigma}\mathcal{V}'(G)$, with

$$R^*_{ss} = \rho - \bar{w}m_{ss}v(g^*_{ss}).$$

Consequently g^*_{ss} need not equal g_n. Bounded stationary continuation satisfies the family transversality condition $\lim_{t\to\infty} e^{-\rho t}\lambda_t a_t = 0$ and firm no-bubble conditions for the positive-return solutions reported here. Numerical convergence is not a uniqueness theorem.

J.4 Interpretation and Verification The ESG calculations integrate over the original score g and the fixed initial-capital density $\phi(g)$. No time-varying re-ranking is used. The family specification is imposed before solving the transition, with the full economy's calibrated preference parameters held fixed.

For any defined scalar outcome,

$$X_t^{\text{full}} - X_t^{\text{Benchmark}} = \left[X_t^{\text{fixed}} - X_t^{\text{Benchmark}}\right] + \left[X_t^{\text{full}} - X_t^{\text{fixed}}\right]. \tag{129}$$

Here Benchmark, fixed and full denote the all-Benchmark, fixed-wealth-share pooled-family and full economies, respectively. The last term includes all induced changes in prices, consumption, saving and allocation. It measures the equilibrium difference from this specified pooled comparison, including its saving and consumption-sharing closure. It is not a unique structural contribution of relative wealth alone. The ownership cutoff is undefined in the all-Benchmark economy, so its table entries compare levels in the full and fixed-share economies rather than fabricate a Benchmark cutoff. No claim of exact equivalence to CPSS preferences or its family objective is required or

made.

The reported counterfactual uses 81 score nodes and a 12,000-year terminal horizon, with the same ownership regularization as the full economy. A replication check of the original pooled-family specification gives a maximum wealth-share error below 1.3×10^{-12}, an independent Euler residual below 1.4×10^{-8}, and household-account budget residuals below 2.0×10^{-7} in model flow units. These checks concern the implemented regularized system. The counterfactual is not recalibrated to the full economy's initial portfolio-return differential.

K Details of the Policy Implementation Exercise

This appendix follows the three steps of Section 5: constructing the policy benchmark in an all-Benchmark economy, implementing its stationary allocation in the All-Advocate economy, and introducing Benchmark investors with Advocate's preferences held fixed. It states the equations used to obtain the stationary allocations and transition paths, explains the reporting conventions, and summarizes the numerical checks accompanying the saved solutions.

K.1 Environment and Measurement Production units are indexed by their initial-capital-weighted cleanliness rank $g \in [0,1]$, increasing from dirtier to cleaner firms. Let H_K denote the initial-capital-weighted distribution of Scope 1 emissions intensity. The corresponding decreasing intensity curve is $e(g) = H_K^{-1}(1-g)$. Integrating over dg incorporates the initial productive-capacity weights; it does not make physical emissions intensity uniform. The empirical construction uses the common sample of 1,170 issuers, the maintained exclusion of RMR, and net PPE weights. The policy exercise uses Scope 1, whereas the main emissions application in Section 4 uses Scope 1 and 2. Firm intensities and initial ranks remain fixed across all experiments.

The supplied monotone, mean-preserving piecewise cubic interpolation has mean intensity $\bar{e} = 427.664$ tonnes CO_2e per million dollars of revenue. Revenue is interpreted as output. All integrals below are over $g \in [0,1]$. The common initial capital allocation and technology parameters are

$$\hat{k} = \left(\frac{\alpha}{\rho+\delta}\right)^{1/(1-\alpha)}, \qquad \alpha = .33, \tag{130}$$

$$\rho = -4\log(.99), \qquad \delta = 1 - .975^4, \qquad \psi = 5.48. \tag{131}$$

Consumption utility is CES, $u(c) = (c^{1-\sigma} - 1)/(1-\sigma)$, with $\sigma = .5$ and EIS equal to two. Sustainable-wealth marginal utility is $(1+\kappa G)^{-1/2}$. The marginal sustainable-wealth value is

therefore

$$m_t = \frac{(c_t^A)^\sigma}{\sqrt{1+\kappa G_t}}. \tag{132}$$

The utility domain requires $1+\kappa G_t > 0$. In the All-Advocate economy, $c_t^A = C_t$.

Let $\mathcal{Q}_t(g)$ be the marginal value of installed capital, distinct from the equity value $q_t(g)$. Net investment per unit of capital is $x_t(g) = \dot{k}_t(g)/k_t(g)$. The maintained adjustment technology gives

$$x_t(g) = \frac{\mathcal{Q}_t(g)-1}{\psi}, \qquad \dot{k}_t(g) = k_t(g)x_t(g). \tag{133}$$

Gross investment includes replacement of depreciated capital. No additional irreversibility, utilization, or within-firm abatement margin is introduced. Aggregates are

$$K_t = \int k_t(g)dg, \quad Y_t = \int k_t(g)^\alpha dg, \quad E_t = \int e(g)k_t(g)^\alpha dg, \quad \mathcal{E}_t = E_t/Y_t. \tag{134}$$

Initially $Y_0 = \hat{k}^\alpha$, $E_0 = \bar{e}Y_0$, and $\mathcal{E}_0 = \bar{e}$.

K.2 Policy in the All-Benchmark Economy The government introduces a permanent emissions charge τ and an output subsidy τe_n at date zero, without prior anticipation. A firm's net payment and aggregate transfers to households are

$$T_t(g) = \tau[e(g)-e_n]k_t(g)^\alpha, \qquad B_t = \int T_t(g)dg = \tau(E_t - e_n Y_t). \tag{135}$$

The charge changes the marginal cost of emissions, while the output subsidy rewards production. This is the incentive structure studied in the output-based rebating and performance-standard literature (Fischer, 2001; Holland et al., 2009). The experiment calibrates a feasible policy to a specified target; it does not solve a welfare-maximization problem.

At stationarity, the financial return equals ρ and $\mathcal{Q}_P(g) = 1$. The firm's investment condition gives

$$k_P(g) = \hat{k}\{1+\tau[e_n - e(g)]\}^{1/(1-\alpha)}. \tag{136}$$

We determine τ and e_n jointly from

$$\int e(g)k_P(g)^\alpha dg = 0.9E_0, \tag{137}$$

$$\tau \int [e(g)-e_n]k_P(g)^\alpha dg = 0. \tag{138}$$

The second condition implies $e_n = E_P/Y_P$. It balances taxes and subsidies at the endogenous stationary allocation, rather than at the initial allocation. The solution is

$$\tau = 0.0001169, \qquad e_n = 386.33.$$

Because output is measured in million dollars, $10^6\tau = 116.9$ is the charge in dollars per tonne. This conversion inherits the revenue/output normalization and is not a social-cost-of-carbon estimate. The implied neutral rank solves $e(g_n) = e_n$ and is $g_n \simeq .233$. The multiplier $1+\tau[e_n - e(g)]$ remains positive for all modeled firms; its recorded minimum is approximately .3130.

For the transition, write R_t^P for the common financial return. Along with (133), the policy equilibrium satisfies

$$\dot{\mathcal{Q}}_t(g) = R_t^P \mathcal{Q}_t(g) + \delta - \{1 + \tau[e_n - e(g)]\}\alpha k_t(g)^{\alpha-1} - \frac{\psi}{2}x_t(g)^2, \tag{139}$$

$$d_t^P(g) = \{1 + \tau[e_n - e(g)]\}k_t(g)^\alpha - \delta k_t(g) - k_t(g)x_t(g) - \frac{\psi}{2}k_t(g)x_t(g)^2, \tag{140}$$

$$C_t = \int d_t^P(g)dg + B_t, \qquad \frac{\dot{C}_t}{C_t} = \frac{R_t^P - \rho}{\sigma}. \tag{141}$$

Substituting B_t into goods clearing cancels the fiscal payments from aggregate resources. The return R_t^P is endogenous during adjustment. The parameters τ and e_n remain fixed, with positive net revenue returned as lump-sum transfers along the computed transition. Budget balance is imposed only in steady state. The policy therefore approaches its emissions target through capital adjustment, rather than imposing an immediate quantity cap.

K.3 All-Advocate Implementation Remove the policy and let Advocate own every firm. Define preferences directly by

$$v(g) = \nu(\rho + \delta)\frac{\tau[e_n - e(g)]}{1 + \tau[e_n - e(g)]}. \tag{142}$$

The policy parameters determine the shape and neutral point of the preference function; $\nu > 0$ determines its scale. At the target capital allocation, an untaxed firm delivers

$$R_{\text{req}}(g) = \alpha k_P(g)^{\alpha-1} - \delta = \frac{\rho + \delta}{1 + \tau[e_n - e(g)]} - \delta = \rho - \frac{v(g)}{\nu}. \tag{143}$$

The stationary Euler condition for each firm is $R_{\text{req}}(g) + mv(g) = \rho$. Substitution gives

$$\left(m - \frac{1}{\nu}\right) v(g) = 0. \tag{144}$$

Consequently, $m\nu = 1$ supports the entire target capital allocation. At the neutral firm, $v(g_n) = 0$ and $R_{\text{req}}(g_n) = \rho$ automatically. Positive nonpecuniary benefits compensate for lower financial returns on cleaner firms; negative nonpecuniary benefits require higher financial returns on dirtier firms.

Choosing the scale consistently. The marginal valuation m depends on sustainable wealth, which itself depends on ν. To solve this consistency condition, first compute the target's untaxed dividends, equity prices, and aggregate consumption:

$$d_P(g) = k_P(g)^\alpha - \delta k_P(g), \qquad q_P(g) = \frac{d_P(g)}{R_{\text{req}}(g)}, \qquad C_P = \int d_P(g) dg. \tag{145}$$

Here q_P denotes equity in the untaxed All-Advocate economy at the target real allocation, not equity under the tax-and-subsidy policy. The required returns and these equity prices are determined by the target allocation before choosing ν. With

$$G_P = \int v(g) q_P(g) dg = \nu \int [\rho - R_{\text{req}}(g)] q_P(g) dg,$$

the condition $m\nu = 1$ becomes the scalar equation

$$C_P^{2\sigma} \nu^2 - \kappa\nu \int [\rho - R_{\text{req}}(g)] q_P(g) dg - 1 = 0. \tag{146}$$

The quadratic has exactly one positive root because its leading coefficient is positive and its constant term is negative. We choose that root and verify the unsquared condition and utility domain. Retaining $\kappa = 0.548$ from the CES Scope 1 calibration with $g_n = .5$ gives $\nu = 0.931$. That calibration targets a 20 bp initial portfolio return differential and a marginal-utility ratio of .90 on the common emissions sample. These two moments are not reimposed in the policy implementation exercise. Policy and All-Advocate match stationary capital, output, emissions, intensity, and aggregate consumption; they need not have the same equity prices.

The actual transition with fixed preferences. Let $R_t^c = \rho + \sigma \dot{C}_t / C_t$ be the common total consumption return. In the untaxed All-Advocate economy,

$$R_t(g) = R_t^c - m_t v(g), \qquad G_t = \int v(g) q_t(g) dg, \tag{147}$$

$$m_t = C_t^\sigma (1 + \kappa G_t)^{-1/2}, \qquad \dot{q}_t(g) = R_t(g) q_t(g) - d_t(g), \tag{148}$$

$$\dot{\mathcal{Q}}_t(g) = R_t(g) \mathcal{Q}_t(g) + \delta - \alpha k_t(g)^{\alpha - 1} - \frac{\psi}{2} x_t(g)^2, \tag{149}$$

$$d_t(g) = k_t(g)^\alpha - \delta k_t(g) - k_t(g) x_t(g) - \frac{\psi}{2} k_t(g) x_t(g)^2, \qquad C_t = \int d_t(g) dg. \tag{150}$$

Capital follows (133). Ownership is one for every firm, so $\int q_t(g) dg$ is Advocate's financial wealth. Initial capital is predetermined, while initial equity prices are part of the equilibrium solution. The same fixed function $v(g)$ applies at every date. In particular, $m_t \nu$ need not equal one outside the target steady state; its recorded initial value is approximately 1.00393.

Exact replication of the policy capital path would also reproduce investment and aggregate consumption, and hence require $R_t^c = R_t^P$. Comparing the two investment equations then yields the necessary pointwise condition

$$m_t v(g) = \tau [e_n - e(g)] \frac{\alpha k_t^P(g)^{\alpha - 1}}{\mathcal{Q}_t^P(g)}. \tag{151}$$

Away from the neutral firm, this can be written

$$m_t \nu = \frac{[k_t^P(g) / k_P(g)]^{\alpha - 1}}{\mathcal{Q}_t^P(g)}.$$

The right-hand side generally differs across firms during adjustment, whereas the left-hand side is common. Matching the stationary allocation therefore does not impose equality of the transition paths. Their close emissions timing in Section 5 is a computed outcome.

K.4 Implementation with Advocate and Benchmark Investors

Introduce Benchmark investors without changing $v(g)$, ν, κ, or initial capital. Let $\bar{s} = .05$ be the per-firm short-sale bound and g_t^* the ownership threshold. In the sharp-cutoff formulation, Advocate's ownership fraction is

$$z_t^A(g) = -\bar{s} + (1 + 2\bar{s}) \mathbf{1}\{g > g_t^*\}. \tag{152}$$

Benchmark owns $1 - z_t^A(g)$. Sustainable wealth and marginal valuation are

$$G_t = \int v(g) z_t^A(g) q_t(g) dg, \qquad m_t = (c_t^A)^\sigma (1 + \kappa G_t)^{-1/2}.$$

The financial return schedule and household Euler equations are

$$R_t(g) = R_t^* - m_t \max\{v(g) - v(g_t^*), 0\}, \tag{153}$$

$$\frac{\dot{c}_t^A}{c_t^A} = \frac{R_t^* + m_t v(g_t^*) - \rho}{\sigma}, \qquad \frac{\dot{c}_t^B}{c_t^B} = \frac{R_t^* - \rho}{\sigma}. \tag{154}$$

Here R_t^* is Benchmark's marginal financial return, rather than necessarily its portfolio-average return. The untaxed firm equations (150) apply with this return schedule, and goods clearing is $c_t^A + c_t^B = \int d_t(g) dg$.

Financial wealth and household budgets satisfy

$$a_t^A = \int z_t^A(g) q_t(g) dg, \qquad \dot{a}_t^A = \int z_t^A(g) q_t(g) R_t(g) dg - c_t^A, \tag{155}$$

$$a_t^B = \int [1 - z_t^A(g)] q_t(g) dg, \qquad \dot{a}_t^B = \int [1 - z_t^A(g)] q_t(g) R_t(g) dg - c_t^B. \tag{156}$$

Initial conditions are $k_0(g) = \hat{k}$ and

$$w_0^A = \frac{a_0^A}{a_0^A + a_0^B} \in \{.33, .66, .99\}.$$

These are net financial wealth shares, not population shares or fixed portfolio mandates. There are no transfers that keep the shares constant during adjustment.

The stationary mixed allocation. With both consumptions positive and constant, the household Euler equations imply $R_M^* = \rho$ and $v(g_M^*) = 0$, hence $g_M^* = g_n$. Write m_M for the stationary

marginal valuation and use (152) with cutoff g_n. Solve

$$R_M(g) = \rho - m_M \max\{v(g), 0\}, \tag{157}$$

$$k_M(g) = \left[\frac{\alpha}{R_M(g) + \delta}\right]^{1/(1-\alpha)}, \tag{158}$$

$$d_M(g) = k_M(g)^\alpha - \delta k_M(g), \qquad q_M(g) = d_M(g)/R_M(g), \tag{159}$$

$$c_M^A = \int z_M^A(g) d_M(g) dg, \tag{160}$$

$$m_M = \frac{(c_M^A)^\sigma}{\sqrt{1 + \kappa \int v(g) z_M^A(g) q_M(g) dg}}. \tag{161}$$

The solution has $m_M \nu \simeq .874$ and an Advocate wealth share of 81.58 percent. The stationary equations contain no initial wealth share, and all three computed paths approach this interior allocation. This does not establish global uniqueness or convergence from every initial condition.

For $g < g_n$, the preference weight is negative, but Benchmark supplies finance at ρ. Thus $k_M(g) = \hat{k}$, whereas the policy requires contraction. Even if $m_M \nu = 1$, the dirty-firm return would remain ρ instead of the higher $\rho - v(g)/\nu$ required for implementation. Cleaner firms expand, reducing aggregate intensity, but output expansion more than offsets that improvement in total emissions.

Why 99 percent initial wealth differs from sole ownership. Let L be the market-value share of firms below the contemporaneous ownership cutoff. The sharp-cutoff rule implies

$$w^B = (1 + \bar{s})L - \bar{s}(1 - L), \qquad L = \frac{w^B + \bar{s}}{1 + 2\bar{s}}.$$

At $w^B = .01$ and $\bar{s} = .05$, $L = 5.45$ percent. Benchmark's gross long holdings are $(1 + \bar{s})L = 5.73$ percent of aggregate equity, financed by one percent net wealth and short positions of 4.73 percent. This is an ownership-region calculation: away from stationarity the cutoff need not equal the policy-neutral rank. Thus a small net Benchmark share can support larger gross financing positions. Removing Benchmark altogether also removes its trading opportunities, and is distinct from giving it a small initial net wealth share.

K.5 Numerical Implementation and Reporting Policy and All-Advocate transitions are solved on a 300-year horizon, with stationary continuation for longer comparisons. Mixed-economy

transitions retain the logistic ownership regularization in equation (106); calculations on 81 and 82 rank nodes are compared in the replication files. Stationary mixed allocations are separately computed with the sharp ownership threshold. The resulting sharp and regularized stationary approximations are close but not identical: the sharp stationary increase in total emissions is .113%, compared with approximately .113% at the endpoint of the 82-node transition. We retain this distinction when checking the saved paths.

Emissions are integrated using the Scope 1 intensity curve and the output quadrature described in Appendix I.6. Changes in K, Y, E, and E/Y are measured relative to their initial all-Benchmark levels. An emissions-reduction milestone of b percent is the first date at which $E_t/E_0 \leq 1 - b/100$. A temporary crossing does not imply sustained implementation; the ten percent objective is a stationary target. Numerical checks cover the stationary target and budget balance, the implementation condition $m\nu = 1$, household Euler equations, goods clearing, and comparison across grids. These checks assess the computed allocations and do not assign a welfare value to the emissions target.